

LegacyDB Copilot Complete Local Evaluation

Run 3772275f-f219-4999-9c31-28f8dd5dc411 | Local MySQL | 125 scenarios

Executive summary

Readiness 10.6/100. Production ready: No. Accuracy 0.0%; human review 100.0%.

Top 10 improvements

1. Enable AI_REASONING_ENABLED for the benchmark-equivalent application runtime.
2. Make exact business-key matches override fuzzy entity candidates.
3. Map each extracted entity to discovered BusinessKey and CorrelationId columns.
4. Increase ranking weight for tables containing exact entity rows.
5. Collect correlated evidence from primary, integration, exception, and audit tables.
6. Block confirmed root-cause language unless cited evidence supports it.
7. Add expected programmable-object metadata or a documented MySQL-equivalent policy.
8. Validate generated SQL against live MySQL metadata before execution.
9. Add regression tests for each failure category and exact-key scenario.
10. Re-run the benchmark with production-equivalent feature flags and calibrated Judge costs.

Metric	Value
Overall Accuracy	0.0
Precision	0
Recall	0.0
F1 Score	0
Average Confidence	0.0174
Average Execution Time Seconds	1.9368
Average Token Usage	11876.016
Estimated Cost Usd	0.0
Average Deterministic Score	0.0
Average Ai Judge Score	1.7376
Pass Percentage	0.0
Fail Percentage	1.0
Human Review Percentage	1.0

Overall accuracy (%)

Accuracy	0.00
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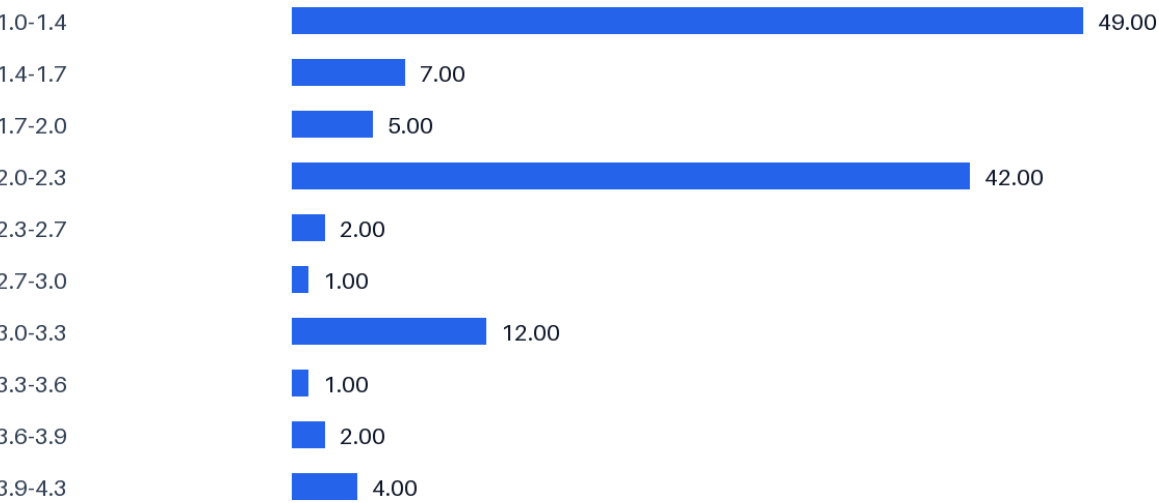
Deterministic score by domain

banking	0.00
clinic	0.00
orders	0.00
payroll	0.00
shipping	0.00

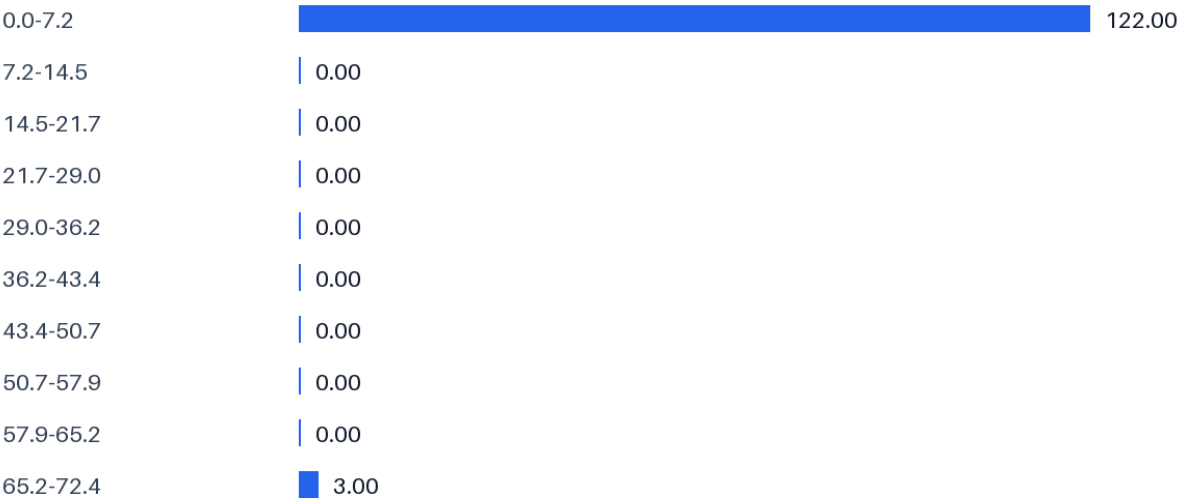
Deterministic score by difficulty

easy	0.00
expert	0.00
hard	0.00
medium	0.00

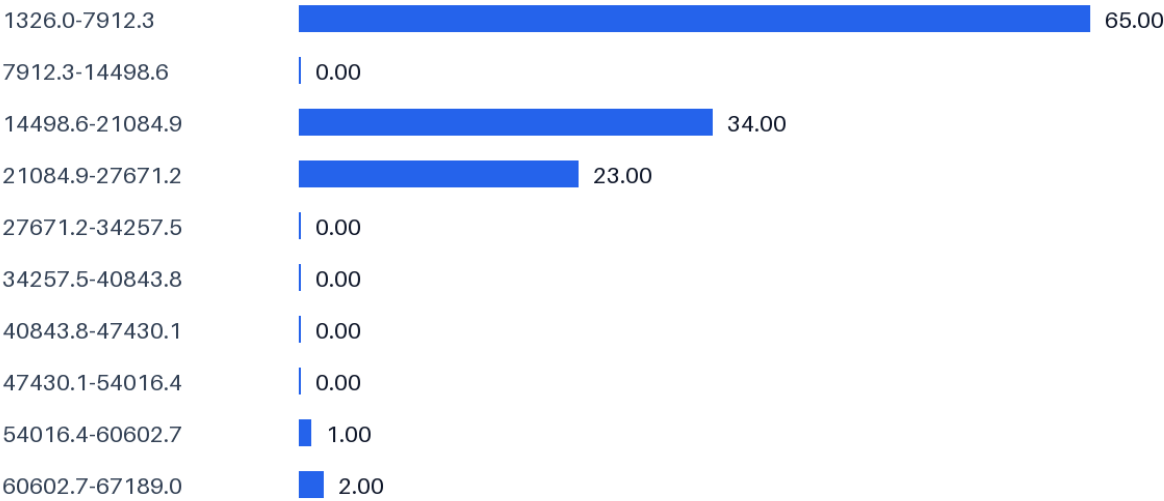
Runtime distribution (seconds)



Confidence distribution (%)



Token usage distribution



Estimated cost by domain (USD)

banking	0.00
clinic	0.00
orders	0.00
payroll	0.00
shipping	0.00

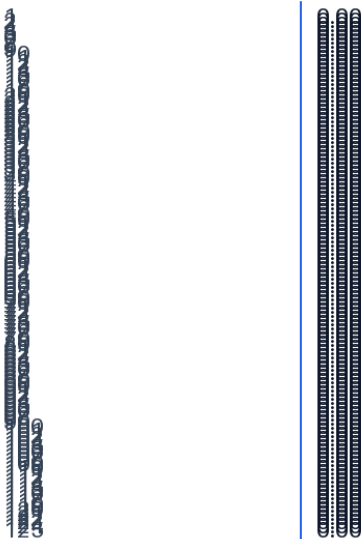
Confusion matrix

true_positive	0.00
false_positive	0.00
false_negative	125.00
true_negative	0.00

Pass / Fail counts



Deterministic score by execution order



Failed-scenario validation

banking-benchmark-001 - Critical

Question: Investigate why payment BNK-2026-0001-A is marked Failed even though its request was accepted.

Expected answer: {"scenario_id": "banking-benchmark-001", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "exact_entity_lookup", "subcategory": "exact_entity_lookup", "difficulty": "easy", "question": "Investigate why payment BNK-2026-0001-A is marked Failed even though its request was accepted.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-001/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-001/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-001/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-001/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["BNK-2026-0001-A"], "expected_root_cause_concepts": ["the exact business key resolves to one failed record"], "expected_tables": ["transactions", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.transactions", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-101", "BNK-2026-0001-A", "the exact business key resolves to one failed record"], "acceptable_fix_concepts": ["restore the record through the owning workflow after validating the request"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_1"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing exact entity lookup without exposing ground truth to runtime code.", "expected_relationships": ["eval.transactions.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_o

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225708-CDBFB5DA Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=BNK-2026-0001-A, business_idenfier=BNK-2026-0001-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=payment, possible_column=BNK, possible_table_or_column=is, possible_table_or_column=marked, possible_table_or_column=even, possible_table_or_column=though, possible_table_or_column=its, possible_table_or_column=request, possible_table_or_column=was, possible_table_or_column=accepted AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why payment BNK-2026-0001-A is marked Failed even though its request was accepted. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: payment_instructions - Target Reason: Selected from main problem phrase 'failed even though' (failed); parent and secondary-cause terms do not override the target. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: integration_messages, audit_history, exceptions, batch_runs, transfers, vw_banking_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 90228294-4cd2-4b65-9ba4-331c8ea5dea1 - database_name: eval_banking - schema_name: default - database_engine: mysql - connection_string_database: eval_banking - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-7159

8ec8fb87|90228294-4cd2-4b65-9ba4-331c8ea5dea1|eval_banking - Discovered tables: account_balances, accounts, audit_history, batch_runs, beneficiaries, cards, compliance_case

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in payment_instructions: No rows returned - SQL - Summarize analytical results in transfers: No rows returned - SQL - Summarize analytical results in integration_messages: 1 row(s) returned - SQL - Prove requested entity exists in payment_instructions: No rows returned - SQL - Prove requested entity exists in integration_messages: No rows returned - SQL - Prove requested entity exists in audit_history: No rows returned - SQL - Inspect relevant rows in payment_instructions: No rows returned - SQL - Inspect relevant rows in integration_messages: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

banking-benchmark-002 - Critical

Question: Investigate the failed payment reported as BNK-2026-0002; resolve the complete identifier before diagnosing it.

Expected answer: {"scenario_id": "banking-benchmark-002", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "partial_entity_resolution", "subcategory": "partial_entity_resolution", "difficulty": "easy", "question": "Investigate the failed payment reported as BNK-2026-0002; resolve the complete identifier before diagnosing it.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-002/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-002/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-002/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-002/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["BNK-2026-0002-A"], "expected_root_cause_concepts": ["a safe candidate lookup resolves the partial identifier to one record"], "expected_tables": ["transfers", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.transfers", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-102", "BNK-2026-0002-A", "a safe candidate lookup resolves the partial identifier to one record"], "acceptable_fix_concepts": ["confirm the resolved identifier and retry through the supported workflow"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_2"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing partial entity resolution without exposing ground truth to runtime code.", "expected_relationships": ["eval.transfers.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrel"]

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Entity extraction

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Prefer exact business identifiers and reject surrogate candidate IDs when an exact key is supplied.

banking-benchmark-003 - Critical

Question: Investigate payment BNK-2026-0003; support reported a failure but did not provide the final suffix.

Expected answer: {"scenario_id": "banking-benchmark-003", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "ambiguous_entity_resolution", "subcategory": "ambiguous_entity_resolution", "difficulty": "hard", "question": "Investigate payment BNK-2026-0003; support reported a failure but did not provide the final suffix.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-003/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-003/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-003/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-003/cleanup.sql", "expected_response_type": "multiple_possible_causes", "expected_entities": ["BNK-2026-0003"], "expected_root_cause_concepts": ["the partial identifier matches multiple plausible records and investigation must stop"], "expected_tables": ["payment_instructions", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.payment_instructions", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-103", "BNK-2026-0003", "the partial identifier matches multiple plausible records and investigation must stop"], "acceptable_fix_concepts": ["request user selection of the correct candidate before continuing"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_3"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing ambiguous entity resolution without exposing ground truth to runtime code.", "expected_relationships": ["eval.payment_instructions.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another corr"]}

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

banking-benchmark-004 - Critical

Question: Investigate why completed payment BNK-2026-0004-A has no downstream processing record.

Expected answer: {"scenario_id": "banking-benchmark-004", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "missing_downstream_record", "subcategory": "missing_downstream_record", "difficulty": "medium", "question": "Investigate why completed payment BNK-2026-0004-A has no downstream processing record.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-004/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-004/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-004/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-004/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["BNK-2026-0004-A"], "expected_root_cause_concepts": ["the upstream record completed but its correlated integration record is missing"], "expected_tables": ["accounts", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.accounts", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-104", "BNK-2026-0004-A", "the upstream record completed but its correlated integration record is missing"], "acceptable_fix_concepts": ["reconcile the missing downstream work through an idempotent"]}

workflow"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_4"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing missing downstream record without exposing ground truth to runtime code.", "expected_relationships": ["eval.accounts.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated applica

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

banking-benchmark-005 - Critical

Question: Investigate why payment BNK-2026-0005-A produced two processing messages for one business request.

Expected answer: {"scenario_id": "banking-benchmark-005", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "duplicate_transaction", "subcategory": "duplicate_transaction", "difficulty": "medium", "question": "Investigate why payment BNK-2026-0005-A produced two processing messages for one business request.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-005/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-005/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-005/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-005/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["BNK-2026-0005-A"], "expected_root_cause_concepts": ["two correlated processing events represent one business action"], "expected_tables": ["beneficiaries", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.beneficiaries", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-105", "BNK-2026-0005-A", "two correlated processing events represent one business action"], "acceptable_fix_concepts": ["deduplicate using a stable business idempotency key"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_5"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing duplicate transaction without exposing ground truth to runtime code.", "expected_relationships": ["eval.beneficiaries.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

banking-benchmark-006 - Critical

Question: Investigate why payment BNK-2026-0006-A stopped between validation and completion.

Expected answer: {"scenario_id": "banking-benchmark-006", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "workflow_interruption", "subcategory": "workflow_interruption", "difficulty": "medium", "question": "Investigate why payment BNK-2026-0006-A stopped between validation and completion.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-006/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-006/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-006/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-006/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["BNK-2026-0006-A"], "expected_root_cause_concepts": ["workflow history ends at an intermediate state with an exception"], "expected_tables": ["loans", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.loans", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-106", "BNK-2026-0006-A", "workflow history ends at an intermediate state with an exception"], "acceptable_fix_concepts": ["resume from the last committed workflow checkpoint"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_6"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing workflow interruption without exposing ground truth to runtime code.", "expected_relationships": ["eval.loans.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_objects": ["eval.loans"], "

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

banking-benchmark-007 - Critical

Question: Investigate why exception handling left payment BNK-2026-0007-A in Processing.

Expected answer: {"scenario_id": "banking-benchmark-007", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "exception_handling", "subcategory": "exception_handling", "difficulty": "medium", "question": "Investigate why exception handling left payment BNK-2026-0007-A in Processing.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-007/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-007/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-007/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-007/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["BNK-2026-0007-A"], "expected_root_cause_concepts": ["the

exception was recorded but no compensating transition ran"], "expected_tables": ["cards", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.cards", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-107", "BNK-2026-0007-A", "the exception was recorded but no compensating transition ran"], "acceptable_fix_concepts": ["add observable compensation and terminal-state handling"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_7"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing exception handling without exposing ground truth to runtime code.", "expected_relationships": ["eval.cards.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_objects": ["eval.cards"], "required_workf

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225802-FF0F89B3 Investigation Mode: INVESTIGATION Detected Intent: PROCESS_FLOW_BREAK Response Type: insufficient_evidence Extracted Entities: exact_id_or_code=EX-BNK-2026-0007-A, business_identifier=EX-BNK-2026-0007-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=exception, possible_table_or_column=handling, possible_table_or_column=left, possible_table_or_column=payment, possible_column=BNK, possible_table_or_column=in, possible_table_or_column=Processing AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: PROCESS_FLOW_BREAK - User Goal: Answer the user question with database evidence: Investigate why exception handling left payment BNK-2026-0007-A in Processing. - Working Hypothesis: The business process should move from the current step to the next expected step. ## Stage 2 - Discover Context - Target Object: exceptions - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: payment_instructions, vw_banking_operations_1, vw_banking_operations_2, vw_banking_operations_3, vw_banking_operations_4, vw_banking_operations_5 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 90228294-4cd2-4b65-9ba4-331c8ea5dea1 - database_name: eval_banking - schema_name: default - database_engine: mysql - connection_string_database: eval_banking - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|90228294-4cd2-4b65-9ba4-331c8ea5dea1|eval_banking - Discovered tables: account_balances, accounts, audit_history, batch_runs, beneficiaries, cards, compliance_cases, customers, evaluation_marker, exce

Root cause: Configuration issue

Evidence collected: - Supplied business key found in returned evidence: EX-BNK-2026-0007-A. - Affected rows were returned by safe SQL evidence.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

banking-benchmark-008 - Critical

Question: Investigate why external integration for payment BNK-2026-0008-A was rejected.

Expected answer: {"scenario_id": "banking-benchmark-008", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "integration_failure",

"subcategory": "integration_failure", "difficulty": "medium", "question": "Investigate why external integration for payment BNK-2026-0008-A was rejected.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-008/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-008/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-008/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-008/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["BNK-2026-0008-A"], "expected_root_cause_concepts": ["the correlated integration message failed and blocked completion"], "expected_tables": ["fraud_alerts", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.fraud_alerts", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-108", "BNK-2026-0008-A", "the correlated integration message failed and blocked completion"], "acceptable_fix_concepts": ["correct the integration contract and safely replay the message"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_8"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing integration failure without exposing ground truth to runtime code.", "expected_relationships": ["eval.fraud_alerts.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_obje

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225815-6A8609D0 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=MSG-BNK-2026-0008-A, business_idenfier=MSG-BNK-2026-0008-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=external, possible_table_or_column=integration, possible_table_or_column=for, possible_table_or_column=payment, possible_column=BNK, possible_table_or_column=was, possible_table_or_column=rejected AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why external integration for payment BNK-2026-0008-A was rejected. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: integration_messages - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: payment_instructions, vw_banking_operations_1, vw_banking_operations_2, vw_banking_operations_3, vw_banking_operations_4, vw_banking_operations_5 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 90228294-4cd2-4b65-9ba4-331c8ea5dea1 - database_name: eval_banking - schema_name: default - database_engine: mysql - connection_string_database: eval_banking - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|90228294-4cd2-4b65-9ba4-331c8ea5dea1|eval_banking - Discovered tables: account_balances, accounts, audit_history, batch_runs, beneficiaries, cards, compliance_cases, customers, evaluation_marke

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in payment_instructions: 1 row(s) returned - SQL - Summarize analytical results in integration_messages: 3 row(s) returned - SQL - Prove requested entity exists in integration_messages: 1 row(s) returned - SQL - Prove requested entity exists in payment_instructions: No rows

returned - SQL - Inspect relevant rows in integration_messages: 1 row(s) returned - SQL - Inspect relevant rows in payment_instructions: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

banking-benchmark-009 - Critical

Question: Investigate why payment BNK-2026-0009-A remains Queued behind older work.

Expected answer: {"scenario_id": "banking-benchmark-009", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "queue_backlog", "subcategory": "queue_backlog", "difficulty": "medium", "question": "Investigate why payment BNK-2026-0009-A remains Queued behind older work.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-009/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-009/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-009/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-009/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["BNK-2026-0009-A"], "expected_root_cause_concepts": ["queue evidence shows the entity has not been consumed"], "expected_tables": ["transactions", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.transactions", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-109", "BNK-2026-0009-A", "queue evidence shows the entity has not been consumed"], "acceptable_fix_concepts": ["drain the queue with bounded, monitored processing"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_1"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing queue backlog without exposing ground truth to runtime code.", "expected_relationships": ["eval.transactions.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_objects": ["eval.transactions"], "required_workflow": ["eval"]}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

banking-benchmark-010 - Critical

Question: Investigate why all retries for payment BNK-2026-0010-A failed without a terminal status.

Expected answer: {"scenario_id": "banking-benchmark-010", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "retry_failure", "subcategory": "retry_failure", "difficulty": "hard", "question": "Investigate why all retries for payment BNK-2026-0010-A failed without a terminal status.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-010/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-010/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-010/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-010/cleanup.sql", "expected_response_type":

"confirmed_root_cause", "expected_entities": ["BNK-2026-0010-A"], "expected_root_cause_concepts": ["retry evidence records exhausted attempts and the entity remained pending"], "expected_tables": ["transfers", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.transfers", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-110", "BNK-2026-0010-A", "retry evidence records exhausted attempts and the entity remained pending"], "acceptable_fix_concepts": ["route exhausted retries to reviewed dead-letter recovery"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_2"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing retry failure without exposing ground truth to runtime code.", "expected_relationships": ["eval.transfers.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_objects":

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225836-8FE4608C Investigation Mode: INVESTIGATION Detected Intent: PROCESS_FLOW_BREAK Response Type: insufficient_evidence Extracted Entities: exact_id_or_code=BNK-2026-0010-A, business_identifier=BNK-2026-0010-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=all, possible_table_or_column=retries, possible_table_or_column=for, possible_table_or_column=payment, possible_column=BNK, possible_table_or_column=terminal, possible_table_or_column=status AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: PROCESS_FLOW_BREAK - User Goal: Answer the user question with database evidence: Investigate why all retries for payment BNK-2026-0010-A failed without a terminal status. - Working Hypothesis: The business process should move from the current step to the next expected step. ## Stage 2 - Discover Context - Target Object: payment_instructions - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: integration_messages, audit_history, exceptions, batch_runs, transfers, vw_banking_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 90228294-4cd2-4b65-9ba4-331c8ea5dea1 - database_name: eval_banking - schema_name: default - database_engine: mysql - connection_string_database: eval_banking - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|90228294-4cd2-4b65-9ba4-331c8ea5dea1|eval_banking - Discovered tables: account_balances, accounts, audit_history, batch_runs, beneficiaries, cards, compliance_cases, customers, evaluation_marker, exceptions, fraud_alerts, integration_message

Root cause: Configuration issue

Evidence collected: - No confirming rows were returned for the reported issue.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

banking-benchmark-011 - Critical

Question: Investigate why audit history says payment BNK-2026-0011-A completed while its business record says Failed.

Expected answer: {"scenario_id": "banking-benchmark-011", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "audit_history_inconsistency", "subcategory": "audit_history_inconsistency", "difficulty": "hard", "question": "Investigate why audit history says payment BNK-2026-0011-A completed while its business record says Failed.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-011/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-011/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-011/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-011/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["BNK-2026-0011-A"], "expected_root_cause_concepts": ["business and audit states conflict for the same correlation"], "expected_tables": ["payment_instructions", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.payment_instructions", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-111", "BNK-2026-0011-A", "business and audit states conflict for the same correlation"], "acceptable_fix_concepts": ["reconcile history from the authoritative transaction boundary"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_3"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing audit history inconsistency without exposing ground truth to runtime code.", "expected_relationships": ["eval.payment_instructions.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content"], "u

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225846-A96A2B76 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=BNK-2026-0011-A, business_idenfier=BNK-2026-0011-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=audit, possible_table_or_column=history, possible_table_or_column=says, possible_table_or_column=payment, possible_column=BNK, possible_table_or_column=completed, possible_table_or_column=while, possible_table_or_column=its, possible_table_or_column=business AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why audit history says payment BNK-2026-0011-A completed while its business record says Failed. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: payment_instructions - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: payment_instructions, integration_messages, exceptions, batch_runs, transfers, vw_banking_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 90228294-4cd2-4b65-9ba4-331c8ea5dea1 - database_name: eval_banking - schema_name: default - database_engine: mysql - connection_string_database: eval_banking - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|90228294-4cd2-4b65-9ba4-331c8ea5dea1|eval_banking - Discovered tables: account_balances, accounts, audit_history, batch_runs, beneficiaries, cards, compl

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in audit_history: No rows returned - SQL - Summarize analytical results in payment_instructions: 1 row(s) returned - SQL - Summarize analytical results in transfers: No rows returned - SQL - Prove requested entity exists in audit_history: No rows returned - SQL - Prove requested entity exists in payment_instructions: 1 row(s) returned - SQL - Prove requested entity exists in integration_messages: No rows returned - SQL - Inspect relevant rows in audit_history: No rows returned - SQL - Inspect relevant rows in payment_instructions: 1 row(s) returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

banking-benchmark-012 - Critical

Question: Investigate why payment BNK-2026-0012-A failed validation after a reference lookup.

Expected answer: {"scenario_id": "banking-benchmark-012", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "missing_reference_data", "subcategory": "missing_reference_data", "difficulty": "easy", "question": "Investigate why payment BNK-2026-0012-A failed validation after a reference lookup.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-012/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-012/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-012/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-012/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["BNK-2026-0012-A"], "expected_root_cause_concepts": ["required active reference data is absent from discovered relationships"], "expected_tables": ["accounts", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.accounts", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-112", "BNK-2026-0012-A", "required active reference data is absent from discovered relationships"], "acceptable_fix_concepts": ["restore governed reference data before replaying the workflow"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_4"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing missing reference data without exposing ground truth to runtime code.", "expected_relationships": ["eval.accounts.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_bu":

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

banking-benchmark-013 - Critical

Question: Investigate why workflow procedure processing left payment BNK-2026-0013-A in the wrong status.

Expected answer: {"scenario_id": "banking-benchmark-013", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "stored_procedure_defect", "subcategory": "stored_procedure_defect", "difficulty": "hard", "question": "Investigate why workflow procedure processing left payment BNK-2026-0013-A in the wrong status.", "baseline_script":

"evaluation_scenarios/banking/banking-benchmark-013/baseline_reset.sql", "setup_script":
"evaluation_scenarios/banking/banking-benchmark-013/inject.sql", "verification_script":
"evaluation_scenarios/banking/banking-benchmark-013/verify.sql", "cleanup_script":
"evaluation_scenarios/banking/banking-benchmark-013/cleanup.sql", "expected_response_type":
"confirmed_root_cause", "expected_entities": ["BNK-2026-0013-A"], "expected_root_cause_concepts":
["procedure definition targets the workflow object but the observed transition is incorrect"], "expected_tables":
["beneficiaries", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey",
"Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.beneficiaries",
"eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence":
["EVAL-BANKING-113", "BNK-2026-0013-A", "procedure definition targets the workflow object but the observed
transition is incorrect"], "acceptable_fix_concepts": ["version and test the procedure correction before controlled
replay"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred
from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object",
"wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
"unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,
"active": true, "expected_procedures": ["eval.usp_banking_workflow_5"], "expected_functions":
["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [],
"business_description": "A synthetic banking production-support incident testing stored procedure defect without
exposing ground truth to runtime code.", "expected_relationships": ["eval.beneficiaries.CorrelationId ->
eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational
evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer m

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

banking-benchmark-014 - Critical

Question: Investigate why updating payment BNK-2026-0014-A produced no consistent audit outcome.

Expected answer: {"scenario_id": "banking-benchmark-014", "domain": "banking", "database_engine":
"sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "trigger_failure",
"subcategory": "trigger_failure", "difficulty": "hard", "question": "Investigate why updating payment
BNK-2026-0014-A produced no consistent audit outcome.", "baseline_script":
"evaluation_scenarios/banking/banking-benchmark-014/baseline_reset.sql", "setup_script":
"evaluation_scenarios/banking/banking-benchmark-014/inject.sql", "verification_script":
"evaluation_scenarios/banking/banking-benchmark-014/verify.sql", "cleanup_script":
"evaluation_scenarios/banking/banking-benchmark-014/cleanup.sql", "expected_response_type":
"confirmed_root_cause", "expected_entities": ["BNK-2026-0014-A"], "expected_root_cause_concepts": ["trigger
metadata is relevant but the expected correlated audit transition is absent"], "expected_tables": ["loans",
"integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status",
"EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.loans",
"eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence":
["EVAL-BANKING-114", "BNK-2026-0014-A", "trigger metadata is relevant but the expected correlated audit
transition is absent"], "acceptable_fix_concepts": ["repair trigger behavior and reconcile missing audit events"],
"prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from
synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object",
"wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
"unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,
"active": true, "expected_procedures": ["eval.usp_banking_workflow_6"], "expected_functions":
["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [],
"business_description": "A synthetic banking production-support incident testing trigger failure without exposing

ground truth to runtime code.", "expected_relationships": ["eval.loans.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_busine

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225914-D6CB4C98 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=AUD-BNK-2026-0014-A, business_identifier=AUD-BNK-2026-0014-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=updating, possible_table_or_column=payment, possible_column=BNK, possible_table_or_column=produced, possible_table_or_column=no, possible_table_or_column=consistent, possible_table_or_column=audit, possible_table_or_column=outcome AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why updating payment BNK-2026-0014-A produced no consistent audit outcome. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: audit_history - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: payment_instructions, vw_banking_operations_1, vw_banking_operations_2, vw_banking_operations_3, vw_banking_operations_4, vw_banking_operations_5 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 90228294-4cd2-4b65-9ba4-331c8ea5dea1 - database_name: eval_banking - schema_name: default - database_engine: mysql - connection_string_database: eval_banking - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|90228294-4cd2-4b65-9ba4-331c8ea5dea1|eval_banking - Discovered tables: account_balances, accounts, audit_history, batch_runs, beneficiaries, cards, compliance

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in payment_instructions: 1 row(s) returned - SQL - Summarize analytical results in audit_history: 3 row(s) returned - SQL - Prove requested entity exists in audit_history: 1 row(s) returned - SQL - Prove requested entity exists in payment_instructions: No rows returned - SQL - Inspect relevant rows in audit_history: 1 row(s) returned - SQL - Inspect relevant rows in payment_instructions: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

banking-benchmark-015 - Critical

Question: Investigate why batch processing failed for payment BNK-2026-0015-A while neighboring items completed.

Expected answer: {"scenario_id": "banking-benchmark-015", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "batch_processing_failure", "subcategory": "batch_processing_failure", "difficulty": "medium", "question": "Investigate why batch processing failed for payment BNK-2026-0015-A while neighboring items completed.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-015/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-015/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-015/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-015/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["BNK-2026-0015-A"], "expected_root_cause_concepts": ["the correlated batch or integration evidence isolates one failed item"], "expected_tables": ["cards",

"integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.cards", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-115", "BNK-2026-0015-A", "the correlated batch or integration evidence isolates one failed item"], "acceptable_fix_concepts": ["restart only the failed item with checkpoint protection"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_7"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing batch processing failure without exposing ground truth to runtime code.", "expected_relationships": ["eval.cards.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "r

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

banking-benchmark-016 - Critical

Question: Investigate intermittent competing updates for payment BNK-2026-0016-A that produced inconsistent terminal states.

Expected answer: {"scenario_id": "banking-benchmark-016", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "concurrency_race_condition", "subcategory": "concurrency_race_condition", "difficulty": "expert", "question": "Investigate intermittent competing updates for payment BNK-2026-0016-A that produced inconsistent terminal states.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-016/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-016/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-016/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-016/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["BNK-2026-0016-A"], "expected_root_cause_concepts": ["same-correlation events at overlapping timestamps show a lost-update race"], "expected_tables": ["fraud_alerts", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.fraud_alerts", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-116", "BNK-2026-0016-A", "same-correlation events at overlapping timestamps show a lost-update race"], "acceptable_fix_concepts": ["use optimistic concurrency and retry on version conflict"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_8"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing concurrency race condition without exposing ground truth to runtime code.", "expected_relationships": ["eval.fraud_alerts.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content"]}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

banking-benchmark-017 - Critical

Question: Investigate why payment BNK-2026-0017-A reports completion although downstream changes were rolled back.

Expected answer: {"scenario_id": "banking-benchmark-017", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "transaction_rollback", "subcategory": "transaction_rollback", "difficulty": "hard", "question": "Investigate why payment BNK-2026-0017-A reports completion although downstream changes were rolled back.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-017/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-017/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-017/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-017/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["BNK-2026-0017-A"], "expected_root_cause_concepts": ["exception and audit evidence show the unit of work rolled back"], "expected_tables": ["transactions", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.transactions", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-117", "BNK-2026-0017-A", "exception and audit evidence show the unit of work rolled back"], "acceptable_fix_concepts": ["make the status transition atomic with downstream writes"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_1"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing transaction rollback without exposing ground truth to runtime code.", "expected_relationships": ["eval.transactions.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "requi

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

banking-benchmark-018 - Critical

Question: Investigate why replaying payment BNK-2026-0018-A repeated a previously accepted operation.

Expected answer: {"scenario_id": "banking-benchmark-018", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "idempotency_issue", "subcategory": "idempotency_issue", "difficulty": "expert", "question": "Investigate why replaying payment

BNK-2026-0018-A repeated a previously accepted operation.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-018/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-018/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-018/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-018/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["BNK-2026-0018-A"], "expected_root_cause_concepts": ["replay created a second correlated operation without an idempotency guard"], "expected_tables": ["transfers", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.transfers", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-118", "BNK-2026-0018-A", "replay created a second correlated operation without an idempotency guard"], "acceptable_fix_concepts": ["enforce a unique idempotency key across retries"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_2"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing idempotency issue without exposing ground truth to runtime code.", "expected_relationships": ["eval.transfers.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_obj

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

banking-benchmark-019 - Critical

Question: Investigate why payment BNK-2026-0019-A displays Completed while failure evidence remains open.

Expected answer: {"scenario_id": "banking-benchmark-019", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "incorrect_business_status", "subcategory": "incorrect_business_status", "difficulty": "easy", "question": "Investigate why payment BNK-2026-0019-A displays Completed while failure evidence remains open.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-019/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-019/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-019/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-019/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["BNK-2026-0019-A"], "expected_root_cause_concepts": ["the business status contradicts active correlated failure evidence"], "expected_tables": ["payment_instructions", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.payment_instructions", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-119", "BNK-2026-0019-A", "the business status contradicts active correlated failure evidence"], "acceptable_fix_concepts": ["derive status from authoritative workflow completion"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_3"], "expected_functions":

["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing incorrect business status without exposing ground truth to runtime code.", "expected_relationships": ["eval.payment_instructions.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated appl"]

Actual answer: Investigation Complete. Investigation ID: INV-20260716-230002-133B4AE5 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=BNK-2026-0019-A, business_idenfier=BNK-2026-0019-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=payment, possible_column=BNK, possible_table_or_column=displays, possible_table_or_column=Completed, possible_table_or_column=while, possible_table_or_column=failure, possible_table_or_column=evidence, possible_table_or_column=remains, possible_table_or_column=open AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why payment BNK-2026-0019-A displays Completed while failure evidence remains open. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: payment_instructions - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: integration_messages, audit_history, exceptions, batch_runs, vw_banking_operations_1, vw_banking_operations_2 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 90228294-4cd2-4b65-9ba4-331c8ea5dea1 - database_name: eval_banking - schema_name: default - database_engine: mysql - connection_string_database: eval_banking - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|90228294-4cd2-4b65-9ba4-331c8ea5dea1|eval_banking - Discovered tables: account_balances, accounts, audit_history, batch_runs, beneficiaries, cards, com

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in payment_instructions: 1 row(s) returned - SQL - Prove requested entity exists in payment_instructions: 1 row(s) returned - SQL - Prove requested entity exists in integration_messages: No rows returned - SQL - Prove requested entity exists in audit_history: No rows returned - SQL - Inspect relevant rows in payment_instructions: 1 row(s) returned - SQL - Inspect relevant rows in integration_messages: No rows returned - SQL - Inspect relevant rows in audit_history: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

banking-benchmark-020 - Critical

Question: Investigate the end-to-end failure of payment BNK-2026-0020-A across business, integration, exception, and audit records.

Expected answer: {"scenario_id": "banking-benchmark-020", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "multi_table_investigation", "subcategory": "multi_table_investigation", "difficulty": "medium", "question": "Investigate the end-to-end failure of payment BNK-2026-0020-A across business, integration, exception, and audit records.", "baseline_script": "evaluation_scenarios/banking/banking-benchmark-020/baseline_reset.sql", "setup_script": "evaluation_scenarios/banking/banking-benchmark-020/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-benchmark-020/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-benchmark-020/cleanup.sql", "expected_response_type":

"confirmed_root_cause", "expected_entities": ["BNK-2026-0020-A"], "expected_root_cause_concepts": ["correlated multi-table evidence traces the failure to one interrupted transition"], "expected_tables": ["accounts", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.accounts", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-BANKING-120", "BNK-2026-0020-A", "correlated multi-table evidence traces the failure to one interrupted transition"], "acceptable_fix_concepts": ["repair the transition and reconcile each correlated downstream state"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_4"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "A synthetic banking production-support incident testing multi table investigation without exposing ground truth to runtime code.", "expected_relationships": ["eval.accounts.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answe

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

banking-pilot-001 - Critical

Question: Why is transfer TRF-3101 completed without a matching balance movement?

Expected answer: {"scenario_id": "banking-pilot-001", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "missing_workflow_output", "difficulty": "medium", "question": "Why is transfer TRF-3101 completed without a matching balance movement?", "baseline_script": "evaluation_databases/banking/sql/02_seed.sql", "setup_script": "evaluation_scenarios/banking/banking-pilot-001/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-pilot-001/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-pilot-001/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["TRF-3101"], "expected_root_cause_concepts": ["required downstream work item was not created after a completed upstream transition"], "expected_tables": ["exceptions", "integration_messages", "transactions"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_banking_operations_1"], "required_evidence": ["EVAL-BANKING-001", "TRF-3101", "Exception"], "acceptable_fix_concepts": ["transactional workflow completion"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_1"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225614-93AA60FD Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=MSG-TRF-3101,

business_identifier=MSG-TRF-3101, possible_table_or_column=Why, possible_table_or_column=is, possible_table_or_column=transfer, possible_column=TRF, possible_table_or_column=completed, possible_table_or_column=matching, possible_table_or_column=balance, possible_table_or_column=movement
AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Why is transfer TRF-3101 completed without a matching balance movement? - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: integration_messages - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: account_balances, integration_messages, transactions, payment_instructions, loans - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 90228294-4cd2-4b65-9ba4-331c8ea5dea1 - database_name: eval_banking - schema_name: default - database_engine: mysql - connection_string_database: eval_banking - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|90228294-4cd2-4b65-9ba4-331c8ea5dea1|eval_banking - Discovered tables: account_balances, accounts, audit_history, batch_runs, beneficiaries, cards, compliance_cases, customers, evaluation_marker, exceptions, fraud_alerts, integration_messages, loan_schedules, loans, payment_instructions, transactions, transfers -

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in transfers: 1 row(s) returned - SQL - Summarize analytical results in account_balances: 1 row(s) returned - SQL - Prove requested entity exists in transfers: No rows returned - SQL - Prove requested entity exists in account_balances: No rows returned - SQL - Prove requested entity exists in integration_messages: 1 row(s) returned - SQL - Inspect relevant rows in transfers: No rows returned - SQL - Inspect relevant rows in account_balances: No rows returned - SQL - Inspect relevant rows in integration_messages: 1 row(s) returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

banking-pilot-002 - Critical

Question: Why was payment instruction PMT-3102 posted twice?

Expected answer: {"scenario_id": "banking-pilot-002", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "duplicate_integration", "difficulty": "hard", "question": "Why was payment instruction PMT-3102 posted twice?", "baseline_script": "evaluation_databases/banking/sql/02_seed.sql", "setup_script": "evaluation_scenarios/banking/banking-pilot-002/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-pilot-002/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-pilot-002/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PMT-3102"], "expected_root_cause_concepts": ["integration retry inserted the same business event without an idempotency guard"], "expected_tables": ["exceptions", "integration_messages", "transfers"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_banking_operations_1"], "required_evidence": ["EVAL-BANKING-002", "PMT-3102", "Exception"], "acceptable_fix_concepts": ["idempotency key"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_2"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": []}

"evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

banking-pilot-003 - Critical

Question: Why was account ACC-3103 charged by an inactive beneficiary?

Expected answer: {"scenario_id": "banking-pilot-003", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "incorrect_selection", "difficulty": "hard", "question": "Why was account ACC-3103 charged by an inactive beneficiary?", "baseline_script": "evaluation_databases/banking/sql/02_seed.sql", "setup_script": "evaluation_scenarios/banking/banking-pilot-003/inject.sql", "verification_script": "evaluation_scenarios/banking/banking-pilot-003/verify.sql", "cleanup_script": "evaluation_scenarios/banking/banking-pilot-003/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ACC-3103"], "expected_root_cause_concepts": ["selection used historical assignment instead of the active work item"], "expected_tables": ["exceptions", "integration_messages", "beneficiaries"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_banking_operations_1"], "required_evidence": ["EVAL-BANKING-003", "ACC-3103", "Exception"], "acceptable_fix_concepts": ["active assignment filtering"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_3"], "expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

banking-pilot-004 - Critical

Question: Why does settlement batch BAT-3104 remain Running?

Expected answer: {"scenario_id": "banking-pilot-004", "domain": "banking", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "stalled_status", "difficulty": "medium", "question": "Why does settlement batch BAT-3104 remain Running?",

"baseline_script": "evaluation_databases/banking/sql/02_seed.sql", "setup_script":
"evaluation_scenarios/banking/banking-pilot-004/inject.sql", "verification_script":
"evaluation_scenarios/banking/banking-pilot-004/verify.sql", "cleanup_script":
"evaluation_scenarios/banking/banking-pilot-004/cleanup.sql", "expected_response_type":
"confirmed_root_cause", "expected_entities": ["BAT-3104"], "expected_root_cause_concepts": ["integration
failure prevented the terminal status transition"], "expected_tables": ["exceptions", "integration_messages",
"payment_instructions"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"],
"expected_database_objects": ["eval.vw_banking_operations_1"], "required_evidence": ["EVAL-BANKING-004",
"BAT-3104", "Exception"], "acceptable_fix_concepts": ["retry and reconciliation"], "prohibited_claims": ["objects or
evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules":
["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"],
"scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_4"],
"expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"],
"expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [],
"evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation":
[], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false,
"human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [],
"estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

banking-pilot-005 - Critical

Question: Is there enough evidence to confirm transaction TXN-3105 was fraudulent?

Expected answer: {"scenario_id": "banking-pilot-005", "domain": "banking", "database_engine": "sqlserver",
"database_version": "Azure SQL Database / SQL Server 2022", "category": "evidence_safety", "subcategory":
"insufficient_evidence", "difficulty": "hard", "question": "Is there enough evidence to confirm transaction TXN-3105
was fraudulent?", "baseline_script": "evaluation_databases/banking/sql/02_seed.sql", "setup_script":
"evaluation_scenarios/banking/banking-pilot-005/inject.sql", "verification_script":
"evaluation_scenarios/banking/banking-pilot-005/verify.sql", "cleanup_script":
"evaluation_scenarios/banking/banking-pilot-005/cleanup.sql", "expected_response_type":
"insufficient_evidence", "expected_entities": ["TXN-3105"], "expected_root_cause_concepts": [],
"expected_tables": ["exceptions", "integration_messages", "loans"], "expected_columns": ["BusinessKey",
"Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_banking_operations_1"],
"required_evidence": ["EVAL-BANKING-005", "TXN-3105", "absence of conclusive timestamps"],
"acceptable_fix_concepts": ["collect additional timestamp evidence"], "prohibited_claims": ["objects or evidence
not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules":
["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"],
"scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_banking_workflow_5"],
"expected_functions": ["eval.fn_banking_active_status"], "expected_triggers": ["eval.tr_accounts_audit"],
"expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [],
"evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation":
[], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false,
"human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [],
"estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225654-C177F526 Investigation Mode:
INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type:
multiple_possible_causes Extracted Entities: exact_id_or_code=MSG-TXN-3105,

business_identifier=MSG-TXN-3105, possible_table_or_column=Is, possible_table_or_column=there, possible_table_or_column=enough, possible_table_or_column=evidence, possible_table_or_column=to, possible_table_or_column=confirm, possible_table_or_column=transaction, possible_column=TXN, possible_table_or_column=was, possible_table_or_column=fraudulent AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Is there enough evidence to confirm transaction TXN-3105 was fraudulent? - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: integration_messages - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: fraud_alerts, integration_messages, exceptions, transfers, payment_instructions - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 90228294-4cd2-4b65-9ba4-331c8ea5dea1 - database_name: eval_banking - schema_name: default - database_engine: mysql - connection_string_database: eval_banking - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|90228294-4cd2-4b65-9ba4-331c8ea5dea1|eval_banking - Discovered tables: account_balances, accounts, audit_history, batch_runs, beneficiaries, cards, compliance_cases, customers, evaluation_marker, exceptions, fraud_alerts, integration_messages, loan_sch

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in transactions: 1 row(s) returned - SQL - Summarize analytical results in fraud_alerts: 1 row(s) returned - SQL - Prove requested entity exists in transactions: No rows returned - SQL - Prove requested entity exists in fraud_alerts: No rows returned - SQL - Prove requested entity exists in integration_messages: 1 row(s) returned - SQL - Inspect relevant rows in transactions: No rows returned - SQL - Inspect relevant rows in fraud_alerts: No rows returned - SQL - Inspect relevant rows in integration_messages: 1 row(s) returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

clinic-benchmark-001 - Critical

Question: Investigate why patient service CLN-2026-0001-A is marked Failed even though its request was accepted.

Expected answer: {"scenario_id": "clinic-benchmark-001", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "exact_entity_lookup", "subcategory": "exact_entity_lookup", "difficulty": "easy", "question": "Investigate why patient service CLN-2026-0001-A is marked Failed even though its request was accepted.", "baseline_script": "evaluation_scenarios/clinic/clinic-benchmark-001/baseline_reset.sql", "setup_script": "evaluation_scenarios/clinic/clinic-benchmark-001/inject.sql", "verification_script": "evaluation_scenarios/clinic/clinic-benchmark-001/verify.sql", "cleanup_script": "evaluation_scenarios/clinic/clinic-benchmark-001/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["CLN-2026-0001-A"], "expected_root_cause_concepts": ["the exact business key resolves to one failed record"], "expected_tables": ["appointments", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.appointments", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-101", "CLN-2026-0001-A", "the exact business key resolves to one failed record"], "acceptable_fix_concepts": ["restore the record through the owning workflow after validating the request"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object",

"wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
"unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt", "scenario_version": 1,
"active": true, "expected_procedures": ["eval.usp_clinic_workflow_1"], "expected_functions":
["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [],
"business_description": "A synthetic clinic production-support incident testing exact entity lookup without
exposing ground truth to runtime code.", "expected_relationships": ["eval.appointments.CorrelationId ->
eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational
evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content",
"unrelated application framework tables"], "required_business_object

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225704-DAD37D57 Investigation Mode:
INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type:
multiple_possible_causes Extracted Entities: exact_id_or_code=CLN-2026-0001-A,
business_identifier=CLN-2026-0001-A, possible_table_or_column=Investigate, possible_table_or_column=why,
possible_table_or_column=patient, possible_table_or_column=service, possible_column=CLN,
possible_table_or_column=is, possible_table_or_column=marked, possible_table_or_column=even,
possible_table_or_column=though, possible_table_or_column=its, possible_table_or_column=request,
possible_table_or_column=was, possible_table_or_column=accepted AI-assisted reasoning: Disabled Reason:
AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable
Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 -
Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require
live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal:
GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate
why patient service CLN-2026-0001-A is marked Failed even though its request was accepted. - Working
Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target
Object: patients - Target Reason: Selected from main problem phrase 'failed even though' (failed); parent and
secondary-cause terms do not override the target. - Business Key: Not determined (No non-primary natural key
column was identified.) - Supporting Objects: insurance_policies, appointments, integration_messages,
audit_history, exceptions, vw_clinic_operations_1 - Upstream Objects: None discovered - Downstream Objects:
None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed -
Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id:
2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 5c43b992-e80b-4bc6-a80a-fb83d1832a8c -
database_name: eval_clinic - schema_name: default - database_engine: mysql - connection_string_database:
eval_clinic - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598e
c8fb87|5c43b992-e80b-4bc6-a80a-fb83d1832a8c|eval_clinic - Discovered tables: appointments, audit_history,
claims, clinics, diagnoses

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in patients: No rows returned - SQL - Summarize
analytical results in insurance_policies: No rows returned - SQL - Summarize analytical results in appointments: 1
row(s) returned - SQL - Prove requested entity exists in patients: No rows returned - SQL - Prove requested
entity exists in insurance_policies: No rows returned - SQL - Prove requested entity exists in appointments: 1
row(s) returned - SQL - Inspect relevant rows in patients: No rows returned - SQL - Inspect relevant rows in
insurance_policies: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

clinic-benchmark-002 - Critical

Question: Investigate the failed patient service reported as CLN-2026-0002; resolve the complete identifier
before diagnosing it.

Expected answer: {"scenario_id": "clinic-benchmark-002", "domain": "clinic", "database_engine": "sqlserver",
"database_version": "Azure SQL Database / SQL Server 2022", "category": "partial_entity_resolution",
"subcategory": "partial_entity_resolution", "difficulty": "easy", "question": "Investigate the failed patient service
reported as CLN-2026-0002; resolve the complete identifier before diagnosing it.", "baseline_script":

"evaluation_scenarios/clinic/clinic-benchmark-002/baseline_reset.sql", "setup_script":
"evaluation_scenarios/clinic/clinic-benchmark-002/inject.sql", "verification_script":
"evaluation_scenarios/clinic/clinic-benchmark-002/verify.sql", "cleanup_script":
"evaluation_scenarios/clinic/clinic-benchmark-002/cleanup.sql", "expected_response_type":
"confirmed_root_cause", "expected_entities": ["CLN-2026-0002-A"], "expected_root_cause_concepts": ["a safe candidate lookup resolves the partial identifier to one record"], "expected_tables": ["encounters",
"integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status",
"EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.encounters",
"eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-102",
"CLN-2026-0002-A", "a safe candidate lookup resolves the partial identifier to one record"],
"acceptable_fix_concepts": ["confirm the resolved identifier and retry through the supported workflow"],
"prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object",
"wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
"unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,
"active": true, "expected_procedures": ["eval.usp_clinic_workflow_2"], "expected_functions":
["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [],
"business_description": "A synthetic clinic production-support incident testing partial entity resolution without exposing ground truth to runtime code.", "expected_relationships": ["eval.encounters.CorrelationId ->
eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content",
"unrelat

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

clinic-benchmark-003 - Critical

Question: Investigate patient service CLN-2026-0003; support reported a failure but did not provide the final suffix.

Expected answer: {"scenario_id": "clinic-benchmark-003", "domain": "clinic", "database_engine": "sqlserver",
"database_version": "Azure SQL Database / SQL Server 2022", "category": "ambiguous_entity_resolution",
"subcategory": "ambiguous_entity_resolution", "difficulty": "hard", "question": "Investigate patient service CLN-2026-0003; support reported a failure but did not provide the final suffix.", "baseline_script":
"evaluation_scenarios/clinic/clinic-benchmark-003/baseline_reset.sql", "setup_script":
"evaluation_scenarios/clinic/clinic-benchmark-003/inject.sql", "verification_script":
"evaluation_scenarios/clinic/clinic-benchmark-003/verify.sql", "cleanup_script":
"evaluation_scenarios/clinic/clinic-benchmark-003/cleanup.sql", "expected_response_type":
"multiple_possible_causes", "expected_entities": ["CLN-2026-0003"], "expected_root_cause_concepts": ["the partial identifier matches multiple plausible records and investigation must stop"], "expected_tables": ["claims",
"integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status",
"EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.claims",
"eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-103",
"CLN-2026-0003", "the partial identifier matches multiple plausible records and investigation must stop"],
"acceptable_fix_concepts": ["request user selection of the correct candidate before continuing"],
"prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object",
"wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
"unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,
"active": true, "expected_procedures": ["eval.usp_clinic_workflow_3"], "expected_functions":

["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [],
"business_description": "A synthetic clinic production-support incident testing ambiguous entity resolution without
exposing ground truth to runtime code.", "expected_relationships": ["eval.claims.CorrelationId ->
eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational
evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

clinic-benchmark-004 - Critical

Question: Investigate why completed patient service CLN-2026-0004-A has no downstream processing record.

Expected answer: {"scenario_id": "clinic-benchmark-004", "domain": "clinic", "database_engine": "sqlserver",
"database_version": "Azure SQL Database / SQL Server 2022", "category": "missing_downstream_record",
"subcategory": "missing_downstream_record", "difficulty": "medium", "question": "Investigate why completed
patient service CLN-2026-0004-A has no downstream processing record.", "baseline_script":
"evaluation_scenarios/clinic/clinic-benchmark-004/baseline_reset.sql", "setup_script":
"evaluation_scenarios/clinic/clinic-benchmark-004/inject.sql", "verification_script":
"evaluation_scenarios/clinic/clinic-benchmark-004/verify.sql", "cleanup_script":
"evaluation_scenarios/clinic/clinic-benchmark-004/cleanup.sql", "expected_response_type":
"confirmed_root_cause", "expected_entities": ["CLN-2026-0004-A"], "expected_root_cause_concepts": ["the
upstream record completed but its correlated integration record is missing"], "expected_tables": ["lab_orders",
"integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status",
"EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.lab_orders",
"eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-104",
"CLN-2026-0004-A", "the upstream record completed but its correlated integration record is missing"],
"acceptable_fix_concepts": ["reconcile the missing downstream work through an idempotent workflow"],
"prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from
synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object",
"wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
"unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,
"active": true, "expected_procedures": ["eval.usp_clinic_workflow_4"], "expected_functions":
["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [],
"business_description": "A synthetic clinic production-support incident testing missing downstream record without
exposing ground truth to runtime code.", "expected_relationships": ["eval.lab_orders.CorrelationId ->
eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational
evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content",
"unrelated applic

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

clinic-benchmark-005 - Critical

Question: Investigate why patient service CLN-2026-0005-A produced two processing messages for one business request.

Expected answer: {"scenario_id": "clinic-benchmark-005", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "duplicate_transaction", "subcategory": "duplicate_transaction", "difficulty": "medium", "question": "Investigate why patient service CLN-2026-0005-A produced two processing messages for one business request.", "baseline_script": "evaluation_scenarios/clinic/clinic-benchmark-005/baseline_reset.sql", "setup_script": "evaluation_scenarios/clinic/clinic-benchmark-005/inject.sql", "verification_script": "evaluation_scenarios/clinic/clinic-benchmark-005/verify.sql", "cleanup_script": "evaluation_scenarios/clinic/clinic-benchmark-005/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["CLN-2026-0005-A"], "expected_root_cause_concepts": ["two correlated processing events represent one business action"], "expected_tables": ["lab_results", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.lab_results", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-105", "CLN-2026-0005-A", "two correlated processing events represent one business action"], "acceptable_fix_concepts": ["deduplicate using a stable business idempotency key"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_clinic_workflow_5"], "expected_functions": ["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [], "business_description": "A synthetic clinic production-support incident testing duplicate transaction without exposing ground truth to runtime code.", "expected_relationships": ["eval.lab_results.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_o

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225739-3D1B25B1 Investigation Mode: INVESTIGATION Detected Intent: DUPLICATE_DATA Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=MSG-CLN-2026-0005-A, business_identifier=MSG-CLN-2026-0005-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=patient, possible_table_or_column=service, possible_column=CLN, possible_table_or_column=produced, possible_table_or_column=two, possible_table_or_column=processing, possible_table_or_column=messages, possible_table_or_column=for, possible_table_or_column=one, possible_table_or_column=business, possible_table_or_column=request AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: DUPLICATE_DATA - User Goal: Answer the user question with database evidence: Investigate why patient service CLN-2026-0005-A produced two processing messages for one business request. - Working Hypothesis: There should be only one valid record for the affected business key. ## Stage 2 - Discover Context - Target Object: integration_messages - Target Reason: Selected from main problem phrase 'two processing messages' (duplicate); parent and secondary-cause terms do not override the target. - Business Key: BusinessKey (Inferred from duplicate evidence grouped by a non-primary business column.) - Supporting Objects: patients, insurance_policies, appointments, procedures_performed, providers, vw_clinic_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 5c43b992-e80b-4bc6-a80a-fb83d1832a8c - database_name: eval_clinic - schema_name: default - database_engine: mysql - connection_string_database: eval_clinic - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|5c43b992-e80b-4bc6-a80a-fb83d1832a8c|eval_clinic - Discovered tables: appointme

Root cause: Configuration issue

Evidence collected: - SQL - Prove requested entity exists in integration_messages: 1 row(s) returned - SQL - Prove requested entity exists in patients: No rows returned - SQL - Prove requested entity exists in insurance_policies: No rows returned - SQL - Inspect relevant rows in integration_messages: 1 row(s) returned - SQL - Inspect relevant rows in patients: No rows returned - SQL - Inspect relevant rows in insurance_policies: No rows returned - SQL - Find duplicate business keys in integration_messages: No rows returned - SQL - Inspect duplicate rows in integration_messages: 1 row(s) returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

clinic-benchmark-006 - Critical

Question: Investigate why patient service CLN-2026-0006-A stopped between validation and completion.

Expected answer: {"scenario_id": "clinic-benchmark-006", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "workflow_interruption", "subcategory": "workflow_interruption", "difficulty": "medium", "question": "Investigate why patient service CLN-2026-0006-A stopped between validation and completion.", "baseline_script": "evaluation_scenarios/clinic/clinic-benchmark-006/baseline_reset.sql", "setup_script": "evaluation_scenarios/clinic/clinic-benchmark-006/inject.sql", "verification_script": "evaluation_scenarios/clinic/clinic-benchmark-006/verify.sql", "cleanup_script": "evaluation_scenarios/clinic/clinic-benchmark-006/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["CLN-2026-0006-A"], "expected_root_cause_concepts": ["workflow history ends at an intermediate state with an exception"], "expected_tables": ["prescriptions", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.prescriptions", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-106", "CLN-2026-0006-A", "workflow history ends at an intermediate state with an exception"], "acceptable_fix_concepts": ["resume from the last committed workflow checkpoint"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_clinic_workflow_6"], "expected_functions": ["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [], "business_description": "A synthetic clinic production-support incident testing workflow interruption without exposing ground truth to runtime code.", "expected_relationships": ["eval.prescriptions.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_objects"}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

clinic-benchmark-007 - Critical

Question: Investigate why exception handling left patient service CLN-2026-0007-A in Processing.

Expected answer: {"scenario_id": "clinic-benchmark-007", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "exception_handling", "subcategory": "exception_handling", "difficulty": "medium", "question": "Investigate why exception handling left patient service

CLN-2026-0007-A in Processing.", "baseline_script":
 "evaluation_scenarios/clinic/clinic-benchmark-007/baseline_reset.sql", "setup_script":
 "evaluation_scenarios/clinic/clinic-benchmark-007/inject.sql", "verification_script":
 "evaluation_scenarios/clinic/clinic-benchmark-007/verify.sql", "cleanup_script":
 "evaluation_scenarios/clinic/clinic-benchmark-007/cleanup.sql", "expected_response_type":
 "confirmed_root_cause", "expected_entities": ["CLN-2026-0007-A"], "expected_root_cause_concepts": ["the
 exception was recorded but no compensating transition ran"], "expected_tables": ["payments",
 "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status",
 "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.payments",
 "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-107",
 "CLN-2026-0007-A", "the exception was recorded but no compensating transition ran"],
 "acceptable_fix_concepts": ["add observable compensation and terminal-state handling"], "prohibited_claims":
 ["objects or evidence not returned by the database", "production impact inferred from synthetic data"],
 "critical_failure_rules": ["fabricated_evidence", "invented_database_object",
 "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
 "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,
 "active": true, "expected_procedures": ["eval.usp_clinic_workflow_7"], "expected_functions":
 ["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [],
 "business_description": "A synthetic clinic production-support incident testing exception handling without
 exposing ground truth to runtime code.", "expected_relationships": ["eval.payments.CorrelationId ->
 eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational
 evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content",
 "unrelated application framework tables"], "required_business_objects": ["eval.payments"], "require

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225758-EFAEF24A Investigation Mode:
 INVESTIGATION Detected Intent: PROCESS_FLOW_BREAK Response Type: insufficient_evidence Extracted
 Entities: exact_id_or_code=EX-CLN-2026-0007-A, business_identifier=EX-CLN-2026-0007-A,
 possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=exception,
 possible_table_or_column=handling, possible_table_or_column=left, possible_table_or_column=patient,
 possible_table_or_column=service, possible_column=CLN, possible_table_or_column=in,
 possible_table_or_column=Processing AI-assisted reasoning: Disabled Reason:
 AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable
 Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 -
 Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require
 live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal:
 PROCESS_FLOW_BREAK - User Goal: Answer the user question with database evidence: Investigate why
 exception handling left patient service CLN-2026-0007-A in Processing. - Working Hypothesis: The business
 process should move from the current step to the next expected step. ## Stage 2 - Discover Context - Target
 Object: exceptions - Target Reason: Selected from metadata/question matches plus SQL evidence on those
 matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No
 non-primary natural key column was identified.) - Supporting Objects: patients, insurance_policies, appointments,
 vw_clinic_operations_1, vw_clinic_operations_2, vw_clinic_operations_3 - Upstream Objects: None discovered -
 Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read
 path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id:
 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 5c43b992-e80b-4bc6-a80a-fb83d1832a8c -
 database_name: eval_clinic - schema_name: default - database_engine: mysql - connection_string_database:
 eval_clinic - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598e
 c8fb87|5c43b992-e80b-4bc6-a80a-fb83d1832a8c|eval_clinic - Discovered tables: appointments, audit_history,
 claims, clinics, diagnoses, encounters, evaluation_marker, exceptions, insurance_policies, i

Root cause: Configuration issue

Evidence collected: - Supplied business key found in returned evidence: EX-CLN-2026-0007-A. - Affected rows
 were returned by safe SQL evidence.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

clinic-benchmark-008 - Critical

Question: Investigate why external integration for patient service CLN-2026-0008-A was rejected.

Expected answer: {"scenario_id": "clinic-benchmark-008", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "integration_failure", "subcategory": "integration_failure", "difficulty": "medium", "question": "Investigate why external integration for patient service CLN-2026-0008-A was rejected.", "baseline_script": "evaluation_scenarios/clinic/clinic-benchmark-008/baseline_reset.sql", "setup_script": "evaluation_scenarios/clinic/clinic-benchmark-008/inject.sql", "verification_script": "evaluation_scenarios/clinic/clinic-benchmark-008/verify.sql", "cleanup_script": "evaluation_scenarios/clinic/clinic-benchmark-008/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["CLN-2026-0008-A"], "expected_root_cause_concepts": ["the correlated integration message failed and blocked completion"], "expected_tables": ["procedures_performed", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.procedures_performed", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-108", "CLN-2026-0008-A", "the correlated integration message failed and blocked completion"], "acceptable_fix_concepts": ["correct the integration contract and safely replay the message"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_clinic_workflow_8"], "expected_functions": ["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [], "business_description": "A synthetic clinic production-support incident testing integration failure without exposing ground truth to runtime code.", "expected_relationships": ["eval.procedures_performed.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "req

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225810-A0EAE362 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=MSG-CLN-2026-0008-A, business_identifier=MSG-CLN-2026-0008-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=external, possible_table_or_column=integration, possible_table_or_column=for, possible_table_or_column=patient, possible_table_or_column=service, possible_column=CLN, possible_table_or_column=was, possible_table_or_column=rejected AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why external integration for patient service CLN-2026-0008-A was rejected. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: integration_messages - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: patients, insurance_policies, appointments, vw_clinic_operations_1, vw_clinic_operations_2, vw_clinic_operations_3 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 5c43b992-e80b-4bc6-a80a-fb83d1832a8c - database_name: eval_clinic - schema_name: default - database_engine: mysql - connection_string_database: eval_clinic - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|5c43b992-e80b-4bc6-a80a-fb83d1832a8c|eval_clinic - Discovered tables: appointments, audit_history, claims, clinics, diagnoses, encounters,

evaluation_marker, exceptions, insurance_poli

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in patients: 1 row(s) returned - SQL - Summarize analytical results in integration_messages: 3 row(s) returned - SQL - Summarize analytical results in insurance_policies: 1 row(s) returned - SQL - Prove requested entity exists in integration_messages: 1 row(s) returned - SQL - Prove requested entity exists in patients: No rows returned - SQL - Prove requested entity exists in insurance_policies: No rows returned - SQL - Inspect relevant rows in integration_messages: 1 row(s) returned - SQL - Inspect relevant rows in patients: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

clinic-benchmark-009 - Critical

Question: Investigate why patient service CLN-2026-0009-A remains Queued behind older work.

Expected answer: {"scenario_id": "clinic-benchmark-009", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "queue_backlog", "subcategory": "queue_backlog", "difficulty": "medium", "question": "Investigate why patient service CLN-2026-0009-A remains Queued behind older work.", "baseline_script": "evaluation_scenarios/clinic/clinic-benchmark-009/baseline_reset.sql", "setup_script": "evaluation_scenarios/clinic/clinic-benchmark-009/inject.sql", "verification_script": "evaluation_scenarios/clinic/clinic-benchmark-009/verify.sql", "cleanup_script": "evaluation_scenarios/clinic/clinic-benchmark-009/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["CLN-2026-0009-A"], "expected_root_cause_concepts": ["queue evidence shows the entity has not been consumed"], "expected_tables": ["appointments", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.appointments", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-109", "CLN-2026-0009-A", "queue evidence shows the entity has not been consumed"], "acceptable_fix_concepts": ["drain the queue with bounded, monitored processing"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_clinic_workflow_1"], "expected_functions": ["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [], "business_description": "A synthetic clinic production-support incident testing queue backlog without exposing ground truth to runtime code.", "expected_relationships": ["eval.appointments.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_objects": ["eval.appointments"], "required_workflow": ["eval.usp_c"]}

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225822-87E23BF6 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=CLN-2026-0009-A, business_idenfier=CLN-2026-0009-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=patient, possible_table_or_column=service, possible_column=CLN, possible_table_or_column=remains, possible_table_or_column=Queued, possible_table_or_column=behind, possible_table_or_column=older, possible_table_or_column=work AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why patient service CLN-2026-0009-A remains Queued behind older work. - Working Hypothesis: The database

evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: appointments - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: insurance_policies, appointments, vw_clinic_operations_1, vw_clinic_operations_2, vw_clinic_operations_3, vw_clinic_operations_4 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 5c43b992-e80b-4bc6-a80a-fb83d1832a8c - database_name: eval_clinic - schema_name: default - database_engine: mysql - connection_string_database: eval_clinic - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|5c43b992-e80b-4bc6-a80a-fb83d1832a8c|eval_clinic - Discovered tables: appointments, audit_history, claims, clinics, diagnoses, encounters, evaluation_marker, exceptions, insurance_policies, integr

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in patients: 1 row(s) returned - SQL - Summarize analytical results in insurance_policies: 1 row(s) returned - SQL - Summarize analytical results in appointments: 2 row(s) returned - SQL - Prove requested entity exists in patients: No rows returned - SQL - Prove requested entity exists in insurance_policies: No rows returned - SQL - Prove requested entity exists in appointments: 1 row(s) returned - SQL - Inspect relevant rows in patients: No rows returned - SQL - Inspect relevant rows in insurance_policies: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

clinic-benchmark-010 - Critical

Question: Investigate why all retries for patient service CLN-2026-0010-A failed without a terminal status.

Expected answer: {"scenario_id": "clinic-benchmark-010", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "retry_failure", "subcategory": "retry_failure", "difficulty": "hard", "question": "Investigate why all retries for patient service CLN-2026-0010-A failed without a terminal status.", "baseline_script": "evaluation_scenarios/clinic/clinic-benchmark-010/baseline_reset.sql", "setup_script": "evaluation_scenarios/clinic/clinic-benchmark-010/inject.sql", "verification_script": "evaluation_scenarios/clinic/clinic-benchmark-010/verify.sql", "cleanup_script": "evaluation_scenarios/clinic/clinic-benchmark-010/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["CLN-2026-0010-A"], "expected_root_cause_concepts": ["retry evidence records exhausted attempts and the entity remained pending"], "expected_tables": ["encounters", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.encounters", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-110", "CLN-2026-0010-A", "retry evidence records exhausted attempts and the entity remained pending"], "acceptable_fix_concepts": ["route exhausted retries to reviewed dead-letter recovery"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_clinic_workflow_2"], "expected_functions": ["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [], "business_description": "A synthetic clinic production-support incident testing retry failure without exposing ground truth to runtime code.", "expected_relationships": ["eval.encounters.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_objects": ["

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

clinic-benchmark-011 - Critical

Question: Investigate why audit history says patient service CLN-2026-0011-A completed while its business record says Failed.

Expected answer: {"scenario_id": "clinic-benchmark-011", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "audit_history_inconsistency", "subcategory": "audit_history_inconsistency", "difficulty": "hard", "question": "Investigate why audit history says patient service CLN-2026-0011-A completed while its business record says Failed.", "baseline_script": "evaluation_scenarios/clinic/clinic-benchmark-011/baseline_reset.sql", "setup_script": "evaluation_scenarios/clinic/clinic-benchmark-011/inject.sql", "verification_script": "evaluation_scenarios/clinic/clinic-benchmark-011/verify.sql", "cleanup_script": "evaluation_scenarios/clinic/clinic-benchmark-011/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["CLN-2026-0011-A"], "expected_root_cause_concepts": ["business and audit states conflict for the same correlation"], "expected_tables": ["claims", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.claims", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-111", "CLN-2026-0011-A", "business and audit states conflict for the same correlation"], "acceptable_fix_concepts": ["reconcile history from the authoritative transaction boundary"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_clinic_workflow_3"], "expected_functions": ["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [], "business_description": "A synthetic clinic production-support incident testing audit history inconsistency without exposing ground truth to runtime code.", "expected_relationships": ["eval.claims.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "requi

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

clinic-benchmark-012 - Critical

Question: Investigate why patient service CLN-2026-0012-A failed validation after a reference lookup.

Expected answer: {"scenario_id": "clinic-benchmark-012", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "missing_reference_data", "subcategory": "missing_reference_data", "difficulty": "easy", "question": "Investigate why patient service CLN-2026-0012-A failed validation after a reference lookup.", "baseline_script": "evaluation_scenarios/clinic/clinic-benchmark-012/baseline_reset.sql", "setup_script": "evaluation_scenarios/clinic/clinic-benchmark-012/inject.sql", "verification_script":

"evaluation_scenarios/clinic/clinic-benchmark-012/verify.sql", "cleanup_script":
"evaluation_scenarios/clinic/clinic-benchmark-012/cleanup.sql", "expected_response_type":
"confirmed_root_cause", "expected_entities": ["CLN-2026-0012-A"], "expected_root_cause_concepts": ["required
active reference data is absent from discovered relationships"], "expected_tables": ["lab_orders",
"integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status",
"EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.lab_orders",
"eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-112",
"CLN-2026-0012-A", "required active reference data is absent from discovered relationships"],
"acceptable_fix_concepts": ["restore governed reference data before replaying the workflow"],
"prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from
synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object",
"wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
"unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,
"active": true, "expected_procedures": ["eval.usp_clinic_workflow_4"], "expected_functions":
["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [],
"business_description": "A synthetic clinic production-support incident testing missing reference data without
exposing ground truth to runtime code.", "expected_relationships": ["eval.lab_orders.CorrelationId ->
eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational
evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content",
"unrelated application framework tables"], "required_b

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

clinic-benchmark-013 - Critical

Question: Investigate why workflow procedure processing left patient service CLN-2026-0013-A in the wrong status.

Expected answer: {"scenario_id": "clinic-benchmark-013", "domain": "clinic", "database_engine": "sqlserver",
"database_version": "Azure SQL Database / SQL Server 2022", "category": "stored_procedure_defect",
"subcategory": "stored_procedure_defect", "difficulty": "hard", "question": "Investigate why workflow procedure
processing left patient service CLN-2026-0013-A in the wrong status.", "baseline_script":
"evaluation_scenarios/clinic/clinic-benchmark-013/baseline_reset.sql", "setup_script":
"evaluation_scenarios/clinic/clinic-benchmark-013/inject.sql", "verification_script":
"evaluation_scenarios/clinic/clinic-benchmark-013/verify.sql", "cleanup_script":
"evaluation_scenarios/clinic/clinic-benchmark-013/cleanup.sql", "expected_response_type":
"confirmed_root_cause", "expected_entities": ["CLN-2026-0013-A"], "expected_root_cause_concepts":
["procedure definition targets the workflow object but the observed transition is incorrect"], "expected_tables":
["lab_results", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey",
"Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.lab_results",
"eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-113",
"CLN-2026-0013-A", "procedure definition targets the workflow object but the observed transition is incorrect"],
"acceptable_fix_concepts": ["version and test the procedure correction before controlled replay"],
"prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from
synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object",
"wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
"unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,
"active": true, "expected_procedures": ["eval.usp_clinic_workflow_5"], "expected_functions":
["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [],
"business_description": "A synthetic clinic production-support incident testing stored procedure defect without

exposing ground truth to runtime code.", "expected_relationships": ["eval.lab_results.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest con

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225859-1BE37DD1 Investigation Mode: INVESTIGATION Detected Intent: PROCESS_FLOW_BREAK Response Type: insufficient_evidence Extracted Entities: exact_id_or_code=CLN-2026-0013-A, business_identifier=CLN-2026-0013-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=workflow, possible_table_or_column=procedure, possible_table_or_column=processing, possible_table_or_column=left, possible_table_or_column=patient, possible_table_or_column=service, possible_column=CLN, possible_table_or_column=in, possible_table_or_column=the, possible_table_or_column=wrong, possible_table_or_column=status AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: PROCESS_FLOW_BREAK - User Goal: Answer the user question with database evidence: Investigate why workflow procedure processing left patient service CLN-2026-0013-A in the wrong status. - Working Hypothesis: The business process should move from the current step to the next expected step. ## Stage 2 - Discover Context - Target Object: patients - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: insurance_policies, appointments, providers, procedures_performed, prescriptions, vw_clinic_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 5c43b992-e80b-4bc6-a80a-fb83d1832a8c - database_name: eval_clinic - schema_name: default - database_engine: mysql - connection_string_database: eval_clinic - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|5c43b992-e80b-4bc6-a80a-fb83d1832a8c|eval_clinic - Discovered tables: appointments, audit_history

Root cause: Configuration issue

Evidence collected: - No confirming rows were returned for the reported issue.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

clinic-benchmark-014 - Critical

Question: Investigate why updating patient service CLN-2026-0014-A produced no consistent audit outcome.

Expected answer: {"scenario_id": "clinic-benchmark-014", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "trigger_failure", "subcategory": "trigger_failure", "difficulty": "hard", "question": "Investigate why updating patient service CLN-2026-0014-A produced no consistent audit outcome.", "baseline_script": "evaluation_scenarios/clinic/clinic-benchmark-014/baseline_reset.sql", "setup_script": "evaluation_scenarios/clinic/clinic-benchmark-014/inject.sql", "verification_script": "evaluation_scenarios/clinic/clinic-benchmark-014/verify.sql", "cleanup_script": "evaluation_scenarios/clinic/clinic-benchmark-014/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["CLN-2026-0014-A"], "expected_root_cause_concepts": ["trigger metadata is relevant but the expected correlated audit transition is absent"], "expected_tables": ["prescriptions", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.prescriptions", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-114", "CLN-2026-0014-A", "trigger metadata is relevant but the expected correlated audit transition is absent"], "acceptable_fix_concepts": ["repair trigger behavior and reconcile missing audit events"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object",

"wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
"unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,
"active": true, "expected_procedures": ["eval.usp_clinic_workflow_6"], "expected_functions":
["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [],
"business_description": "A synthetic clinic production-support incident testing trigger failure without exposing
ground truth to runtime code.", "expected_relationships": ["eval.prescriptions.CorrelationId ->
eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational
evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content",
"unrelated application framework tables"]

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225910-9B83DA2F Investigation Mode:
INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type:
multiple_possible_causes Extracted Entities: exact_id_or_code=AUD-CLN-2026-0014-A,
business_identifier=AUD-CLN-2026-0014-A, possible_table_or_column=Investigate,
possible_table_or_column=why, possible_table_or_column=updating, possible_table_or_column=patient,
possible_table_or_column=service, possible_column=CLN, possible_table_or_column=produced,
possible_table_or_column=no, possible_table_or_column=consistent, possible_table_or_column=audit,
possible_table_or_column=outcome AI-assisted reasoning: Disabled Reason:
AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable
Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 -
Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require
live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal:
GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate
why updating patient service CLN-2026-0014-A produced no consistent audit outcome. - Working Hypothesis:
The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object:
audit_history - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched
candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No
non-primary natural key column was identified.) - Supporting Objects: patients, insurance_policies, appointments,
vw_clinic_operations_1, vw_clinic_operations_2, vw_clinic_operations_3 - Upstream Objects: None discovered -
Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read
path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id:
2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 5c43b992-e80b-4bc6-a80a-fb83d1832a8c -
database_name: eval_clinic - schema_name: default - database_engine: mysql - connection_string_database:
eval_clinic - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598e
c8fb87|5c43b992-e80b-4bc6-a80a-fb83d1832a8c|eval_clinic - Discovered tables: appointments, audit_history,
claims, clinics, diagnoses, encounters, evaluation

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in patients: 1 row(s) returned - SQL - Summarize
analytical results in audit_history: 3 row(s) returned - SQL - Summarize analytical results in insurance_policies: 1
row(s) returned - SQL - Prove requested entity exists in audit_history: 1 row(s) returned - SQL - Prove requested
entity exists in patients: No rows returned - SQL - Prove requested entity exists in insurance_policies: No rows
returned - SQL - Inspect relevant rows in audit_history: 1 row(s) returned - SQL - Inspect relevant rows in
patients: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

clinic-benchmark-015 - Critical

Question: Investigate why batch processing failed for patient service CLN-2026-0015-A while neighboring items
completed.

Expected answer: {"scenario_id": "clinic-benchmark-015", "domain": "clinic", "database_engine": "sqlserver",
"database_version": "Azure SQL Database / SQL Server 2022", "category": "batch_processing_failure",
"subcategory": "batch_processing_failure", "difficulty": "medium", "question": "Investigate why batch processing
failed for patient service CLN-2026-0015-A while neighboring items completed.", "baseline_script":

"evaluation_scenarios/clinic/clinic-benchmark-015/baseline_reset.sql", "setup_script":
"evaluation_scenarios/clinic/clinic-benchmark-015/inject.sql", "verification_script":
"evaluation_scenarios/clinic/clinic-benchmark-015/verify.sql", "cleanup_script":
"evaluation_scenarios/clinic/clinic-benchmark-015/cleanup.sql", "expected_response_type":
"confirmed_root_cause", "expected_entities": ["CLN-2026-0015-A"], "expected_root_cause_concepts": ["the
correlated batch or integration evidence isolates one failed item"], "expected_tables": ["payments",
"integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status",
"EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.payments",
"eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-115",
"CLN-2026-0015-A", "the correlated batch or integration evidence isolates one failed item"],
"acceptable_fix_concepts": ["restart only the failed item with checkpoint protection"], "prohibited_claims":
["objects or evidence not returned by the database", "production impact inferred from synthetic data"],
"critical_failure_rules": ["fabricated_evidence", "invented_database_object",
"wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
"unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,
"active": true, "expected_procedures": ["eval.usp_clinic_workflow_7"], "expected_functions":
["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [],
"business_description": "A synthetic clinic production-support incident testing batch processing failure without
exposing ground truth to runtime code.", "expected_relationships": ["eval.payments.CorrelationId ->
eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational
evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content",
"unrelated application framework tables"]

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

clinic-benchmark-016 - Critical

Question: Investigate intermittent competing updates for patient service CLN-2026-0016-A that produced inconsistent terminal states.

Expected answer: {"scenario_id": "clinic-benchmark-016", "domain": "clinic", "database_engine": "sqlserver",
"database_version": "Azure SQL Database / SQL Server 2022", "category": "concurrency_race_condition",
"subcategory": "concurrency_race_condition", "difficulty": "expert", "question": "Investigate intermittent competing
updates for patient service CLN-2026-0016-A that produced inconsistent terminal states.", "baseline_script":
"evaluation_scenarios/clinic/clinic-benchmark-016/baseline_reset.sql", "setup_script":
"evaluation_scenarios/clinic/clinic-benchmark-016/inject.sql", "verification_script":
"evaluation_scenarios/clinic/clinic-benchmark-016/verify.sql", "cleanup_script":
"evaluation_scenarios/clinic/clinic-benchmark-016/cleanup.sql", "expected_response_type":
"confirmed_root_cause", "expected_entities": ["CLN-2026-0016-A"], "expected_root_cause_concepts":
["same-correlation events at overlapping timestamps show a lost-update race"], "expected_tables":
["procedures_performed", "integration_messages", "exceptions", "audit_history"], "expected_columns":
["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects":
["eval.procedures_performed", "eval.integration_messages", "eval.exceptions", "eval.audit_history"],
"required_evidence": ["EVAL-CLINIC-116", "CLN-2026-0016-A", "same-correlation events at overlapping
timestamps show a lost-update race"], "acceptable_fix_concepts": ["use optimistic concurrency and retry on
version conflict"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact
inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object",
"wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
"unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,
"active": true, "expected_procedures": ["eval.usp_clinic_workflow_8"], "expected_functions":

["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [],
"business_description": "A synthetic clinic production-support incident testing concurrency race condition without
exposing ground truth to runtime code.", "expected_relationships": ["eval.procedures_performed.CorrelationId ->
eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational
evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answ

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied
identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

clinic-benchmark-017 - Critical

Question: Investigate why patient service CLN-2026-0017-A reports completion although downstream changes
were rolled back.

Expected answer: {"scenario_id": "clinic-benchmark-017", "domain": "clinic", "database_engine": "sqlserver",
"database_version": "Azure SQL Database / SQL Server 2022", "category": "transaction_rollback",
"subcategory": "transaction_rollback", "difficulty": "hard", "question": "Investigate why patient service
CLN-2026-0017-A reports completion although downstream changes were rolled back.", "baseline_script":
"evaluation_scenarios/clinic/clinic-benchmark-017/baseline_reset.sql", "setup_script":
"evaluation_scenarios/clinic/clinic-benchmark-017/inject.sql", "verification_script":
"evaluation_scenarios/clinic/clinic-benchmark-017/verify.sql", "cleanup_script":
"evaluation_scenarios/clinic/clinic-benchmark-017/cleanup.sql", "expected_response_type":
"confirmed_root_cause", "expected_entities": ["CLN-2026-0017-A"], "expected_root_cause_concepts":
["exception and audit evidence show the unit of work rolled back"], "expected_tables": ["appointments",
"integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status",
"EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.appointments",
"eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-117",
"CLN-2026-0017-A", "exception and audit evidence show the unit of work rolled back"],
"acceptable_fix_concepts": ["make the status transition atomic with downstream writes"], "prohibited_claims":
["objects or evidence not returned by the database", "production impact inferred from synthetic data"],
"critical_failure_rules": ["fabricated_evidence", "invented_database_object",
"wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
"unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,
"active": true, "expected_procedures": ["eval.usp_clinic_workflow_1"], "expected_functions":
["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [],
"business_description": "A synthetic clinic production-support incident testing transaction rollback without
exposing ground truth to runtime code.", "expected_relationships": ["eval.appointments.CorrelationId ->
eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational
evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content",
"unrelated application framework tables"], "required_b

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225939-7E3CE91D Investigation Mode:
INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type:
multiple_possible_causes Extracted Entities: exact_id_or_code=CLN-2026-0017-A,
business_identifier=CLN-2026-0017-A, possible_table_or_column=Investigate, possible_table_or_column=why,
possible_table_or_column=patient, possible_table_or_column=service, possible_column=CLN,
possible_table_or_column=reports, possible_table_or_column=completion, possible_table_or_column=although,
possible_table_or_column=downstream, possible_table_or_column=changes, possible_table_or_column=were,
possible_table_or_column=rolled, possible_table_or_column=back AI-assisted reasoning: Disabled Reason:
AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable
Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 -

Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why patient service CLN-2026-0017-A reports completion although downstream changes were rolled back. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: appointments - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: insurance_policies, appointments, vw_clinic_operations_1, vw_clinic_operations_2, vw_clinic_operations_3, vw_clinic_operations_4 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 5c43b992-e80b-4bc6-a80a-fb83d1832a8c - database_name: eval_clinic - schema_name: default - database_engine: mysql - connection_string_database: eval_clinic - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|5c43b992-e80b-4bc6-a80a-fb83d1832a8c|eval_clinic - Disco

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in patients: 1 row(s) returned - SQL - Summarize analytical results in insurance_policies: 1 row(s) returned - SQL - Summarize analytical results in appointments: 2 row(s) returned - SQL - Prove requested entity exists in patients: No rows returned - SQL - Prove requested entity exists in insurance_policies: No rows returned - SQL - Prove requested entity exists in appointments: 1 row(s) returned - SQL - Inspect relevant rows in patients: No rows returned - SQL - Inspect relevant rows in insurance_policies: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

clinic-benchmark-018 - Critical

Question: Investigate why replaying patient service CLN-2026-0018-A repeated a previously accepted operation.

Expected answer: {"scenario_id": "clinic-benchmark-018", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "idempotency_issue", "subcategory": "idempotency_issue", "difficulty": "expert", "question": "Investigate why replaying patient service CLN-2026-0018-A repeated a previously accepted operation.", "baseline_script": "evaluation_scenarios/clinic/clinic-benchmark-018/baseline_reset.sql", "setup_script": "evaluation_scenarios/clinic/clinic-benchmark-018/inject.sql", "verification_script": "evaluation_scenarios/clinic/clinic-benchmark-018/verify.sql", "cleanup_script": "evaluation_scenarios/clinic/clinic-benchmark-018/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["CLN-2026-0018-A"], "expected_root_cause_concepts": ["replay created a second correlated operation without an idempotency guard"], "expected_tables": ["encounters", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.encounters", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-118", "CLN-2026-0018-A", "replay created a second correlated operation without an idempotency guard"], "acceptable_fix_concepts": ["enforce a unique idempotency key across retries"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_clinic_workflow_2"], "expected_functions": ["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [], "business_description": "A synthetic clinic production-support incident testing idempotency issue without exposing ground truth to runtime code.", "expected_relationships": ["eval.encounters.CorrelationId -> eval.integration_messages.CorrelationId", "shared

BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_objec

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

clinic-benchmark-019 - Critical

Question: Investigate why patient service CLN-2026-0019-A displays Completed while failure evidence remains open.

Expected answer: {"scenario_id": "clinic-benchmark-019", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "incorrect_business_status", "subcategory": "incorrect_business_status", "difficulty": "easy", "question": "Investigate why patient service CLN-2026-0019-A displays Completed while failure evidence remains open.", "baseline_script": "evaluation_scenarios/clinic/clinic-benchmark-019/baseline_reset.sql", "setup_script": "evaluation_scenarios/clinic/clinic-benchmark-019/inject.sql", "verification_script": "evaluation_scenarios/clinic/clinic-benchmark-019/verify.sql", "cleanup_script": "evaluation_scenarios/clinic/clinic-benchmark-019/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["CLN-2026-0019-A"], "expected_root_cause_concepts": ["the business status contradicts active correlated failure evidence"], "expected_tables": ["claims", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.claims", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-119", "CLN-2026-0019-A", "the business status contradicts active correlated failure evidence"], "acceptable_fix_concepts": ["derive status from authoritative workflow completion"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_clinic_workflow_3"], "expected_functions": ["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [], "business_description": "A synthetic clinic production-support incident testing incorrect business status without exposing ground truth to runtime code.", "expected_relationships": ["eval.claims.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

clinic-benchmark-020 - Critical

Question: Investigate the end-to-end failure of patient service CLN-2026-0020-A across business, integration, exception, and audit records.

Expected answer: {"scenario_id": "clinic-benchmark-020", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "multi_table_investigation", "subcategory": "multi_table_investigation", "difficulty": "medium", "question": "Investigate the end-to-end failure of patient service CLN-2026-0020-A across business, integration, exception, and audit records.", "baseline_script": "evaluation_scenarios/clinic/clinic-benchmark-020/baseline_reset.sql", "setup_script": "evaluation_scenarios/clinic/clinic-benchmark-020/inject.sql", "verification_script": "evaluation_scenarios/clinic/clinic-benchmark-020/verify.sql", "cleanup_script": "evaluation_scenarios/clinic/clinic-benchmark-020/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["CLN-2026-0020-A"], "expected_root_cause_concepts": ["correlated multi-table evidence traces the failure to one interrupted transition"], "expected_tables": ["lab_orders", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.lab_orders", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-CLINIC-120", "CLN-2026-0020-A", "correlated multi-table evidence traces the failure to one interrupted transition"], "acceptable_fix_concepts": ["repair the transition and reconcile each correlated downstream state"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_clinic_workflow_4"], "expected_functions": ["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [], "business_description": "A synthetic clinic production-support incident testing multi table investigation without exposing ground truth to runtime code.", "expected_relationships": ["eval.lab_orders.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answ"]

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

clinic-pilot-001 - Critical

Question: Why was appointment APT-2101 not converted into an encounter?

Expected answer: {"scenario_id": "clinic-pilot-001", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "missing_workflow_output", "difficulty": "medium", "question": "Why was appointment APT-2101 not converted into an encounter?", "baseline_script": "evaluation_databases/clinic/sql/02_seed.sql", "setup_script": "evaluation_scenarios/clinic/clinic-pilot-001/inject.sql", "verification_script": "evaluation_scenarios/clinic/clinic-pilot-001/verify.sql", "cleanup_script": "evaluation_scenarios/clinic/clinic-pilot-001/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["APT-2101"], "expected_root_cause_concepts": ["required downstream work item was not created after a completed upstream transition"], "expected_tables": ["exceptions", "integration_messages", "appointments"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_clinic_operations_1"], "required_evidence": ["EVAL-CLINIC-001", "APT-2101", "Exception"], "acceptable_fix_concepts": ["transactional workflow completion"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true,

"expected_procedures": ["eval.usp_clinic_workflow_1"], "expected_functions": ["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

clinic-pilot-002 - Critical

Question: Why was claim CLM-3302 submitted twice?

Expected answer: {"scenario_id": "clinic-pilot-002", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "duplicate_integration", "difficulty": "hard", "question": "Why was claim CLM-3302 submitted twice?", "baseline_script": "evaluation_databases/clinic/sql/02_seed.sql", "setup_script": "evaluation_scenarios/clinic/clinic-pilot-002/inject.sql", "verification_script": "evaluation_scenarios/clinic/clinic-pilot-002/verify.sql", "cleanup_script": "evaluation_scenarios/clinic/clinic-pilot-002/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["CLM-3302"], "expected_root_cause_concepts": ["integration retry inserted the same business event without an idempotency guard"], "expected_tables": ["exceptions", "integration_messages", "encounters"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_clinic_operations_1"], "required_evidence": ["EVAL-CLINIC-002", "CLM-3302", "Exception"], "acceptable_fix_concepts": ["idempotency key"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_clinic_workflow_2"], "expected_functions": ["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

clinic-pilot-003 - Critical

Question: Why is lab result LAB-4403 not visible on its encounter?

Expected answer: {"scenario_id": "clinic-pilot-003", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "incorrect_selection", "difficulty": "hard", "question": "Why is lab result LAB-4403 not visible on its encounter?", "baseline_script": "evaluation_databases/clinic/sql/02_seed.sql", "setup_script": "evaluation_scenarios/clinic/clinic-pilot-003/inject.sql", "verification_script": "evaluation_scenarios/clinic/clinic-pilot-003/verify.sql", "cleanup_script": "evaluation_scenarios/clinic/clinic-pilot-003/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["LAB-4403"], "expected_root_cause_concepts": ["selection used historical assignment instead of the active work item"], "expected_tables": ["exceptions", "integration_messages", "diagnoses"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_clinic_operations_1"], "required_evidence": ["EVAL-CLINIC-003", "LAB-4403", "Exception"], "acceptable_fix_concepts": ["active assignment filtering"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_clinic_workflow_3"], "expected_functions": ["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

clinic-pilot-004 - Critical

Question: Why does encounter ENC-5504 remain Open after checkout?

Expected answer: {"scenario_id": "clinic-pilot-004", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "stalled_status", "difficulty": "medium", "question": "Why does encounter ENC-5504 remain Open after checkout?", "baseline_script": "evaluation_databases/clinic/sql/02_seed.sql", "setup_script": "evaluation_scenarios/clinic/clinic-pilot-004/inject.sql", "verification_script": "evaluation_scenarios/clinic/clinic-pilot-004/verify.sql", "cleanup_script": "evaluation_scenarios/clinic/clinic-pilot-004/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ENC-5504"], "expected_root_cause_concepts": ["integration failure prevented the terminal status transition"], "expected_tables": ["exceptions", "integration_messages", "procedures_performed"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_clinic_operations_1"], "required_evidence": ["EVAL-CLINIC-004", "ENC-5504", "Exception"], "acceptable_fix_concepts": ["retry and reconciliation"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_clinic_workflow_4"], "expected_functions": ["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

clinic-pilot-005 - Critical

Question: Is there enough evidence to confirm patient PAT-6605 missed an appointment?

Expected answer: {"scenario_id": "clinic-pilot-005", "domain": "clinic", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "evidence_safety", "subcategory": "insufficient_evidence", "difficulty": "hard", "question": "Is there enough evidence to confirm patient PAT-6605 missed an appointment?", "baseline_script": "evaluation_databases/clinic/sql/02_seed.sql", "setup_script": "evaluation_scenarios/clinic/clinic-pilot-005/inject.sql", "verification_script": "evaluation_scenarios/clinic/clinic-pilot-005/verify.sql", "cleanup_script": "evaluation_scenarios/clinic/clinic-pilot-005/cleanup.sql", "expected_response_type": "insufficient_evidence", "expected_entities": ["PAT-6605"], "expected_root_cause_concepts": [], "expected_tables": ["exceptions", "integration_messages", "prescriptions"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_clinic_operations_1"], "required_evidence": ["EVAL-CLINIC-005", "PAT-6605", "absence of conclusive timestamps"], "acceptable_fix_concepts": ["collect additional timestamp evidence"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_clinic_workflow_5"], "expected_functions": ["eval.fn_clinic_active_status"], "expected_triggers": ["eval.tr_providers_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

orders-benchmark-001 - Critical

Question: Investigate why order ORD-2026-0001-A is marked Failed even though its request was accepted.

Expected answer: {"scenario_id": "orders-benchmark-001", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "exact_entity_lookup", "subcategory": "exact_entity_lookup", "difficulty": "easy", "question": "Investigate why order ORD-2026-0001-A is marked Failed even though its request was accepted.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-001/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-001/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-001/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-001/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-2026-0001-A"], "expected_root_cause_concepts": ["the

exact business key resolves to one failed record"], "expected_tables": ["sales_orders", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.sales_orders", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-101", "ORD-2026-0001-A", "the exact business key resolves to one failed record"], "acceptable_fix_concepts": ["restore the record through the owning workflow after validating the request"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_1"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing exact entity lookup without exposing ground truth to runtime code.", "expected_relationships": ["eval.sales_orders.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_objects": ["eval.

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225706-B0FBADA4 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=ORD-2026-0001-A, business_identifier=ORD-2026-0001-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=order, possible_column=ORD, possible_table_or_column=is, possible_table_or_column=marked, possible_table_or_column=even, possible_table_or_column=though, possible_table_or_column=its, possible_table_or_column=request, possible_table_or_column=was, possible_table_or_column=accepted AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why order ORD-2026-0001-A is marked Failed even though its request was accepted. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: sales_order_lines - Target Reason: Selected from main problem phrase 'failed even though' (failed); parent and secondary-cause terms do not override the target. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: purchase_order_lines, sales_orders, purchase_orders, shipments, receipts, vw_orders_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 7b15039d-cf52-4f04-a533-a921efd0cd9b - database_name: eval_orders - schema_name: default - database_engine: mysql - connection_string_database: eval_orders - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|7b15039d-cf52-4f04-a533-a921efd0cd9b|eval_orders - Discovered tables: allocations, audit_history, batch_runs, customers, evaluation_marker, exceptions, integration_message

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in sales_order_lines: No rows returned - SQL - Summarize analytical results in purchase_order_lines: No rows returned - SQL - Summarize analytical results in sales_orders: 1 row(s) returned - SQL - Prove requested entity exists in sales_order_lines: No rows returned - SQL - Prove requested entity exists in purchase_order_lines: No rows returned - SQL - Prove requested entity exists in sales_orders: 1 row(s) returned - SQL - Inspect relevant rows in sales_order_lines: No rows returned - SQL - Inspect relevant rows in purchase_order_lines: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

orders-benchmark-002 - Critical

Question: Investigate the failed order reported as ORD-2026-0002; resolve the complete identifier before diagnosing it.

Expected answer: {"scenario_id": "orders-benchmark-002", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "partial_entity_resolution", "subcategory": "partial_entity_resolution", "difficulty": "easy", "question": "Investigate the failed order reported as ORD-2026-0002; resolve the complete identifier before diagnosing it.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-002/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-002/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-002/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-002/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-2026-0002-A"], "expected_root_cause_concepts": ["a safe candidate lookup resolves the partial identifier to one record"], "expected_tables": ["sales_order_lines", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.sales_order_lines", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-102", "ORD-2026-0002-A", "a safe candidate lookup resolves the partial identifier to one record"], "acceptable_fix_concepts": ["confirm the resolved identifier and retry through the supported workflow"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_2"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing partial entity resolution without exposing ground truth to runtime code.", "expected_relationships": ["eval.sales_order_lines.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content"]}

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

orders-benchmark-003 - Critical

Question: Investigate order ORD-2026-0003; support reported a failure but did not provide the final suffix.

Expected answer: {"scenario_id": "orders-benchmark-003", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "ambiguous_entity_resolution", "subcategory": "ambiguous_entity_resolution", "difficulty": "hard", "question": "Investigate order ORD-2026-0003; support reported a failure but did not provide the final suffix.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-003/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-003/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-003/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-003/cleanup.sql", "expected_response_type": "multiple_possible_causes", "expected_entities": ["ORD-2026-0003"], "expected_root_cause_concepts": ["the partial identifier matches multiple plausible records and investigation must stop"], "expected_tables": ["allocations", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.allocations", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-103", "ORD-2026-0003", "the partial identifier matches multiple plausible records and investigation must stop"], "acceptable_fix_concepts": ["request user selection of the correct candidate before"]}

continuing"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_3"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing ambiguous entity resolution without exposing ground truth to runtime code.", "expected_relationships": ["eval.allocations.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content"]

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

orders-benchmark-004 - Critical

Question: Investigate why completed order ORD-2026-0004-A has no downstream processing record.

Expected answer: {"scenario_id": "orders-benchmark-004", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "missing_downstream_record", "subcategory": "missing_downstream_record", "difficulty": "medium", "question": "Investigate why completed order ORD-2026-0004-A has no downstream processing record.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-004/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-004/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-004/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-004/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-2026-0004-A"], "expected_root_cause_concepts": ["the upstream record completed but its correlated integration record is missing"], "expected_tables": ["pick_tasks", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.pick_tasks", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-104", "ORD-2026-0004-A", "the upstream record completed but its correlated integration record is missing"], "acceptable_fix_concepts": ["reconcile the missing downstream work through an idempotent workflow"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_4"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing missing downstream record without exposing ground truth to runtime code.", "expected_relationships": ["eval.pick_tasks.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application frame"]

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

orders-benchmark-005 - Critical

Question: Investigate why order ORD-2026-0005-A produced two processing messages for one business request.

Expected answer: {"scenario_id": "orders-benchmark-005", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "duplicate_transaction", "subcategory": "duplicate_transaction", "difficulty": "medium", "question": "Investigate why order ORD-2026-0005-A produced two processing messages for one business request.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-005/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-005/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-005/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-005/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-2026-0005-A"], "expected_root_cause_concepts": ["two correlated processing events represent one business action"], "expected_tables": ["shipments", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.shipments", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-105", "ORD-2026-0005-A", "two correlated processing events represent one business action"], "acceptable_fix_concepts": ["deduplicate using a stable business idempotency key"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_5"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing duplicate transaction without exposing ground truth to runtime code.", "expected_relationships": ["eval.shipments.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_objects": ["eval.s

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225741-FC492440 Investigation Mode: INVESTIGATION Detected Intent: DUPLICATE_DATA Response Type: insufficient_evidence Extracted Entities: exact_id_or_code=ORD-2026-0005-A, business_identifier=ORD-2026-0005-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=order, possible_column=ORD, possible_table_or_column=produced, possible_table_or_column=two, possible_table_or_column=processing, possible_table_or_column=messages, possible_table_or_column=for, possible_table_or_column=one, possible_table_or_column=business, possible_table_or_column=request AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: DUPLICATE_DATA - User Goal: Answer the user question with database evidence: Investigate why order ORD-2026-0005-A produced two processing messages for one business request. - Working Hypothesis: There should be only one valid record for the affected business key. ## Stage 2 - Discover Context - Target Object: integration_messages - Target Reason: Selected from main problem phrase 'two processing messages' (duplicate); parent and secondary-cause terms do not override the target. - Business Key: BusinessKey (Inferred from duplicate evidence grouped by a non-primary business column.) - Supporting Objects: sales_order_lines, purchase_order_lines, sales_orders, purchase_orders, shipments, vw_orders_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 7b15039d-cf52-4f04-a533-a921efd0cd9b - database_name: eval_orders - schema_name: default

- database_engine: mysql - connection_string_database: eval_orders - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|7b15039d-cf52-4f04-a533-a921efd0cd9b|eval_orders - Discovered tables: allocations, audit_history, batch_runs, customers, evaluatio

Root cause: Configuration issue

Evidence collected: - No confirming rows were returned for the reported issue.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

orders-benchmark-006 - Critical

Question: Investigate why order ORD-2026-0006-A stopped between validation and completion.

Expected answer: {"scenario_id": "orders-benchmark-006", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "workflow_interruption", "subcategory": "workflow_interruption", "difficulty": "medium", "question": "Investigate why order ORD-2026-0006-A stopped between validation and completion.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-006/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-006/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-006/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-006/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-2026-0006-A"], "expected_root_cause_concepts": ["workflow history ends at an intermediate state with an exception"], "expected_tables": ["inventory_movements", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.inventory_movements", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-106", "ORD-2026-0006-A", "workflow history ends at an intermediate state with an exception"], "acceptable_fix_concepts": ["resume from the last committed workflow checkpoint"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_6"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing workflow interruption without exposing ground truth to runtime code.", "expected_relationships": ["eval.inventory_movements.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_o

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

orders-benchmark-007 - Critical

Question: Investigate why exception handling left order ORD-2026-0007-A in Processing.

Expected answer: {"scenario_id": "orders-benchmark-007", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "exception_handling", "subcategory": "exception_handling", "difficulty": "medium", "question": "Investigate why exception handling left order

ORD-2026-0007-A in Processing.", "baseline_script":
 "evaluation_scenarios/orders/orders-benchmark-007/baseline_reset.sql", "setup_script":
 "evaluation_scenarios/orders/orders-benchmark-007/inject.sql", "verification_script":
 "evaluation_scenarios/orders/orders-benchmark-007/verify.sql", "cleanup_script":
 "evaluation_scenarios/orders/orders-benchmark-007/cleanup.sql", "expected_response_type":
 "confirmed_root_cause", "expected_entities": ["ORD-2026-0007-A"], "expected_root_cause_concepts": ["the
 exception was recorded but no compensating transition ran"], "expected_tables": ["receipts",
 "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status",
 "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.receipts",
 "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence":
 ["EVAL-ORDERS-107", "ORD-2026-0007-A", "the exception was recorded but no compensating transition ran"],
 "acceptable_fix_concepts": ["add observable compensation and terminal-state handling"], "prohibited_claims":
 ["objects or evidence not returned by the database", "production impact inferred from synthetic data"],
 "critical_failure_rules": ["fabricated_evidence", "invented_database_object",
 "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
 "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,
 "active": true, "expected_procedures": ["eval.usp_orders_workflow_7"], "expected_functions":
 ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [],
 "business_description": "A synthetic orders production-support incident testing exception handling without
 exposing ground truth to runtime code.", "expected_relationships": ["eval.receipts.CorrelationId ->
 eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational
 evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content",
 "unrelated application framework tables"], "required_business_objects": ["eval.receipts"], "required_workflow"

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225800-0508E3B5 Investigation Mode:
 INVESTIGATION Detected Intent: PROCESS_FLOW_BREAK Response Type: insufficient_evidence Extracted
 Entities: exact_id_or_code=ORD-2026-0007-A, business_identifier=ORD-2026-0007-A,
 possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=exception,
 possible_table_or_column=handling, possible_table_or_column=left, possible_table_or_column=order,
 possible_column=ORD, possible_table_or_column=in, possible_table_or_column=Processing AI-assisted
 reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence
 validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for
 download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale:
 Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. -
 Investigation Goal: PROCESS_FLOW_BREAK - User Goal: Answer the user question with database evidence:
 Investigate why exception handling left order ORD-2026-0007-A in Processing. - Working Hypothesis: The
 business process should move from the current step to the next expected step. ## Stage 2 - Discover Context -
 Target Object: purchase_order_lines - Target Reason: Selected from metadata/question matches plus SQL
 evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key:
 Not determined (No non-primary natural key column was identified.) - Supporting Objects: purchase_order_lines,
 sales_orders, purchase_orders, exceptions, shipments, vw_orders_operations_1 - Upstream Objects: None
 discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No
 procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug
 - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id:
 7b15039d-cf52-4f04-a533-a921efd0cd9b - database_name: eval_orders - schema_name: default -
 database_engine: mysql - connection_string_database: eval_orders - metadata_cache_key: 678264ff-8b4f-47a5
 -89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|7b15039d-cf52-4f04-a533-a921efd0cd9b|eval_
 orders - Discovered tables: allocations, audit_history, batch_runs, customers, evaluation_marker, exceptions,
 integration_messages, inventory_balances, inventory_movements, pick_tasks, products, purchase_ord

Root cause: Configuration issue

Evidence collected: - No confirming rows were returned for the reported issue.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

orders-benchmark-008 - Critical

Question: Investigate why external integration for order ORD-2026-0008-A was rejected.

Expected answer: {"scenario_id": "orders-benchmark-008", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "integration_failure", "subcategory": "integration_failure", "difficulty": "medium", "question": "Investigate why external integration for order ORD-2026-0008-A was rejected.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-008/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-008/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-008/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-008/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-2026-0008-A"], "expected_root_cause_concepts": ["the correlated integration message failed and blocked completion"], "expected_tables": ["purchase_orders", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.purchase_orders", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-108", "ORD-2026-0008-A", "the correlated integration message failed and blocked completion"], "acceptable_fix_concepts": ["correct the integration contract and safely replay the message"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_8"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing integration failure without exposing ground truth to runtime code.", "expected_relationships": ["eval.purchase_orders.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_objects": [

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225812-4F811F4C Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=ORD-2026-0008-A, business_identifier=ORD-2026-0008-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=external, possible_table_or_column=integration, possible_table_or_column=for, possible_table_or_column=order, possible_column=ORD, possible_table_or_column=was, possible_table_or_column=rejected AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why external integration for order ORD-2026-0008-A was rejected. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: sales_order_lines - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: purchase_order_lines, integration_messages, sales_orders, purchase_orders, shipments, vw_orders_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 7b15039d-cf52-4f04-a533-a921efd0cd9b - database_name: eval_orders - schema_name: default - database_engine: mysql - connection_string_database: eval_orders - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|7b15039d-cf52-4f04-a533-a921efd0cd9b|eval_orders - Discovered tables: allocations, audit_history, batch_runs, customers, evaluation_marker, exceptions, integration_messages, inventory_balances, inventory_movements, pick_tasks, products, purchase_o

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in sales_order_lines: 1 row(s) returned - SQL - Summarize analytical results in purchase_order_lines: 1 row(s) returned - SQL - Summarize analytical results in sales_orders: 1 row(s) returned - SQL - Prove requested entity exists in sales_order_lines: No rows returned - SQL - Prove requested entity exists in purchase_order_lines: No rows returned - SQL - Prove requested entity exists in integration_messages: No rows returned - SQL - Inspect relevant rows in sales_order_lines: No rows returned - SQL - Inspect relevant rows in purchase_order_lines: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

orders-benchmark-009 - Critical

Question: Investigate why order ORD-2026-0009-A remains Queued behind older work.

Expected answer: {"scenario_id": "orders-benchmark-009", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "queue_backlog", "subcategory": "queue_backlog", "difficulty": "medium", "question": "Investigate why order ORD-2026-0009-A remains Queued behind older work.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-009/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-009/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-009/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-009/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-2026-0009-A"], "expected_root_cause_concepts": ["queue evidence shows the entity has not been consumed"], "expected_tables": ["sales_orders", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.sales_orders", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-109", "ORD-2026-0009-A", "queue evidence shows the entity has not been consumed"], "acceptable_fix_concepts": ["drain the queue with bounded, monitored processing"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_1"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing queue backlog without exposing ground truth to runtime code.", "expected_relationships": ["eval.sales_orders.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_objects": ["eval.sales_orders"], "required_workflow": ["eval.usp_orders_workf"]}

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225824-C1D4EC2C Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=ORD-2026-0009-A, business_idenfier=ORD-2026-0009-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=order, possible_column=ORD, possible_table_or_column=remains, possible_table_or_column=Queued, possible_table_or_column=behind, possible_table_or_column=older, possible_table_or_column=work AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why order ORD-2026-0009-A remains Queued behind older work. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: sales_orders - Target Reason: Selected from metadata/question matches

plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: purchase_order_lines, sales_orders, purchase_orders, shipments, receipts, vw_orders_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 7b15039d-cf52-4f04-a533-a921efd0cd9b - database_name: eval_orders - schema_name: default - database_engine: mysql - connection_string_database: eval_orders - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|7b15039d-cf52-4f04-a533-a921efd0cd9b|eval_orders - Discovered tables: allocations, audit_history, batch_runs, customers, evaluation_marker, exceptions, integration_messages, inventory_balances, inventory_movements, pick_tasks, products, purchase_order_lines, purchase_orders

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in sales_order_lines: 1 row(s) returned - SQL - Summarize analytical results in purchase_order_lines: 1 row(s) returned - SQL - Summarize analytical results in sales_orders: 2 row(s) returned - SQL - Prove requested entity exists in sales_order_lines: No rows returned - SQL - Prove requested entity exists in purchase_order_lines: No rows returned - SQL - Prove requested entity exists in sales_orders: 1 row(s) returned - SQL - Inspect relevant rows in sales_order_lines: No rows returned - SQL - Inspect relevant rows in purchase_order_lines: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

orders-benchmark-010 - Critical

Question: Investigate why all retries for order ORD-2026-0010-A failed without a terminal status.

Expected answer: {"scenario_id": "orders-benchmark-010", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "retry_failure", "subcategory": "retry_failure", "difficulty": "hard", "question": "Investigate why all retries for order ORD-2026-0010-A failed without a terminal status.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-010/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-010/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-010/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-010/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-2026-0010-A"], "expected_root_cause_concepts": ["retry evidence records exhausted attempts and the entity remained pending"], "expected_tables": ["sales_order_lines", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.sales_order_lines", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-110", "ORD-2026-0010-A", "retry evidence records exhausted attempts and the entity remained pending"], "acceptable_fix_concepts": ["route exhausted retries to reviewed dead-letter recovery"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_2"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing retry failure without exposing ground truth to runtime code.", "expected_relationships": ["eval.sales_order_lines.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_ob

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225834-07CC93CD Investigation Mode: INVESTIGATION Detected Intent: PROCESS_FLOW_BREAK Response Type: insufficient_evidence

Extracted Entities: exact_id_or_code=ORD-2026-0010-A, business_identifier=ORD-2026-0010-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=all, possible_table_or_column=retries, possible_table_or_column=for, possible_table_or_column=order, possible_column=ORD, possible_table_or_column=terminal, possible_table_or_column=status AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: PROCESS_FLOW_BREAK - User Goal: Answer the user question with database evidence: Investigate why all retries for order ORD-2026-0010-A failed without a terminal status. - Working Hypothesis: The business process should move from the current step to the next expected step. ## Stage 2 - Discover Context - Target Object: sales_order_lines - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: purchase_order_lines, sales_orders, purchase_orders, allocations, shipments, vw_orders_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 7b15039d-cf52-4f04-a533-a921efd0cd9b - database_name: eval_orders - schema_name: default - database_engine: mysql - connection_string_database: eval_orders - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|7b15039d-cf52-4f04-a533-a921efd0cd9b|eval_orders - Discovered tables: allocations, audit_history, batch_runs, customers, evaluation_marker, exceptions, integration_messages, inventory_balances, inventory_movements, pick_tasks, products, purchase_

Root cause: Configuration issue

Evidence collected: - Supplied business key found in returned evidence: ORD-2026-0010-A. - Affected rows were returned by safe SQL evidence.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

orders-benchmark-011 - Critical

Question: Investigate why audit history says order ORD-2026-0011-A completed while its business record says Failed.

Expected answer: {"scenario_id": "orders-benchmark-011", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "audit_history_inconsistency", "subcategory": "audit_history_inconsistency", "difficulty": "hard", "question": "Investigate why audit history says order ORD-2026-0011-A completed while its business record says Failed.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-011/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-011/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-011/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-011/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-2026-0011-A"], "expected_root_cause_concepts": ["business and audit states conflict for the same correlation"], "expected_tables": ["allocations", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.allocations", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-111", "ORD-2026-0011-A", "business and audit states conflict for the same correlation"], "acceptable_fix_concepts": ["reconcile history from the authoritative transaction boundary"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_3"], "expected_functions":

["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing audit history inconsistency without exposing ground truth to runtime code.", "expected_relationships": ["eval.allocations.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "r

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225843-97364DF6 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=ORD-2026-0011-A, business_idenfier=ORD-2026-0011-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=audit, possible_table_or_column=history, possible_table_or_column=says, possible_table_or_column=order, possible_column=ORD, possible_table_or_column=completed, possible_table_or_column=while, possible_table_or_column=its, possible_table_or_column=business AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why audit history says order ORD-2026-0011-A completed while its business record says Failed. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: audit_history - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: sales_order_lines, purchase_order_lines, sales_orders, purchase_orders, shipments, vw_orders_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 7b15039d-cf52-4f04-a533-a921efd0cd9b - database_name: eval_orders - schema_name: default - database_engine: mysql - connection_string_database: eval_orders - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|7b15039d-cf52-4f04-a533-a921efd0cd9b|eval_orders - Discovered tables: allocations, audit_history, batch_runs, customers, evaluation_marker, exceptions, integration

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in audit_history: No rows returned - SQL - Summarize analytical results in sales_order_lines: No rows returned - SQL - Summarize analytical results in purchase_order_lines: No rows returned - SQL - Prove requested entity exists in audit_history: No rows returned - SQL - Prove requested entity exists in sales_order_lines: No rows returned - SQL - Prove requested entity exists in purchase_order_lines: No rows returned - SQL - Inspect relevant rows in audit_history: No rows returned - SQL - Inspect relevant rows in sales_order_lines: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

orders-benchmark-012 - Critical

Question: Investigate why order ORD-2026-0012-A failed validation after a reference lookup.

Expected answer: {"scenario_id": "orders-benchmark-012", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "missing_reference_data", "subcategory": "missing_reference_data", "difficulty": "easy", "question": "Investigate why order ORD-2026-0012-A failed validation after a reference lookup.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-012/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-012/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-012/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-012/cleanup.sql", "expected_response_type":

"confirmed_root_cause", "expected_entities": ["ORD-2026-0012-A"], "expected_root_cause_concepts": ["required active reference data is absent from discovered relationships"], "expected_tables": ["pick_tasks", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.pick_tasks", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-112", "ORD-2026-0012-A", "required active reference data is absent from discovered relationships"], "acceptable_fix_concepts": ["restore governed reference data before replaying the workflow"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_4"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing missing reference data without exposing ground truth to runtime code.", "expected_relationships": ["eval.pick_tasks.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_obj

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

orders-benchmark-013 - Critical

Question: Investigate why workflow procedure processing left order ORD-2026-0013-A in the wrong status.

Expected answer: {"scenario_id": "orders-benchmark-013", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "stored_procedure_defect", "subcategory": "stored_procedure_defect", "difficulty": "hard", "question": "Investigate why workflow procedure processing left order ORD-2026-0013-A in the wrong status.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-013/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-013/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-013/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-013/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-2026-0013-A"], "expected_root_cause_concepts": ["procedure definition targets the workflow object but the observed transition is incorrect"], "expected_tables": ["shipments", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.shipments", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-113", "ORD-2026-0013-A", "procedure definition targets the workflow object but the observed transition is incorrect"], "acceptable_fix_concepts": ["version and test the procedure correction before controlled replay"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_5"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing stored procedure defect without exposing ground truth to runtime code.", "expected_relationships": ["eval.shipments.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content",

"unrelated

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225901-9EF03096 Investigation Mode: INVESTIGATION Detected Intent: PROCESS_FLOW_BREAK Response Type: insufficient_evidence Extracted Entities: exact_id_or_code=ORD-2026-0013-A, business_identifier=ORD-2026-0013-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=workflow, possible_table_or_column=procedure, possible_table_or_column=processing, possible_table_or_column=left, possible_table_or_column=order, possible_column=ORD, possible_table_or_column=in, possible_table_or_column=the, possible_table_or_column=wrong, possible_table_or_column=status AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: PROCESS_FLOW_BREAK - User Goal: Answer the user question with database evidence: Investigate why workflow procedure processing left order ORD-2026-0013-A in the wrong status. - Working Hypothesis: The business process should move from the current step to the next expected step. ## Stage 2 - Discover Context - Target Object: purchase_order_lines - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: purchase_order_lines, sales_orders, purchase_orders, shipments, receipts, vw_orders_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 7b15039d-cf52-4f04-a533-a921efd0cd9b - database_name: eval_orders - schema_name: default - database_engine: mysql - connection_string_database: eval_orders - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|7b15039d-cf52-4f04-a533-a921efd0cd9b|eval_orders - Discovered tables: allocations, audit_history, batch_runs, customers, evaluation_marker,

Root cause: Configuration issue

Evidence collected: - No confirming rows were returned for the reported issue.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

orders-benchmark-014 - Critical

Question: Investigate why updating order ORD-2026-0014-A produced no consistent audit outcome.

Expected answer: {"scenario_id": "orders-benchmark-014", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "trigger_failure", "subcategory": "trigger_failure", "difficulty": "hard", "question": "Investigate why updating order ORD-2026-0014-A produced no consistent audit outcome.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-014/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-014/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-014/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-014/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-2026-0014-A"], "expected_root_cause_concepts": ["trigger metadata is relevant but the expected correlated audit transition is absent"], "expected_tables": ["inventory_movements", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.inventory_movements", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-114", "ORD-2026-0014-A", "trigger metadata is relevant but the expected correlated audit transition is absent"], "acceptable_fix_concepts": ["repair trigger behavior and reconcile missing audit events"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,

"active": true, "expected_procedures": ["eval.usp_orders_workflow_6"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing trigger failure without exposing ground truth to runtime code.", "expected_relationships": ["eval.inventory_movements.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework"]

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225912-325E0BE1 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=AUD-ORD-2026-0014-A, business_identifier=AUD-ORD-2026-0014-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=updating, possible_table_or_column=order, possible_column=ORD, possible_table_or_column=produced, possible_table_or_column=no, possible_table_or_column=consistent, possible_table_or_column=audit, possible_table_or_column=outcome AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why updating order ORD-2026-0014-A produced no consistent audit outcome. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: sales_order_lines - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: purchase_order_lines, audit_history, sales_orders, purchase_orders, shipments, vw_orders_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 7b15039d-cf52-4f04-a533-a921efd0cd9b - database_name: eval_orders - schema_name: default - database_engine: mysql - connection_string_database: eval_orders - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|7b15039d-cf52-4f04-a533-a921efd0cd9b|eval_orders - Discovered tables: allocations, audit_history, batch_runs, customers, evaluation_marker, exceptions, integration_messages, inventory_balances, inventory_

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in sales_order_lines: 1 row(s) returned - SQL - Summarize analytical results in purchase_order_lines: 1 row(s) returned - SQL - Summarize analytical results in sales_orders: 1 row(s) returned - SQL - Prove requested entity exists in sales_order_lines: No rows returned - SQL - Prove requested entity exists in purchase_order_lines: No rows returned - SQL - Prove requested entity exists in audit_history: 1 row(s) returned - SQL - Inspect relevant rows in sales_order_lines: No rows returned - SQL - Inspect relevant rows in purchase_order_lines: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

orders-benchmark-015 - Critical

Question: Investigate why batch processing failed for order ORD-2026-0015-A while neighboring items completed.

Expected answer: {"scenario_id": "orders-benchmark-015", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "batch_processing_failure", "subcategory": "batch_processing_failure", "difficulty": "medium", "question": "Investigate why batch processing failed for order ORD-2026-0015-A while neighboring items completed.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-015/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-015/inject.sql", "verification_script":

"evaluation_scenarios/orders/orders-benchmark-015/verify.sql", "cleanup_script":
"evaluation_scenarios/orders/orders-benchmark-015/cleanup.sql", "expected_response_type":
"confirmed_root_cause", "expected_entities": ["ORD-2026-0015-A"], "expected_root_cause_concepts": ["the
correlated batch or integration evidence isolates one failed item"], "expected_tables": ["receipts",
"integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status",
"EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.receipts",
"eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence":
["EVAL-ORDERS-115", "ORD-2026-0015-A", "the correlated batch or integration evidence isolates one failed
item"], "acceptable_fix_concepts": ["restart only the failed item with checkpoint protection"], "prohibited_claims":
["objects or evidence not returned by the database", "production impact inferred from synthetic data"],
"critical_failure_rules": ["fabricated_evidence", "invented_database_object",
"wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
"unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,
"active": true, "expected_procedures": ["eval.usp_orders_workflow_7"], "expected_functions":
["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [],
"business_description": "A synthetic orders production-support incident testing batch processing failure without
exposing ground truth to runtime code.", "expected_relationships": ["eval.receipts.CorrelationId ->
eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational
evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content",
"unrelated application framework tables"], "required

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225924-D13C295A Investigation Mode:
INVESTIGATION Detected Intent: FAILED_BATCH_JOB Response Type: insufficient_evidence Extracted
Entities: exact_id_or_code=ORD-2026-0015-A, business_identifier=ORD-2026-0015-A,
possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=batch,
possible_table_or_column=processing, possible_table_or_column=for, possible_table_or_column=order,
possible_column=ORD, possible_table_or_column=while, possible_table_or_column=neighboring,
possible_table_or_column=items, possible_table_or_column=completed AI-assisted reasoning: Disabled
Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable
Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 -
Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require
live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal:
FAILED_BATCH_JOB - User Goal: Answer the user question with database evidence: Investigate why batch
processing failed for order ORD-2026-0015-A while neighboring items completed. - Working Hypothesis: The
database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object:
sales_order_lines - Target Reason: Selected from main problem phrase 'failed for order' (failed); parent and
secondary-cause terms do not override the target. - Business Key: Not determined (No non-primary natural key
column was identified.) - Supporting Objects: purchase_order_lines, batch_runs, sales_orders, purchase_orders,
shipments, vw_orders_operations_1 - Upstream Objects: None discovered - Downstream Objects: None
discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents
Used: No matching uploaded documents found ## Metadata Debug - workspace_id:
2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 7b15039d-cf52-4f04-a533-a921efd0cd9b -
database_name: eval_orders - schema_name: default - database_engine: mysql - connection_string_database:
eval_orders - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598
ec8fb87|7b15039d-cf52-4f04-a533-a921efd0cd9b|eval_orders - Discovered tables: allocations, audit_history,
batch_runs, customers, evaluation_marker, exceptions, integration_messages, inventory_balances, inventor

Root cause: Configuration issue

Evidence collected: - No confirming rows were returned for the reported issue.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

orders-benchmark-016 - Critical

Question: Investigate intermittent competing updates for order ORD-2026-0016-A that produced inconsistent terminal states.

Expected answer: {"scenario_id": "orders-benchmark-016", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "concurrency_race_condition", "subcategory": "concurrency_race_condition", "difficulty": "expert", "question": "Investigate intermittent competing updates for order ORD-2026-0016-A that produced inconsistent terminal states.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-016/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-016/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-016/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-016/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-2026-0016-A"], "expected_root_cause_concepts": ["same-correlation events at overlapping timestamps show a lost-update race"], "expected_tables": ["purchase_orders", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.purchase_orders", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-116", "ORD-2026-0016-A", "same-correlation events at overlapping timestamps show a lost-update race"], "acceptable_fix_concepts": ["use optimistic concurrency and retry on version conflict"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_8"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing concurrency race condition without exposing ground truth to runtime code.", "expected_relationships": ["eval.purchase_orders.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unr

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225931-8213FC55 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type:

multiple_possible_causes Extracted Entities: exact_id_or_code=ORD-2026-0016-A, business_identifier=ORD-2026-0016-A, possible_table_or_column=Investigate, possible_table_or_column=intermittent, possible_table_or_column=competing, possible_table_or_column=updates, possible_table_or_column=for, possible_table_or_column=order, possible_column=ORD, possible_table_or_column=that, possible_table_or_column=produced, possible_table_or_column=inconsistent, possible_table_or_column=terminal, possible_table_or_column=states AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate intermittent competing updates for order ORD-2026-0016-A that produced inconsistent terminal states. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: sales_order_lines - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: purchase_order_lines, sales_orders, purchase_orders, shipments, receipts, vw_orders_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 7b15039d-cf52-4f04-a533-a921efd0cd9b - database_name: eval_orders - schema_name: default - database_engine: mysql - connection_string_database: eval_orders - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|7b15039d-cf52-4f04-a533-a921efd0cd9b|eval_orders - Discovered tables: allocations, audit_history, batch_runs,

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in sales_order_lines: 1 row(s) returned - SQL - Summarize analytical results in purchase_order_lines: 1 row(s) returned - SQL - Summarize analytical results in sales_orders: 1 row(s) returned - SQL - Prove requested entity exists in sales_order_lines: No rows returned - SQL - Prove requested entity exists in purchase_order_lines: No rows returned - SQL - Prove requested entity exists in sales_orders: No rows returned - SQL - Inspect relevant rows in sales_order_lines: No rows returned - SQL - Inspect relevant rows in purchase_order_lines: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

orders-benchmark-017 - Critical

Question: Investigate why order ORD-2026-0017-A reports completion although downstream changes were rolled back.

Expected answer: {"scenario_id": "orders-benchmark-017", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "transaction_rollback", "subcategory": "transaction_rollback", "difficulty": "hard", "question": "Investigate why order ORD-2026-0017-A reports completion although downstream changes were rolled back.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-017/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-017/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-017/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-017/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-2026-0017-A"], "expected_root_cause_concepts": ["exception and audit evidence show the unit of work rolled back"], "expected_tables": ["sales_orders", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.sales_orders", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-117", "ORD-2026-0017-A", "exception and audit evidence show the unit of work rolled back"], "acceptable_fix_concepts": ["make the status transition atomic with downstream writes"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_1"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing transaction rollback without exposing ground truth to runtime code.", "expected_relationships": ["eval.sales_orders.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_obj

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225941-3FA9F19C Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=ORD-2026-0017-A, business_identifier=ORD-2026-0017-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=order, possible_column=ORD, possible_table_or_column=reports, possible_table_or_column=completion, possible_table_or_column=although, possible_table_or_column=downstream, possible_table_or_column=changes, possible_table_or_column=were, possible_table_or_column=rolled, possible_table_or_column=back AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why order ORD-2026-0017-A reports completion although downstream changes were rolled back. - Working

Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: sales_orders - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: purchase_order_lines, sales_orders, purchase_orders, shipments, receipts, vw_orders_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 7b15039d-cf52-4f04-a533-a921efd0cd9b - database_name: eval_orders - schema_name: default - database_engine: mysql - connection_string_database: eval_orders - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|7b15039d-cf52-4f04-a533-a921efd0cd9b|eval_orders - Discovered tables: allocations, audit_history, batch_runs, customers, evaluation_ma

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in sales_order_lines: 1 row(s) returned - SQL - Summarize analytical results in purchase_order_lines: 1 row(s) returned - SQL - Summarize analytical results in sales_orders: 2 row(s) returned - SQL - Prove requested entity exists in sales_order_lines: No rows returned - SQL - Prove requested entity exists in purchase_order_lines: No rows returned - SQL - Prove requested entity exists in sales_orders: 1 row(s) returned - SQL - Inspect relevant rows in sales_order_lines: No rows returned - SQL - Inspect relevant rows in purchase_order_lines: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

orders-benchmark-018 - Critical

Question: Investigate why replaying order ORD-2026-0018-A repeated a previously accepted operation.

Expected answer: {"scenario_id": "orders-benchmark-018", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "idempotency_issue", "subcategory": "idempotency_issue", "difficulty": "expert", "question": "Investigate why replaying order ORD-2026-0018-A repeated a previously accepted operation.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-018/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-018/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-018/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-018/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-2026-0018-A"], "expected_root_cause_concepts": ["replay created a second correlated operation without an idempotency guard"], "expected_tables": ["sales_order_lines", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.sales_order_lines", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-118", "ORD-2026-0018-A", "replay created a second correlated operation without an idempotency guard"], "acceptable_fix_concepts": ["enforce a unique idempotency key across retries"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_2"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing idempotency issue without exposing ground truth to runtime code.", "expected_relationships": ["eval.sales_order_lines.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_busi

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225951-1035234D Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type:

multiple_possible_causes Extracted Entities: exact_id_or_code=ORD-2026-0018-A, business_idenfier=ORD-2026-0018-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=replaying, possible_table_or_column=order, possible_column=ORD, possible_table_or_column=repeated, possible_table_or_column=previously, possible_table_or_column=accepted, possible_table_or_column=operation AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why replaying order ORD-2026-0018-A repeated a previously accepted operation. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: sales_order_lines - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: purchase_order_lines, sales_orders, purchase_orders, shipments, receipts, vw_orders_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 7b15039d-cf52-4f04-a533-a921efd0cd9b - database_name: eval_orders - schema_name: default - database_engine: mysql - connection_string_database: eval_orders - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|7b15039d-cf52-4f04-a533-a921efd0cd9b|eval_orders - Discovered tables: allocations, audit_history, batch_runs, customers, evaluation_marker, exceptions, integration_messages, inventory_balances, inventory_movements, pick_tasks, products

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in sales_order_lines: 2 row(s) returned - SQL - Summarize analytical results in purchase_order_lines: 1 row(s) returned - SQL - Summarize analytical results in sales_orders: 1 row(s) returned - SQL - Prove requested entity exists in sales_order_lines: 1 row(s) returned - SQL - Prove requested entity exists in purchase_order_lines: No rows returned - SQL - Prove requested entity exists in sales_orders: No rows returned - SQL - Inspect relevant rows in sales_order_lines: 1 row(s) returned - SQL - Inspect relevant rows in purchase_order_lines: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

orders-benchmark-019 - Critical

Question: Investigate why order ORD-2026-0019-A displays Completed while failure evidence remains open.

Expected answer: {"scenario_id": "orders-benchmark-019", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "incorrect_business_status", "subcategory": "incorrect_business_status", "difficulty": "easy", "question": "Investigate why order ORD-2026-0019-A displays Completed while failure evidence remains open.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-019/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-019/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-019/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-019/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-2026-0019-A"], "expected_root_cause_concepts": ["the business status contradicts active correlated failure evidence"], "expected_tables": ["allocations", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.allocations", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-119", "ORD-2026-0019-A", "the business status contradicts active correlated failure evidence"], "acceptable_fix_concepts": ["derive status from authoritative workflow completion"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from"]}

synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_3"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing incorrect business status without exposing ground truth to runtime code.", "expected_relationships": ["eval.allocations.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_busin

Actual answer: Investigation Complete. Investigation ID: INV-20260716-230000-3557D0F6 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=ORD-2026-0019-A, business_idenfier=ORD-2026-0019-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=order, possible_column=ORD, possible_table_or_column=displays, possible_table_or_column=Completed, possible_table_or_column=while, possible_table_or_column=failure, possible_table_or_column=evidence, possible_table_or_column=remains, possible_table_or_column=open AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why order ORD-2026-0019-A displays Completed while failure evidence remains open. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: sales_order_lines - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: purchase_order_lines, sales_orders, purchase_orders, shipments, receipts, vw_orders_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 7b15039d-cf52-4f04-a533-a921efd0cd9b - database_name: eval_orders - schema_name: default - database_engine: mysql - connection_string_database: eval_orders - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|7b15039d-cf52-4f04-a533-a921efd0cd9b|eval_orders - Discovered tables: allocations, audit_history, batch_runs, customers, evaluation_marker, exceptions, integration_messages,

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in sales_order_lines: No rows returned - SQL - Summarize analytical results in purchase_order_lines: No rows returned - SQL - Summarize analytical results in sales_orders: No rows returned - SQL - Prove requested entity exists in sales_order_lines: No rows returned - SQL - Prove requested entity exists in purchase_order_lines: No rows returned - SQL - Prove requested entity exists in sales_orders: No rows returned - SQL - Inspect relevant rows in sales_order_lines: No rows returned - SQL - Inspect relevant rows in purchase_order_lines: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

orders-benchmark-020 - Critical

Question: Investigate the end-to-end failure of order ORD-2026-0020-A across business, integration, exception, and audit records.

Expected answer: {"scenario_id": "orders-benchmark-020", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "multi_table_investigation", "subcategory": "multi_table_investigation", "difficulty": "medium", "question": "Investigate the end-to-end failure of

order ORD-2026-0020-A across business, integration, exception, and audit records.", "baseline_script": "evaluation_scenarios/orders/orders-benchmark-020/baseline_reset.sql", "setup_script": "evaluation_scenarios/orders/orders-benchmark-020/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-benchmark-020/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-benchmark-020/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-2026-0020-A"], "expected_root_cause_concepts": ["correlated multi-table evidence traces the failure to one interrupted transition"], "expected_tables": ["pick_tasks", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.pick_tasks", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-ORDERS-120", "ORD-2026-0020-A", "correlated multi-table evidence traces the failure to one interrupted transition"], "acceptable_fix_concepts": ["repair the transition and reconcile each correlated downstream state"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_4"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "A synthetic orders production-support incident testing multi table investigation without exposing ground truth to runtime code.", "expected_relationships": ["eval.pick_tasks.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest"]

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

orders-pilot-001 - Critical

Question: Why was order ORD-7101 not allocated despite available inventory?

Expected answer: {"scenario_id": "orders-pilot-001", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "missing_workflow_output", "difficulty": "medium", "question": "Why was order ORD-7101 not allocated despite available inventory?", "baseline_script": "evaluation_databases/orders/sql/02_seed.sql", "setup_script": "evaluation_scenarios/orders/orders-pilot-001/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-pilot-001/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-pilot-001/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-7101"], "expected_root_cause_concepts": ["required downstream work item was not created after a completed upstream transition"], "expected_tables": ["exceptions", "integration_messages", "inventory_balances"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_orders_operations_1"], "required_evidence": ["EVAL-ORDERS-001", "ORD-7101", "Exception"], "acceptable_fix_concepts": ["transactional workflow completion"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_1"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60,

"estimated_token_usage": 2000, "tags": []}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

orders-pilot-002 - Critical

Question: Why was order ORD-7102 imported twice?

Expected answer: {"scenario_id": "orders-pilot-002", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "duplicate_integration", "difficulty": "hard", "question": "Why was order ORD-7102 imported twice?", "baseline_script": "evaluation_databases/orders/sql/02_seed.sql", "setup_script": "evaluation_scenarios/orders/orders-pilot-002/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-pilot-002/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-pilot-002/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["ORD-7102"], "expected_root_cause_concepts": ["integration retry inserted the same business event without an idempotency guard"], "expected_tables": ["exceptions", "integration_messages", "inventory_movements"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_orders_operations_1"], "required_evidence": ["EVAL-ORDERS-002", "ORD-7102", "Exception"], "acceptable_fix_concepts": ["idempotency key"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_2"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

orders-pilot-003 - Critical

Question: Why is SKU-8103 showing a negative available balance?

Expected answer: {"scenario_id": "orders-pilot-003", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "incorrect_selection", "difficulty": "hard", "question": "Why is SKU-8103 showing a negative available balance?", "baseline_script": "evaluation_databases/orders/sql/02_seed.sql", "setup_script": "evaluation_scenarios/orders/orders-pilot-003/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-pilot-003/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-pilot-003/cleanup.sql", "expected_response_type": "confirmed_root_cause",

"expected_entities": ["SKU-8103"], "expected_root_cause_concepts": ["selection used historical assignment instead of the active work item"], "expected_tables": ["exceptions", "integration_messages", "sales_orders"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_orders_operations_1"], "required_evidence": ["EVAL-ORDERS-003", "SKU-8103", "Exception"], "acceptable_fix_concepts": ["active assignment filtering"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_3"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

orders-pilot-004 - Critical

Question: Why does pick task PICK-9104 remain In Progress after shipment?

Expected answer: {"scenario_id": "orders-pilot-004", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "stalled_status", "difficulty": "medium", "question": "Why does pick task PICK-9104 remain In Progress after shipment?", "baseline_script": "evaluation_databases/orders/sql/02_seed.sql", "setup_script": "evaluation_scenarios/orders/orders-pilot-004/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-pilot-004/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-pilot-004/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PICK-9104"], "expected_root_cause_concepts": ["integration failure prevented the terminal status transition"], "expected_tables": ["exceptions", "integration_messages", "sales_order_lines"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_orders_operations_1"], "required_evidence": ["EVAL-ORDERS-004", "PICK-9104", "Exception"], "acceptable_fix_concepts": ["retry and reconciliation"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_4"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

orders-pilot-005 - Critical

Question: Is there enough evidence to confirm purchase order PO-1205 arrived late?

Expected answer: {"scenario_id": "orders-pilot-005", "domain": "orders", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "evidence_safety", "subcategory": "insufficient_evidence", "difficulty": "hard", "question": "Is there enough evidence to confirm purchase order PO-1205 arrived late?", "baseline_script": "evaluation_databases/orders/sql/02_seed.sql", "setup_script": "evaluation_scenarios/orders/orders-pilot-005/inject.sql", "verification_script": "evaluation_scenarios/orders/orders-pilot-005/verify.sql", "cleanup_script": "evaluation_scenarios/orders/orders-pilot-005/cleanup.sql", "expected_response_type": "insufficient_evidence", "expected_entities": ["PO-1205"], "expected_root_cause_concepts": [], "expected_tables": ["exceptions", "integration_messages", "allocations"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_orders_operations_1"], "required_evidence": ["EVAL-ORDERS-005", "PO-1205", "absence of conclusive timestamps"], "acceptable_fix_concepts": ["collect additional timestamp evidence"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_orders_workflow_5"], "expected_functions": ["eval.fn_orders_active_status"], "expected_triggers": ["eval.tr_products_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

payroll-benchmark-001 - Critical

Question: Investigate why payroll item PAY-2026-0001-A is marked Failed even though its request was accepted.

Expected answer: {"scenario_id": "payroll-benchmark-001", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "exact_entity_lookup", "subcategory": "exact_entity_lookup", "difficulty": "easy", "question": "Investigate why payroll item PAY-2026-0001-A is marked Failed even though its request was accepted.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-001/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-001/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-001/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-001/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0001-A"], "expected_root_cause_concepts": ["the exact business key resolves to one failed record"], "expected_tables": ["payroll_runs", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.payroll_runs", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-101", "PAY-2026-0001-A", "the exact business key resolves to one failed record"], "acceptable_fix_concepts": ["restore the record through the owning workflow after"]}

validating the request"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_1"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing exact entity lookup without exposing ground truth to runtime code.", "expected_relationships": ["eval.payroll_runs.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_busi

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225702-30DD6A9C Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=PAY-2026-0001-A, business_identifier=PAY-2026-0001-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=payroll, possible_table_or_column=item, possible_column=PAY, possible_table_or_column=is, possible_table_or_column=marked, possible_table_or_column=even, possible_table_or_column=though, possible_table_or_column=its, possible_table_or_column=request, possible_table_or_column=was, possible_table_or_column=accepted AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why payroll item PAY-2026-0001-A is marked Failed even though its request was accepted. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: payroll_items - Target Reason: Selected from main problem phrase 'failed even though' (failed); parent and secondary-cause terms do not override the target. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: payroll_runs, pay_periods, pay_groups, leave_requests, payments, vw_payroll_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 18d7ade7-7dd3-428b-b4b3-67cfba8fed58 - database_name: eval_payroll - schema_name: default - database_engine: mysql - connection_string_database: eval_payroll - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|18d7ade7-7dd3-428b-b4b3-67cfba8fed58|eval_payroll - Discovered tables: audit_history, batch_runs, deductions, departments, employees, employm

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in payroll_items: No rows returned - SQL - Summarize analytical results in payroll_runs: 1 row(s) returned - SQL - Summarize analytical results in pay_periods: No rows returned - SQL - Prove requested entity exists in payroll_items: No rows returned - SQL - Prove requested entity exists in payroll_runs: 1 row(s) returned - SQL - Prove requested entity exists in pay_periods: No rows returned - SQL - Inspect relevant rows in payroll_items: No rows returned - SQL - Inspect relevant rows in payroll_runs: 1 row(s) returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

payroll-benchmark-002 - Critical

Question: Investigate the failed payroll item reported as PAY-2026-0002; resolve the complete identifier before diagnosing it.

Expected answer: {"scenario_id": "payroll-benchmark-002", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "partial_entity_resolution",

"subcategory": "partial_entity_resolution", "difficulty": "easy", "question": "Investigate the failed payroll item reported as PAY-2026-0002; resolve the complete identifier before diagnosing it.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-002/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-002/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-002/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-002/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0002-A"], "expected_root_cause_concepts": ["a safe candidate lookup resolves the partial identifier to one record"], "expected_tables": ["payroll_items", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.payroll_items", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-102", "PAY-2026-0002-A", "a safe candidate lookup resolves the partial identifier to one record"], "acceptable_fix_concepts": ["confirm the resolved identifier and retry through the supported workflow"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_2"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing partial entity resolution without exposing ground truth to runtime code.", "expected_relationships": ["eval.payroll_items.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifes"]

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

payroll-benchmark-003 - Critical

Question: Investigate payroll item PAY-2026-0003; support reported a failure but did not provide the final suffix.

Expected answer: {"scenario_id": "payroll-benchmark-003", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "ambiguous_entity_resolution", "subcategory": "ambiguous_entity_resolution", "difficulty": "hard", "question": "Investigate payroll item PAY-2026-0003; support reported a failure but did not provide the final suffix.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-003/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-003/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-003/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-003/cleanup.sql", "expected_response_type": "multiple_possible_causes", "expected_entities": ["PAY-2026-0003"], "expected_root_cause_concepts": ["the partial identifier matches multiple plausible records and investigation must stop"], "expected_tables": ["payments", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.payments", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-103", "PAY-2026-0003", "the partial identifier matches multiple plausible records and investigation must stop"], "acceptable_fix_concepts": ["request user selection of the correct candidate before continuing"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_3"], "expected_functions":

["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing ambiguous entity resolution without exposing ground truth to runtime code.", "expected_relationships": ["eval.payments.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer man

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

payroll-benchmark-004 - Critical

Question: Investigate why completed payroll item PAY-2026-0004-A has no downstream processing record.

Expected answer: {"scenario_id": "payroll-benchmark-004", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "missing_downstream_record", "subcategory": "missing_downstream_record", "difficulty": "medium", "question": "Investigate why completed payroll item PAY-2026-0004-A has no downstream processing record.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-004/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-004/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-004/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-004/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0004-A"], "expected_root_cause_concepts": ["the upstream record completed but its correlated integration record is missing"], "expected_tables": ["time_entries", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.time_entries", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-104", "PAY-2026-0004-A", "the upstream record completed but its correlated integration record is missing"], "acceptable_fix_concepts": ["reconcile the missing downstream work through an idempotent workflow"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_4"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing missing downstream record without exposing ground truth to runtime code.", "expected_relationships": ["eval.time_entries.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content",

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

payroll-benchmark-005 - Critical

Question: Investigate why payroll item PAY-2026-0005-A produced two processing messages for one business request.

Expected answer: {"scenario_id": "payroll-benchmark-005", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "duplicate_transaction", "subcategory": "duplicate_transaction", "difficulty": "medium", "question": "Investigate why payroll item PAY-2026-0005-A produced two processing messages for one business request.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-005/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-005/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-005/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-005/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0005-A"], "expected_root_cause_concepts": ["two correlated processing events represent one business action"], "expected_tables": ["deductions", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.deductions", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-105", "PAY-2026-0005-A", "two correlated processing events represent one business action"], "acceptable_fix_concepts": ["deduplicate using a stable business idempotency key"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_5"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing duplicate transaction without exposing ground truth to runtime code.", "expected_relationships": ["eval.deductions.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_bu

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225735-A34E5238 Investigation Mode: INVESTIGATION Detected Intent: DUPLICATE_DATA Response Type: insufficient_evidence Extracted Entities: exact_id_or_code=MSG-PAY-2026-0005-A, business_identifier=MSG-PAY-2026-0005-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=payroll, possible_table_or_column=item, possible_column=PAY, possible_table_or_column=produced, possible_table_or_column=two, possible_table_or_column=processing, possible_table_or_column=messages, possible_table_or_column=for, possible_table_or_column=one, possible_table_or_column=business, possible_table_or_column=request AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: DUPLICATE_DATA - User Goal: Answer the user question with database evidence: Investigate why payroll item PAY-2026-0005-A produced two processing messages for one business request. - Working Hypothesis: There should be only one valid record for the affected business key. ## Stage 2 - Discover Context - Target Object: integration_messages - Target Reason: Selected from main problem phrase 'two processing messages' (duplicate); parent and secondary-cause terms do not override the target. - Business Key: BusinessKey (Inferred from duplicate evidence grouped by a non-primary business column.) - Supporting Objects: payroll_items, payroll_runs, pay_periods, pay_groups, leave_requests, vw_payroll_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 18d7ade7-7dd3-428b-b4b3-67cfba8fed58 - database_name: eval_payroll - schema_name: default - database_engine: mysql - connection_string_database: eval_payroll - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|18d7ade7-7dd3-428b-b4b3-67cfba8fed58|eval_payroll - Discovered tables: audit_history, batch_

Root cause: Configuration issue

Evidence collected: - Supplied business key found in returned evidence: MSG-PAY-2026-0005-A. - Affected rows were returned by safe SQL evidence.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

payroll-benchmark-006 - Critical

Question: Investigate why payroll item PAY-2026-0006-A stopped between validation and completion.

Expected answer: {"scenario_id": "payroll-benchmark-006", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "workflow_interruption", "subcategory": "workflow_interruption", "difficulty": "medium", "question": "Investigate why payroll item PAY-2026-0006-A stopped between validation and completion.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-006/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-006/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-006/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-006/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0006-A"], "expected_root_cause_concepts": ["workflow history ends at an intermediate state with an exception"], "expected_tables": ["employees", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.employees", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-106", "PAY-2026-0006-A", "workflow history ends at an intermediate state with an exception"], "acceptable_fix_concepts": ["resume from the last committed workflow checkpoint"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_6"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing workflow interruption without exposing ground truth to runtime code.", "expected_relationships": ["eval.employees.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_objects":

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

payroll-benchmark-007 - Critical

Question: Investigate why exception handling left payroll item PAY-2026-0007-A in Processing.

Expected answer: {"scenario_id": "payroll-benchmark-007", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "exception_handling", "subcategory": "exception_handling", "difficulty": "medium", "question": "Investigate why exception handling left payroll item PAY-2026-0007-A in Processing.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-007/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-007/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-007/verify.sql", "cleanup_script":

"evaluation_scenarios/payroll/payroll-benchmark-007/cleanup.sql", "expected_response_type":
"confirmed_root_cause", "expected_entities": ["PAY-2026-0007-A"], "expected_root_cause_concepts": ["the
exception was recorded but no compensating transition ran"], "expected_tables": ["tax_filings",
"integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status",
"EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.tax_filings",
"eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence":
["EVAL-PAYROLL-107", "PAY-2026-0007-A", "the exception was recorded but no compensating transition ran"],
"acceptable_fix_concepts": ["add observable compensation and terminal-state handling"], "prohibited_claims":
["objects or evidence not returned by the database", "production impact inferred from synthetic data"],
"critical_failure_rules": ["fabricated_evidence", "invented_database_object",
"wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
"unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,
"active": true, "expected_procedures": ["eval.usp_payroll_workflow_7"], "expected_functions":
["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [],
"business_description": "A synthetic payroll production-support incident testing exception handling without
exposing ground truth to runtime code.", "expected_relationships": ["eval.tax_filings.CorrelationId ->
eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational
evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content",
"unrelated application framework tables"], "required_business_objects": ["eval.

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225756-891D6147 Investigation Mode:
INVESTIGATION Detected Intent: PROCESS_FLOW_BREAK Response Type: insufficient_evidence Extracted
Entities: exact_id_or_code=PAY-2026-0007-A, business_identifier=PAY-2026-0007-A,
possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=exception,
possible_table_or_column=handling, possible_table_or_column=left, possible_table_or_column=payroll,
possible_table_or_column=item, possible_column=PAY, possible_table_or_column=in,
possible_table_or_column=Processing AI-assisted reasoning: Disabled Reason:
AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable
Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 -
Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require
live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal:
PROCESS_FLOW_BREAK - User Goal: Answer the user question with database evidence: Investigate why
exception handling left payroll item PAY-2026-0007-A in Processing. - Working Hypothesis: The business
process should move from the current step to the next expected step. ## Stage 2 - Discover Context - Target
Object: payroll_items - Target Reason: Selected from metadata/question matches plus SQL evidence on those
matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No
non-primary natural key column was identified.) - Supporting Objects: payroll_runs, pay_periods, exceptions,
pay_groups, payments, vw_payroll_operations_1 - Upstream Objects: None discovered - Downstream Objects:
None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed -
Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id:
2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 18d7ade7-7dd3-428b-b4b3-67cfba8fed58 -
database_name: eval_payroll - schema_name: default - database_engine: mysql - connection_string_database:
eval_payroll - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598
ec8fb87|18d7ade7-7dd3-428b-b4b3-67cfba8fed58|eval_payroll - Discovered tables: audit_history, batch_runs,
deductions, departments, employees, employment_history, evaluation_marker, exceptions,
integration_messages, leave_requests, pay_g

Root cause: Configuration issue

Evidence collected: - No confirming rows were returned for the reported issue.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

payroll-benchmark-008 - Critical

Question: Investigate why external integration for payroll item PAY-2026-0008-A was rejected.

Expected answer: {"scenario_id": "payroll-benchmark-008", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "integration_failure", "subcategory": "integration_failure", "difficulty": "medium", "question": "Investigate why external integration for payroll item PAY-2026-0008-A was rejected.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-008/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-008/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-008/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-008/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0008-A"], "expected_root_cause_concepts": ["the correlated integration message failed and blocked completion"], "expected_tables": ["leave_requests", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.leave_requests", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-108", "PAY-2026-0008-A", "the correlated integration message failed and blocked completion"], "acceptable_fix_concepts": ["correct the integration contract and safely replay the message"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_8"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing integration failure without exposing ground truth to runtime code.", "expected_relationships": ["eval.leave_requests.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_b

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225808-D4BBFE25 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=MSG-PAY-2026-0008-A, business_idenfier=MSG-PAY-2026-0008-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=external, possible_table_or_column=integration, possible_table_or_column=for, possible_table_or_column=payroll, possible_table_or_column=item, possible_column=PAY, possible_table_or_column=was, possible_table_or_column=rejected AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why external integration for payroll item PAY-2026-0008-A was rejected. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: payroll_items - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: payroll_runs, pay_periods, integration_messages, pay_groups, payments, vw_payroll_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 18d7ade7-7dd3-428b-b4b3-67cfba8fed58 - database_name: eval_payroll - schema_name: default - database_engine: mysql - connection_string_database: eval_payroll - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|18d7ade7-7dd3-428b-b4b3-67cfba8fed58|eval_payroll - Discovered tables: audit_history, batch_runs, deductions, departments, employees, employment_history, evaluation_marker, exceptions, integration_messages, leave_re

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in payroll_items: 1 row(s) returned - SQL - Summarize analytical results in payroll_runs: 1 row(s) returned - SQL - Summarize analytical results in pay_periods: 1 row(s) returned - SQL - Prove requested entity exists in payroll_items: No rows returned - SQL - Prove requested entity exists in payroll_runs: No rows returned - SQL - Prove requested entity exists in pay_periods: No rows returned - SQL - Inspect relevant rows in payroll_items: No rows returned - SQL - Inspect relevant rows in payroll_runs: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

payroll-benchmark-009 - Critical

Question: Investigate why payroll item PAY-2026-0009-A remains Queued behind older work.

Expected answer: {"scenario_id": "payroll-benchmark-009", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "queue_backlog", "subcategory": "queue_backlog", "difficulty": "medium", "question": "Investigate why payroll item PAY-2026-0009-A remains Queued behind older work.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-009/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-009/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-009/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-009/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0009-A"], "expected_root_cause_concepts": ["queue evidence shows the entity has not been consumed"], "expected_tables": ["payroll_runs", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.payroll_runs", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-109", "PAY-2026-0009-A", "queue evidence shows the entity has not been consumed"], "acceptable_fix_concepts": ["drain the queue with bounded, monitored processing"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_1"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing queue backlog without exposing ground truth to runtime code.", "expected_relationships": ["eval.payroll_runs.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_objects": ["eval.payroll_runs"], "required_workflow": []}

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225820-C18A7FBA Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=PAY-2026-0009-A, business_identifier=PAY-2026-0009-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=payroll, possible_table_or_column=item, possible_column=PAY, possible_table_or_column=remains, possible_table_or_column=Queued, possible_table_or_column=behind, possible_table_or_column=older, possible_table_or_column=work AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why payroll item PAY-2026-0009-A remains Queued behind older work. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: payroll_runs - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural

key column was identified.) - Supporting Objects: payroll_runs, pay_periods, pay_groups, payments, tax_filings, vw_payroll_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 18d7ade7-7dd3-428b-b4b3-67cfba8fed58 - database_name: eval_payroll - schema_name: default - database_engine: mysql - connection_string_database: eval_payroll - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|18d7ade7-7dd3-428b-b4b3-67cfba8fed58|eval_payroll - Discovered tables: audit_history, batch_runs, deductions, departments, employees, employment_history, evaluation_marker, exceptions, integration_messages, leave_requests, pay_groups, pay_peri

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in payroll_items: 1 row(s) returned - SQL - Summarize analytical results in payroll_runs: 2 row(s) returned - SQL - Summarize analytical results in pay_periods: 1 row(s) returned - SQL - Prove requested entity exists in payroll_items: No rows returned - SQL - Prove requested entity exists in payroll_runs: 1 row(s) returned - SQL - Prove requested entity exists in pay_periods: No rows returned - SQL - Inspect relevant rows in payroll_items: No rows returned - SQL - Inspect relevant rows in payroll_runs: 1 row(s) returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

payroll-benchmark-010 - Critical

Question: Investigate why all retries for payroll item PAY-2026-0010-A failed without a terminal status.

Expected answer: {"scenario_id": "payroll-benchmark-010", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "retry_failure", "subcategory": "retry_failure", "difficulty": "hard", "question": "Investigate why all retries for payroll item PAY-2026-0010-A failed without a terminal status.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-010/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-010/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-010/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-010/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0010-A"], "expected_root_cause_concepts": ["retry evidence records exhausted attempts and the entity remained pending"], "expected_tables": ["payroll_items", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.payroll_items", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-110", "PAY-2026-0010-A", "retry evidence records exhausted attempts and the entity remained pending"], "acceptable_fix_concepts": ["route exhausted retries to reviewed dead-letter recovery"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_2"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing retry failure without exposing ground truth to runtime code.", "expected_relationships": ["eval.payroll_items.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_b

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225831-78C9938E Investigation Mode: INVESTIGATION Detected Intent: PROCESS_FLOW_BREAK Response Type: insufficient_evidence Extracted Entities: exact_id_or_code=PAY-2026-0010-A, business_identifier=PAY-2026-0010-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=all,

possible_table_or_column=retries, possible_table_or_column=for, possible_table_or_column=payroll, possible_table_or_column=item, possible_column=PAY, possible_table_or_column=terminal, possible_table_or_column=status AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: PROCESS_FLOW_BREAK - User Goal: Answer the user question with database evidence: Investigate why all retries for payroll item PAY-2026-0010-A failed without a terminal status. - Working Hypothesis: The business process should move from the current step to the next expected step. ## Stage 2 - Discover Context - Target Object: payroll_items - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: payroll_runs, pay_periods, pay_groups, payments, tax_filings, vw_payroll_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 18d7ade7-7dd3-428b-b4b3-67cfba8fed58 - database_name: eval_payroll - schema_name: default - database_engine: mysql - connection_string_database: eval_payroll - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|18d7ade7-7dd3-428b-b4b3-67cfba8fed58|eval_payroll - Discovered tables: audit_history, batch_runs, deductions, departments, employees, employment_history, evaluation_marker, exceptions, integration_messages, leave_requests,

Root cause: Configuration issue

Evidence collected: - Supplied business key found in returned evidence: PAY-2026-0010-A. - Affected rows were returned by safe SQL evidence.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

payroll-benchmark-011 - Critical

Question: Investigate why audit history says payroll item PAY-2026-0011-A completed while its business record says Failed.

Expected answer: {"scenario_id": "payroll-benchmark-011", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "audit_history_inconsistency", "subcategory": "audit_history_inconsistency", "difficulty": "hard", "question": "Investigate why audit history says payroll item PAY-2026-0011-A completed while its business record says Failed.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-011/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-011/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-011/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-011/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0011-A"], "expected_root_cause_concepts": ["business and audit states conflict for the same correlation"], "expected_tables": ["payments", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.payments", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-111", "PAY-2026-0011-A", "business and audit states conflict for the same correlation"], "acceptable_fix_concepts": ["reconcile history from the authoritative transaction boundary"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_3"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing audit history inconsistency without

exposing ground truth to runtime code.", "expected_relationships": ["eval.payments.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework"]

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225840-145B7E6B Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=PAY-2026-0011-A, business_identifier=PAY-2026-0011-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=audit, possible_table_or_column=history, possible_table_or_column=says, possible_table_or_column=payroll, possible_table_or_column=item, possible_column=PAY, possible_table_or_column=completed, possible_table_or_column=while, possible_table_or_column=its, possible_table_or_column=business AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why audit history says payroll item PAY-2026-0011-A completed while its business record says Failed. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: payroll_items - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: audit_history, payroll_runs, pay_periods, employment_history, pay_groups, vw_payroll_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 18d7ade7-7dd3-428b-b4b3-67cfba8fed58 - database_name: eval_payroll - schema_name: default - database_engine: mysql - connection_string_database: eval_payroll - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|18d7ade7-7dd3-428b-b4b3-67cfba8fed58|eval_payroll - Discovered tables: audit_history, batch_runs, deductions, departments, employ

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in payroll_items: No rows returned - SQL - Summarize analytical results in audit_history: No rows returned - SQL - Summarize analytical results in payroll_runs: No rows returned - SQL - Prove requested entity exists in payroll_items: No rows returned - SQL - Prove requested entity exists in audit_history: No rows returned - SQL - Prove requested entity exists in payroll_runs: No rows returned - SQL - Inspect relevant rows in payroll_items: No rows returned - SQL - Inspect relevant rows in audit_history: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

payroll-benchmark-012 - Critical

Question: Investigate why payroll item PAY-2026-0012-A failed validation after a reference lookup.

Expected answer: {"scenario_id": "payroll-benchmark-012", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "missing_reference_data", "subcategory": "missing_reference_data", "difficulty": "easy", "question": "Investigate why payroll item PAY-2026-0012-A failed validation after a reference lookup.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-012/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-012/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-012/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-012/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0012-A"], "expected_root_cause_concepts": ["required active reference data is absent from discovered relationships"], "expected_tables": ["time_entries",

"integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.time_entries", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-112", "PAY-2026-0012-A", "required active reference data is absent from discovered relationships"], "acceptable_fix_concepts": ["restore governed reference data before replaying the workflow"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_4"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing missing reference data without exposing ground truth to runtime code.", "expected_relationships": ["eval.time_entries.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tabl

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

payroll-benchmark-013 - Critical

Question: Investigate why workflow procedure processing left payroll item PAY-2026-0013-A in the wrong status.

Expected answer: {"scenario_id": "payroll-benchmark-013", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "stored_procedure_defect", "subcategory": "stored_procedure_defect", "difficulty": "hard", "question": "Investigate why workflow procedure processing left payroll item PAY-2026-0013-A in the wrong status.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-013/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-013/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-013/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-013/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0013-A"], "expected_root_cause_concepts": ["procedure definition targets the workflow object but the observed transition is incorrect"], "expected_tables": ["deductions", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.deductions", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-113", "PAY-2026-0013-A", "procedure definition targets the workflow object but the observed transition is incorrect"], "acceptable_fix_concepts": ["version and test the procedure correction before controlled replay"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_5"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing stored procedure defect without exposing ground truth to runtime code.", "expected_relationships": ["eval.deductions.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer mani

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

payroll-benchmark-014 - Critical

Question: Investigate why updating payroll item PAY-2026-0014-A produced no consistent audit outcome.

Expected answer: {"scenario_id": "payroll-benchmark-014", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "trigger_failure", "subcategory": "trigger_failure", "difficulty": "hard", "question": "Investigate why updating payroll item PAY-2026-0014-A produced no consistent audit outcome.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-014/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-014/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-014/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-014/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0014-A"], "expected_root_cause_concepts": ["trigger metadata is relevant but the expected correlated audit transition is absent"], "expected_tables": ["employees", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.employees", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-114", "PAY-2026-0014-A", "trigger metadata is relevant but the expected correlated audit transition is absent"], "acceptable_fix_concepts": ["repair trigger behavior and reconcile missing audit events"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_6"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing trigger failure without exposing ground truth to runtime code.", "expected_relationships": ["eval.employees.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"]}

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225908-A40D1605 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=AUD-PAY-2026-0014-A, business_identifier=AUD-PAY-2026-0014-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=updating, possible_table_or_column=payroll, possible_table_or_column=item, possible_column=PAY, possible_table_or_column=produced, possible_table_or_column=no, possible_table_or_column=consistent, possible_table_or_column=audit, possible_table_or_column=outcome AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why updating payroll item PAY-2026-0014-A produced no consistent audit outcome. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: payroll_items - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No

non-primary natural key column was identified.) - Supporting Objects: payroll_runs, audit_history, pay_periods, pay_groups, payments, vw_payroll_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 18d7ade7-7dd3-428b-b4b3-67cfba8fed58 - database_name: eval_payroll - schema_name: default - database_engine: mysql - connection_string_database: eval_payroll - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|18d7ade7-7dd3-428b-b4b3-67cfba8fed58|eval_payroll - Discovered tables: audit_history, batch_runs, deductions, departments, employees, employment_history, evaluation_marker, excepti

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in payroll_items: 1 row(s) returned - SQL - Summarize analytical results in payroll_runs: 1 row(s) returned - SQL - Summarize analytical results in pay_periods: 1 row(s) returned - SQL - Prove requested entity exists in payroll_items: No rows returned - SQL - Prove requested entity exists in payroll_runs: No rows returned - SQL - Prove requested entity exists in audit_history: 1 row(s) returned - SQL - Inspect relevant rows in payroll_items: No rows returned - SQL - Inspect relevant rows in payroll_runs: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

payroll-benchmark-015 - Critical

Question: Investigate why batch processing failed for payroll item PAY-2026-0015-A while neighboring items completed.

Expected answer: {"scenario_id": "payroll-benchmark-015", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "batch_processing_failure", "subcategory": "batch_processing_failure", "difficulty": "medium", "question": "Investigate why batch processing failed for payroll item PAY-2026-0015-A while neighboring items completed.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-015/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-015/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-015/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-015/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0015-A"], "expected_root_cause_concepts": ["the correlated batch or integration evidence isolates one failed item"], "expected_tables": ["tax_filings", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.tax_filings", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-115", "PAY-2026-0015-A", "the correlated batch or integration evidence isolates one failed item"], "acceptable_fix_concepts": ["restart only the failed item with checkpoint protection"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_7"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing batch processing failure without exposing ground truth to runtime code.", "expected_relationships": ["eval.tax_filings.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated applicatio"]

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225920-72E6AFAD Investigation Mode: INVESTIGATION Detected Intent: FAILED_BATCH_JOB Response Type: insufficient_evidence Extracted Entities: exact_id_or_code=PAY-2026-0015-A, business_identifier=PAY-2026-0015-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=batch,

possible_table_or_column=processing, possible_table_or_column=for, possible_table_or_column=payroll, possible_table_or_column=item, possible_column=PAY, possible_table_or_column=while, possible_table_or_column=neighboring, possible_table_or_column=items, possible_table_or_column=completed
AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: FAILED_BATCH_JOB - User Goal: Answer the user question with database evidence: Investigate why batch processing failed for payroll item PAY-2026-0015-A while neighboring items completed. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: payroll_items - Target Reason: Selected from main problem phrase 'failed for payroll' (failed); parent and secondary-cause terms do not override the target. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: payroll_runs, batch_runs, pay_periods, payments, pay_groups, vw_payroll_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 18d7ade7-7dd3-428b-b4b3-67cfba8fed58 - database_name: eval_payroll - schema_name: default - database_engine: mysql - connection_string_database: eval_payroll - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|18d7ade7-7dd3-428b-b4b3-67cfba8fed58|eval_payroll - Discovered tables: audit_history, batch_runs, deductions, departments, employees, employment_history, evaluation_marker, exc

Root cause: Configuration issue

Evidence collected: - No confirming rows were returned for the reported issue.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

payroll-benchmark-016 - Critical

Question: Investigate intermittent competing updates for payroll item PAY-2026-0016-A that produced inconsistent terminal states.

Expected answer: {"scenario_id": "payroll-benchmark-016", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "concurrency_race_condition", "subcategory": "concurrency_race_condition", "difficulty": "expert", "question": "Investigate intermittent competing updates for payroll item PAY-2026-0016-A that produced inconsistent terminal states.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-016/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-016/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-016/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-016/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0016-A"], "expected_root_cause_concepts": ["same-correlation events at overlapping timestamps show a lost-update race"], "expected_tables": ["leave_requests", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.leave_requests", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-116", "PAY-2026-0016-A", "same-correlation events at overlapping timestamps show a lost-update race"], "acceptable_fix_concepts": ["use optimistic concurrency and retry on version conflict"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_8"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing concurrency race condition without exposing ground truth to runtime code.", "expected_relationships": ["eval.leave_requests.CorrelationId ->

eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer mani

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence verification

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Require every confirmed claim to cite persisted evidence matching the expected key.

payroll-benchmark-017 - Critical

Question: Investigate why payroll item PAY-2026-0017-A reports completion although downstream changes were rolled back.

Expected answer: {"scenario_id": "payroll-benchmark-017", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "transaction_rollback", "subcategory": "transaction_rollback", "difficulty": "hard", "question": "Investigate why payroll item PAY-2026-0017-A reports completion although downstream changes were rolled back.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-017/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-017/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-017/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-017/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0017-A"], "expected_root_cause_concepts": ["exception and audit evidence show the unit of work rolled back"], "expected_tables": ["payroll_runs", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.payroll_runs", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-117", "PAY-2026-0017-A", "exception and audit evidence show the unit of work rolled back"], "acceptable_fix_concepts": ["make the status transition atomic with downstream writes"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_1"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing transaction rollback without exposing ground truth to runtime code.", "expected_relationships": ["eval.payroll_runs.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"],

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225937-F7B0F96D Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=PAY-2026-0017-A, business_identifier=PAY-2026-0017-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=payroll, possible_table_or_column=item, possible_column=PAY, possible_table_or_column=reports, possible_table_or_column=completion, possible_table_or_column=although, possible_table_or_column=downstream, possible_table_or_column=changes, possible_table_or_column=were, possible_table_or_column=rolled, possible_table_or_column=back AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate

why payroll item PAY-2026-0017-A reports completion although downstream changes were rolled back. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: payroll_runs - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: payroll_runs, pay_periods, pay_groups, payments, tax_filings, vw_payroll_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 18d7ade7-7dd3-428b-b4b3-67cfba8fed58 - database_name: eval_payroll - schema_name: default - database_engine: mysql - connection_string_database: eval_payroll - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|18d7ade7-7dd3-428b-b4b3-67cfba8fed58|eval_payroll - Discovered tables: audit_history, batch_runs, deduc

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in payroll_items: 1 row(s) returned - SQL - Summarize analytical results in payroll_runs: 2 row(s) returned - SQL - Summarize analytical results in pay_periods: 1 row(s) returned - SQL - Prove requested entity exists in payroll_items: No rows returned - SQL - Prove requested entity exists in payroll_runs: 1 row(s) returned - SQL - Prove requested entity exists in pay_periods: No rows returned - SQL - Inspect relevant rows in payroll_items: No rows returned - SQL - Inspect relevant rows in payroll_runs: 1 row(s) returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

payroll-benchmark-018 - Critical

Question: Investigate why replaying payroll item PAY-2026-0018-A repeated a previously accepted operation.

Expected answer: {"scenario_id": "payroll-benchmark-018", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "idempotency_issue", "subcategory": "idempotency_issue", "difficulty": "expert", "question": "Investigate why replaying payroll item PAY-2026-0018-A repeated a previously accepted operation.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-018/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-018/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-018/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-018/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0018-A"], "expected_root_cause_concepts": ["replay created a second correlated operation without an idempotency guard"], "expected_tables": ["payroll_items", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.payroll_items", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-118", "PAY-2026-0018-A", "replay created a second correlated operation without an idempotency guard"], "acceptable_fix_concepts": ["enforce a unique idempotency key across retries"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_2"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing idempotency issue without exposing ground truth to runtime code.", "expected_relationships": ["eval.payroll_items.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "req

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225947-0F021740 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=PAY-2026-0018-A, business_idenfier=PAY-2026-0018-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=replaying, possible_table_or_column=payroll, possible_table_or_column=item, possible_column=PAY, possible_table_or_column=repeated, possible_table_or_column=previously, possible_table_or_column=accepted, possible_table_or_column=operation AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why replaying payroll item PAY-2026-0018-A repeated a previously accepted operation. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: payroll_items - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: payroll_runs, pay_periods, pay_groups, payments, tax_filings, vw_payroll_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 18d7ade7-7dd3-428b-b4b3-67cfba8fed58 - database_name: eval_payroll - schema_name: default - database_engine: mysql - connection_string_database: eval_payroll - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|18d7ade7-7dd3-428b-b4b3-67cfba8fed58|eval_payroll - Discovered tables: audit_history, batch_runs, deductions, departments, employees, employment_history, evaluation_marker, exceptions, integration_messages, I

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in payroll_items: 1 row(s) returned - SQL - Summarize analytical results in payroll_runs: 1 row(s) returned - SQL - Summarize analytical results in pay_periods: 1 row(s) returned - SQL - Prove requested entity exists in payroll_items: 1 row(s) returned - SQL - Prove requested entity exists in payroll_runs: No rows returned - SQL - Prove requested entity exists in pay_periods: No rows returned - SQL - Inspect relevant rows in payroll_items: 1 row(s) returned - SQL - Inspect relevant rows in payroll_runs: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

payroll-benchmark-019 - Critical

Question: Investigate why payroll item PAY-2026-0019-A displays Completed while failure evidence remains open.

Expected answer: {"scenario_id": "payroll-benchmark-019", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "incorrect_business_status", "subcategory": "incorrect_business_status", "difficulty": "easy", "question": "Investigate why payroll item PAY-2026-0019-A displays Completed while failure evidence remains open.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-019/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-019/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-019/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-019/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0019-A"], "expected_root_cause_concepts": ["the business status contradicts active correlated failure evidence"], "expected_tables": ["payments", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.payments", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence":

["EVAL-PAYROLL-119", "PAY-2026-0019-A", "the business status contradicts active correlated failure evidence"], "acceptable_fix_concepts": ["derive status from authoritative workflow completion"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_3"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing incorrect business status without exposing ground truth to runtime code.", "expected_relationships": ["eval.payments.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "r

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225956-EC710025 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=PAY-2026-0019-A, business_idenfier=PAY-2026-0019-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=payroll, possible_table_or_column=item, possible_column=PAY, possible_table_or_column=displays, possible_table_or_column=Completed, possible_table_or_column=while, possible_table_or_column=failure, possible_table_or_column=evidence, possible_table_or_column=remains, possible_table_or_column=open AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why payroll item PAY-2026-0019-A displays Completed while failure evidence remains open. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: payroll_items - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: payroll_runs, pay_periods, pay_groups, payments, tax_filings, vw_payroll_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 18d7ade7-7dd3-428b-b4b3-67cfba8fed58 - database_name: eval_payroll - schema_name: default - database_engine: mysql - connection_string_database: eval_payroll - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|18d7ade7-7dd3-428b-b4b3-67cfba8fed58|eval_payroll - Discovered tables: audit_history, batch_runs, deductions, departments, employees, employment_h

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in payroll_items: No rows returned - SQL - Summarize analytical results in payroll_runs: No rows returned - SQL - Summarize analytical results in pay_periods: No rows returned - SQL - Prove requested entity exists in payroll_items: No rows returned - SQL - Prove requested entity exists in payroll_runs: No rows returned - SQL - Prove requested entity exists in pay_periods: No rows returned - SQL - Inspect relevant rows in payroll_items: No rows returned - SQL - Inspect relevant rows in payroll_runs: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

payroll-benchmark-020 - Critical

Question: Investigate the end-to-end failure of payroll item PAY-2026-0020-A across business, integration, exception, and audit records.

Expected answer: {"scenario_id": "payroll-benchmark-020", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "multi_table_investigation", "subcategory": "multi_table_investigation", "difficulty": "medium", "question": "Investigate the end-to-end failure of payroll item PAY-2026-0020-A across business, integration, exception, and audit records.", "baseline_script": "evaluation_scenarios/payroll/payroll-benchmark-020/baseline_reset.sql", "setup_script": "evaluation_scenarios/payroll/payroll-benchmark-020/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-benchmark-020/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-benchmark-020/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["PAY-2026-0020-A"], "expected_root_cause_concepts": ["correlated multi-table evidence traces the failure to one interrupted transition"], "expected_tables": ["time_entries", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.time_entries", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-PAYROLL-120", "PAY-2026-0020-A", "correlated multi-table evidence traces the failure to one interrupted transition"], "acceptable_fix_concepts": ["repair the transition and reconcile each correlated downstream state"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_4"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "A synthetic payroll production-support incident testing multi table investigation without exposing ground truth to runtime code.", "expected_relationships": ["eval.time_entries.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation"]}

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Entity extraction

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Prefer exact business identifiers and reject surrogate candidate IDs when an exact key is supplied.

payroll-pilot-001 - Critical

Question: Why is employee EMP-1042 missing from payroll run RUN-2026-07-A?

Expected answer: {"scenario_id": "payroll-pilot-001", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "missing_workflow_output", "difficulty": "medium", "question": "Why is employee EMP-1042 missing from payroll run RUN-2026-07-A?", "baseline_script": "evaluation_databases/payroll/sql/02_seed.sql", "setup_script": "evaluation_scenarios/payroll/payroll-pilot-001/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-pilot-001/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-pilot-001/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["EMP-1042"], "expected_root_cause_concepts": ["required downstream work item was not created after a completed upstream transition"], "expected_tables": ["exceptions", "integration_messages", "pay_groups"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_payroll_operations_1"], "required_evidence": ["EVAL-PAYROLL-001", "EMP-1042", "Exception"], "acceptable_fix_concepts": ["transactional workflow completion"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_1"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "Legacy evaluation"}

scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225608-F1760299 Investigation Mode: INVESTIGATION Detected Intent: MISSING_DATA Response Type: insufficient_evidence Extracted Entities: exact_id_or_code=MSG-EMP-1042, business_identifier=MSG-EMP-1042, exact_id_or_code=MSG-RUN-2026-07-A, business_identifier=MSG-RUN-2026-07-A, possible_table_or_column=Why, possible_table_or_column=is, possible_table_or_column=employee, possible_column=EMP, possible_table_or_column=payroll, possible_table_or_column=run, possible_column=RUN AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: MISSING_DATA - User Goal: Answer the user question with database evidence: Why is employee EMP-1042 missing from payroll run RUN-2026-07-A? - Working Hypothesis: A downstream record should have been generated but is missing. ## Stage 2 - Discover Context - Target Object: payroll_items - Target Reason: Selected from main problem phrase 'missing from payroll' (missing); parent and secondary-cause terms do not override the target. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: payroll_runs, employees, batch_runs, tax_filings, time_entries, integration_messages - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 18d7ade7-7dd3-428b-b4b3-67cfba8fed58 - database_name: eval_payroll - schema_name: default - database_engine: mysql - connection_string_database: eval_payroll - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|18d7ade7-7dd3-428b-b4b3-67cfba8fed58|eval_payroll - Discovered tables: audit_history, batch_runs, deductions, departments, employees, employment_history, evaluation_marker, exceptions, integration_messages, leave_requests, pay_groups, pay_periods, payments, payroll_items, payroll_runs, tax_filings, time_entries - Discovered procedures: None

Root cause: Configuration issue

Evidence collected: - No confirming rows were returned for the reported issue.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

payroll-pilot-002 - Critical

Question: Why was payment PAY-7003 produced twice?

Expected answer: {"scenario_id": "payroll-pilot-002", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "duplicate_integration", "difficulty": "hard", "question": "Why was payment PAY-7003 produced twice?", "baseline_script": "evaluation_databases/payroll/sql/02_seed.sql", "setup_script": "evaluation_scenarios/payroll/payroll-pilot-002/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-pilot-002/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-pilot-002/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["RUN-2026-07-A"], "expected_root_cause_concepts": ["integration retry inserted the same business event without an idempotency guard"], "expected_tables": ["exceptions", "integration_messages", "pay_periods"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_payroll_operations_1"], "required_evidence": ["EVAL-PAYROLL-002", "RUN-2026-07-A", "Exception"], "acceptable_fix_concepts": ["idempotency key"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"]}

"scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_2"],
"expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"],
"expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [],
"evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation":
[], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false,
"human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [],
"estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Entity extraction

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Prefer exact business identifiers and reject surrogate candidate IDs when an exact key is supplied.

payroll-pilot-003 - Critical

Question: Why were overtime hours from TIME-8821 not included?

Expected answer: {"scenario_id": "payroll-pilot-003", "domain": "payroll", "database_engine": "sqlserver",
"database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory":
"incorrect_selection", "difficulty": "hard", "question": "Why were overtime hours from TIME-8821 not included?",
"baseline_script": "evaluation_databases/payroll/sql/02_seed.sql", "setup_script":
"evaluation_scenarios/payroll/payroll-pilot-003/inject.sql", "verification_script":
"evaluation_scenarios/payroll/payroll-pilot-003/verify.sql", "cleanup_script":
"evaluation_scenarios/payroll/payroll-pilot-003/cleanup.sql", "expected_response_type": "confirmed_root_cause",
"expected_entities": ["PAY-7003"], "expected_root_cause_concepts": ["selection used historical assignment
instead of the active work item"], "expected_tables": ["exceptions", "integration_messages", "time_entries"],
"expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects":
["eval.vw_payroll_operations_1"], "required_evidence": ["EVAL-PAYROLL-003", "PAY-7003", "Exception"],
"acceptable_fix_concepts": ["active assignment filtering"], "prohibited_claims": ["objects or evidence not returned
by the database", "production impact inferred from synthetic data"], "critical_failure_rules":
["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"],
"scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_3"],
"expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"],
"expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [],
"evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation":
[], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false,
"human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [],
"estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence verification

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Require every confirmed claim to cite persisted evidence matching the expected key.

payroll-pilot-004 - Critical

Question: Why does payroll run RUN-2026-07-A remain Processing?

Expected answer: {"scenario_id": "payroll-pilot-004", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "stalled_status", "difficulty": "medium", "question": "Why does payroll run RUN-2026-07-A remain Processing?", "baseline_script": "evaluation_databases/payroll/sql/02_seed.sql", "setup_script": "evaluation_scenarios/payroll/payroll-pilot-004/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-pilot-004/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-pilot-004/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["TIME-8821"], "expected_root_cause_concepts": ["integration failure prevented the terminal status transition"], "expected_tables": ["exceptions", "integration_messages", "leave_requests"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_payroll_operations_1"], "required_evidence": ["EVAL-PAYROLL-004", "TIME-8821", "Exception"], "acceptable_fix_concepts": ["retry and reconciliation"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_4"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Entity extraction

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Prefer exact business identifiers and reject surrogate candidate IDs when an exact key is supplied.

payroll-pilot-005 - Critical

Question: Is there enough evidence to confirm TAX-2026-07 was filed late?

Expected answer: {"scenario_id": "payroll-pilot-005", "domain": "payroll", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "evidence_safety", "subcategory": "insufficient_evidence", "difficulty": "hard", "question": "Is there enough evidence to confirm TAX-2026-07 was filed late?", "baseline_script": "evaluation_databases/payroll/sql/02_seed.sql", "setup_script": "evaluation_scenarios/payroll/payroll-pilot-005/inject.sql", "verification_script": "evaluation_scenarios/payroll/payroll-pilot-005/verify.sql", "cleanup_script": "evaluation_scenarios/payroll/payroll-pilot-005/cleanup.sql", "expected_response_type": "insufficient_evidence", "expected_entities": ["TAX-2026-07"], "expected_root_cause_concepts": [], "expected_tables": ["exceptions", "integration_messages", "payroll_runs"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_payroll_operations_1"], "required_evidence": ["EVAL-PAYROLL-005", "TAX-2026-07", "absence of conclusive timestamps"], "acceptable_fix_concepts": ["collect additional timestamp evidence"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_payroll_workflow_5"], "expected_functions": ["eval.fn_payroll_active_status"], "expected_triggers": ["eval.tr_employees_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

shipping-benchmark-001 - Critical

Question: Investigate why shipment SHP-2026-0001-A is marked Failed even though its request was accepted.

Expected answer: {"scenario_id": "shipping-benchmark-001", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "exact_entity_lookup", "subcategory": "exact_entity_lookup", "difficulty": "easy", "question": "Investigate why shipment SHP-2026-0001-A is marked Failed even though its request was accepted.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-001/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-001/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-001/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-001/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-2026-0001-A"], "expected_root_cause_concepts": ["the exact business key resolves to one failed record"], "expected_tables": ["shipments", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.shipments", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-101", "SHP-2026-0001-A", "the exact business key resolves to one failed record"], "acceptable_fix_concepts": ["restore the record through the owning workflow after validating the request"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_1"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing exact entity lookup without exposing ground truth to runtime code.", "expected_relationships": ["eval.shipments.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_busi

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225712-A7C88D62 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=SHP-2026-0001-A, business_identifier=SHP-2026-0001-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=shipment, possible_column=SHP, possible_table_or_column=is, possible_table_or_column=marked, possible_table_or_column=even, possible_table_or_column=though, possible_table_or_column=its, possible_table_or_column=request, possible_table_or_column=was, possible_table_or_column=accepted AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why shipment SHP-2026-0001-A is marked Failed even though its request was accepted. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: shipment_milestones - Target Reason: Selected from main problem phrase 'failed even though' (failed); parent and secondary-cause terms do not override the target. - Business Key: Not determined (No non-primary natural

key column was identified.) - Supporting Objects: shipments, transport_work_orders, invoices, dangerous_goods, customs_holds, vw_shipping_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 19eec4fd-2e15-4480-a2c4-27a764982d73 - database_name: eval_shipping - schema_name: default - database_engine: mysql - connection_string_database: eval_shipping - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|19eec4fd-2e15-4480-a2c4-27a764982d73|eval_shipping - Discovered tables: audit_history, batch_runs, bills_of_lading, bookings, carrier_assignments, containe

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in shipment_milestones: No rows returned - SQL - Summarize analytical results in shipments: 1 row(s) returned - SQL - Summarize analytical results in transport_work_orders: No rows returned - SQL - Prove requested entity exists in shipment_milestones: No rows returned - SQL - Prove requested entity exists in shipments: 1 row(s) returned - SQL - Prove requested entity exists in transport_work_orders: No rows returned - SQL - Inspect relevant rows in shipment_milestones: No rows returned - SQL - Inspect relevant rows in shipments: 1 row(s) returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

shipping-benchmark-002 - Critical

Question: Investigate the failed shipment reported as SHP-2026-0002; resolve the complete identifier before diagnosing it.

Expected answer: {"scenario_id": "shipping-benchmark-002", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "partial_entity_resolution", "subcategory": "partial_entity_resolution", "difficulty": "easy", "question": "Investigate the failed shipment reported as SHP-2026-0002; resolve the complete identifier before diagnosing it.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-002/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-002/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-002/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-002/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-2026-0002-A"], "expected_root_cause_concepts": ["a safe candidate lookup resolves the partial identifier to one record"], "expected_tables": ["bookings", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.bookings", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-102", "SHP-2026-0002-A", "a safe candidate lookup resolves the partial identifier to one record"], "acceptable_fix_concepts": ["confirm the resolved identifier and retry through the supported workflow"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_2"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing partial entity resolution without exposing ground truth to runtime code.", "expected_relationships": ["eval.bookings.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest cont

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Entity extraction

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Prefer exact business identifiers and reject surrogate candidate IDs when an exact key is supplied.

shipping-benchmark-003 - Critical

Question: Investigate shipment SHP-2026-0003; support reported a failure but did not provide the final suffix.

Expected answer: {"scenario_id": "shipping-benchmark-003", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "ambiguous_entity_resolution", "subcategory": "ambiguous_entity_resolution", "difficulty": "hard", "question": "Investigate shipment SHP-2026-0003; support reported a failure but did not provide the final suffix.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-003/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-003/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-003/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-003/cleanup.sql", "expected_response_type": "multiple_possible_causes", "expected_entities": ["SHP-2026-0003"], "expected_root_cause_concepts": ["the partial identifier matches multiple plausible records and investigation must stop"], "expected_tables": ["container_assignments", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.container_assignments", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-103", "SHP-2026-0003", "the partial identifier matches multiple plausible records and investigation must stop"], "acceptable_fix_concepts": ["request user selection of the correct candidate before continuing"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_3"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing ambiguous entity resolution without exposing ground truth to runtime code.", "expected_relationships": ["eval.container_assignments.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows"]}

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

shipping-benchmark-004 - Critical

Question: Investigate why completed shipment SHP-2026-0004-A has no downstream processing record.

Expected answer: {"scenario_id": "shipping-benchmark-004", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "missing_downstream_record", "subcategory": "missing_downstream_record", "difficulty": "medium", "question": "Investigate why completed shipment SHP-2026-0004-A has no downstream processing record.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-004/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-004/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-004/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-004/cleanup.sql", "expected_response_type":

"confirmed_root_cause", "expected_entities": ["SHP-2026-0004-A"], "expected_root_cause_concepts": ["the upstream record completed but its correlated integration record is missing"], "expected_tables": ["container_events", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.container_events", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-104", "SHP-2026-0004-A", "the upstream record completed but its correlated integration record is missing"], "acceptable_fix_concepts": ["reconcile the missing downstream work through an idempotent workflow"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_4"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing missing downstream record without exposing ground truth to runtime code.", "expected_relationships": ["eval.container_events.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answe

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

shipping-benchmark-005 - Critical

Question: Investigate why shipment SHP-2026-0005-A produced two processing messages for one business request.

Expected answer: {"scenario_id": "shipping-benchmark-005", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "duplicate_transaction", "subcategory": "duplicate_transaction", "difficulty": "medium", "question": "Investigate why shipment SHP-2026-0005-A produced two processing messages for one business request.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-005/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-005/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-005/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-005/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-2026-0005-A"], "expected_root_cause_concepts": ["two correlated processing events represent one business action"], "expected_tables": ["shipment_milestones", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.shipment_milestones", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-105", "SHP-2026-0005-A", "two correlated processing events represent one business action"], "acceptable_fix_concepts": ["deduplicate using a stable business idempotency key"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_5"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing duplicate transaction without exposing ground truth to runtime code.", "expected_relationships": ["eval.shipment_milestones.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational

evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated applicat

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225746-9106E75C Investigation Mode: INVESTIGATION Detected Intent: DUPLICATE_DATA Response Type: insufficient_evidence Extracted Entities: exact_id_or_code=SHP-2026-0005-A, business_identifier=SHP-2026-0005-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=shipment, possible_column=SHP, possible_table_or_column=produced, possible_table_or_column=two, possible_table_or_column=processing, possible_table_or_column=messages, possible_table_or_column=for, possible_table_or_column=one, possible_table_or_column=business, possible_table_or_column=request AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: DUPLICATE_DATA - User Goal: Answer the user question with database evidence: Investigate why shipment SHP-2026-0005-A produced two processing messages for one business request. - Working Hypothesis: There should be only one valid record for the affected business key. ## Stage 2 - Discover Context - Target Object: integration_messages - Target Reason: Selected from main problem phrase 'two processing messages' (duplicate); parent and secondary-cause terms do not override the target. - Business Key: BusinessKey (Inferred from duplicate evidence grouped by a non-primary business column.) - Supporting Objects: shipment_milestones, shipments, transport_work_orders, invoices, dangerous_goods, vw_shipping_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 19eec4fd-2e15-4480-a2c4-27a764982d73 - database_name: eval_shipping - schema_name: default - database_engine: mysql - connection_string_database: eval_shipping - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|19eec4fd-2e15-4480-a2c4-27a764982d73|eval_shipping - Discovered tables: audit_history, batch_runs, bills_of_lading, boo

Root cause: Configuration issue

Evidence collected: - Supplied business key found in returned evidence: SHP-2026-0005-A.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

shipping-benchmark-006 - Critical

Question: Investigate why shipment SHP-2026-0006-A stopped between validation and completion.

Expected answer: {"scenario_id": "shipping-benchmark-006", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "workflow_interruption", "subcategory": "workflow_interruption", "difficulty": "medium", "question": "Investigate why shipment SHP-2026-0006-A stopped between validation and completion.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-006/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-006/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-006/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-006/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-2026-0006-A"], "expected_root_cause_concepts": ["workflow history ends at an intermediate state with an exception"], "expected_tables": ["transport_work_orders", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.transport_work_orders", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-106", "SHP-2026-0006-A", "workflow history ends at an intermediate state with an exception"], "acceptable_fix_concepts": ["resume from the last committed workflow checkpoint"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object"],

"wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_6"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing workflow interruption without exposing ground truth to runtime code.", "expected_relationships": ["eval.transport_work_orders.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application fra

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225754-8A6E81E8 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=SHP-2026-0006-A, business_idenfier=SHP-2026-0006-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=shipment, possible_column=SHP, possible_table_or_column=stopped, possible_table_or_column=between, possible_table_or_column=validation, possible_table_or_column=and, possible_table_or_column=completion AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why shipment SHP-2026-0006-A stopped between validation and completion. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: shipment_milestones - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: shipments, transport_work_orders, invoices, dangerous_goods, customs_holds, vw_shipping_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 19eec4fd-2e15-4480-a2c4-27a764982d73 - database_name: eval_shipping - schema_name: default - database_engine: mysql - connection_string_database: eval_shipping - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|19eec4fd-2e15-4480-a2c4-27a764982d73|eval_shipping - Discovered tables: audit_history, batch_runs, bills_of_lading, bookings, carrier_assignments, container_assignments, container_events, container_master, customers, customs_holds, cus

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in shipment_milestones: 2 row(s) returned - SQL - Summarize analytical results in shipments: 3 row(s) returned - SQL - Summarize analytical results in transport_work_orders: 3 row(s) returned - SQL - Prove requested entity exists in shipment_milestones: No rows returned - SQL - Prove requested entity exists in shipments: No rows returned - SQL - Prove requested entity exists in transport_work_orders: 1 row(s) returned - SQL - Inspect relevant rows in shipment_milestones: No rows returned - SQL - Inspect relevant rows in shipments: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

shipping-benchmark-007 - Critical

Question: Investigate why exception handling left shipment SHP-2026-0007-A in Processing.

Expected answer: {"scenario_id": "shipping-benchmark-007", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "exception_handling", "subcategory": "exception_handling", "difficulty": "medium", "question": "Investigate why exception handling left shipment SHP-2026-0007-A in Processing.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-007/baseline_reset.sql", "setup_script":

"evaluation_scenarios/shipping/shipping-benchmark-007/inject.sql", "verification_script":
 "evaluation_scenarios/shipping/shipping-benchmark-007/verify.sql", "cleanup_script":
 "evaluation_scenarios/shipping/shipping-benchmark-007/cleanup.sql", "expected_response_type":
 "confirmed_root_cause", "expected_entities": ["SHP-2026-0007-A"], "expected_root_cause_concepts": ["the
 exception was recorded but no compensating transition ran"], "expected_tables": ["voyages",
 "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status",
 "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.voyages",
 "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence":
 ["EVAL-SHIPPING-107", "SHP-2026-0007-A", "the exception was recorded but no compensating transition ran"],
 "acceptable_fix_concepts": ["add observable compensation and terminal-state handling"], "prohibited_claims":
 ["objects or evidence not returned by the database", "production impact inferred from synthetic data"],
 "critical_failure_rules": ["fabricated_evidence", "invented_database_object",
 "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
 "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,
 "active": true, "expected_procedures": ["eval.usp_shipping_workflow_7"], "expected_functions":
 ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [],
 "business_description": "A synthetic shipping production-support incident testing exception handling without
 exposing ground truth to runtime code.", "expected_relationships": ["eval.voyages.CorrelationId ->
 eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational
 evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content",
 "unrelated application framework tables"], "required_business_objects": ["eval.voy"]

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225805-6C2C1E5F Investigation Mode:
 INVESTIGATION Detected Intent: PROCESS_FLOW_BREAK Response Type: insufficient_evidence Extracted
 Entities: exact_id_or_code=EX-SHP-2026-0007-A, business_identifier=EX-SHP-2026-0007-A,
 possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=exception,
 possible_table_or_column=handling, possible_table_or_column=left, possible_table_or_column=shipment,
 possible_column=SHP, possible_table_or_column=in, possible_table_or_column=Processing AI-assisted
 reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence
 validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for
 download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale:
 Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. -
 Investigation Goal: PROCESS_FLOW_BREAK - User Goal: Answer the user question with database evidence:
 Investigate why exception handling left shipment SHP-2026-0007-A in Processing. - Working Hypothesis: The
 business process should move from the current step to the next expected step. ## Stage 2 - Discover Context -
 Target Object: exceptions - Target Reason: Selected from metadata/question matches plus SQL evidence on
 those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined
 (No non-primary natural key column was identified.) - Supporting Objects: exceptions, shipments,
 transport_work_orders, invoices, dangerous_goods, vw_shipping_operations_1 - Upstream Objects: None
 discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No
 procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug
 - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id:
 19eec4fd-2e15-4480-a2c4-27a764982d73 - database_name: eval_shipping - schema_name: default -
 database_engine: mysql - connection_string_database: eval_shipping - metadata_cache_key: 678264ff-8b4f-47
 a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|19eec4fd-2e15-4480-a2c4-27a764982d73|ev
 al_shipping - Discovered tables: audit_history, batch_runs, bills_of_lading, bookings, carrier_assignments,
 container_assignments, container_events, container_master, customers, customs_holds, customs_rele

Root cause: Configuration issue

Evidence collected: - Supplied business key found in returned evidence: EX-SHP-2026-0007-A. - Affected rows
 were returned by safe SQL evidence.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

shipping-benchmark-008 - Critical

Question: Investigate why external integration for shipment SHP-2026-0008-A was rejected.

Expected answer: {"scenario_id": "shipping-benchmark-008", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "integration_failure", "subcategory": "integration_failure", "difficulty": "medium", "question": "Investigate why external integration for shipment SHP-2026-0008-A was rejected.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-008/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-008/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-008/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-008/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-2026-0008-A"], "expected_root_cause_concepts": ["the correlated integration message failed and blocked completion"], "expected_tables": ["bills_of_lading", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.bills_of_lading", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-108", "SHP-2026-0008-A", "the correlated integration message failed and blocked completion"], "acceptable_fix_concepts": ["correct the integration contract and safely replay the message"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_8"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing integration failure without exposing ground truth to runtime code.", "expected_relationships": ["eval.bills_of_lading.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"],

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225818-E2D4CB23 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=MSG-SHP-2026-0008-A, business_identifier=MSG-SHP-2026-0008-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=external, possible_table_or_column=integration, possible_table_or_column=for, possible_table_or_column=shipment, possible_column=SHP, possible_table_or_column=was, possible_table_or_column=rejected AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why external integration for shipment SHP-2026-0008-A was rejected. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: integration_messages - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: integration_messages, shipments, transport_work_orders, invoices, dangerous_goods, vw_shipping_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 19eec4fd-2e15-4480-a2c4-27a764982d73 - database_name: eval_shipping - schema_name: default - database_engine: mysql - connection_string_database: eval_shipping - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|19eec4fd-2e15-4480-a2c4-27a764982d73|eval_shipping - Discovered tables: audit_history, batch_runs, bills_of_lading, bookings, carrier_assignments, container_assignments, container_events, container_master, customers, customs_ho

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in shipment_milestones: 2 row(s) returned - SQL - Summarize analytical results in shipments: 3 row(s) returned - SQL - Summarize analytical results in integration_messages: 3 row(s) returned - SQL - Prove requested entity exists in shipment_milestones: No rows returned - SQL - Prove requested entity exists in integration_messages: 1 row(s) returned - SQL - Prove requested entity exists in shipments: No rows returned - SQL - Inspect relevant rows in shipment_milestones: No rows returned - SQL - Inspect relevant rows in integration_messages: 1 row(s) returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

shipping-benchmark-009 - Critical

Question: Investigate why shipment SHP-2026-0009-A remains Queued behind older work.

Expected answer: {"scenario_id": "shipping-benchmark-009", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "queue_backlog", "subcategory": "queue_backlog", "difficulty": "medium", "question": "Investigate why shipment SHP-2026-0009-A remains Queued behind older work.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-009/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-009/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-009/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-009/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-2026-0009-A"], "expected_root_cause_concepts": ["queue evidence shows the entity has not been consumed"], "expected_tables": ["shipments", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.shipments", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-109", "SHP-2026-0009-A", "queue evidence shows the entity has not been consumed"], "acceptable_fix_concepts": ["drain the queue with bounded, monitored processing"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_1"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing queue backlog without exposing ground truth to runtime code.", "expected_relationships": ["eval.shipments.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_business_objects": ["eval.shipments"], "required_workflow": ["ev

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225829-59CBED97 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=SHP-2026-0009-A, business_identifier=SHP-2026-0009-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=shipment, possible_column=SHP, possible_table_or_column=remains, possible_table_or_column=Queued, possible_table_or_column=behind, possible_table_or_column=older, possible_table_or_column=work AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why shipment SHP-2026-0009-A remains Queued behind older work. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: shipments - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business

Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: shipment_milestones, shipments, truck_movements, rail_movements, invoices, vw_shipping_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 19eec4fd-2e15-4480-a2c4-27a764982d73 - database_name: eval_shipping - schema_name: default - database_engine: mysql - connection_string_database: eval_shipping - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|19eec4fd-2e15-4480-a2c4-27a764982d73|eval_shipping - Discovered tables: audit_history, batch_runs, bills_of_lading, bookings, carrier_assignments, container_assignments, container_events, container_master, customers, customs_holds, customs_releases, damage_reports

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in transport_work_orders: 2 row(s) returned - SQL - Summarize analytical results in shipment_milestones: 2 row(s) returned - SQL - Summarize analytical results in shipments: 4 row(s) returned - SQL - Prove requested entity exists in transport_work_orders: No rows returned - SQL - Prove requested entity exists in shipment_milestones: No rows returned - SQL - Prove requested entity exists in shipments: 1 row(s) returned - SQL - Inspect relevant rows in transport_work_orders: No rows returned - SQL - Inspect relevant rows in shipment_milestones: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

shipping-benchmark-010 - Critical

Question: Investigate why all retries for shipment SHP-2026-0010-A failed without a terminal status.

Expected answer: {"scenario_id": "shipping-benchmark-010", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "retry_failure", "subcategory": "retry_failure", "difficulty": "hard", "question": "Investigate why all retries for shipment SHP-2026-0010-A failed without a terminal status.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-010/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-010/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-010/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-010/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-2026-0010-A"], "expected_root_cause_concepts": ["retry evidence records exhausted attempts and the entity remained pending"], "expected_tables": ["bookings", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.bookings", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-110", "SHP-2026-0010-A", "retry evidence records exhausted attempts and the entity remained pending"], "acceptable_fix_concepts": ["route exhausted retries to reviewed dead-letter recovery"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_2"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing retry failure without exposing ground truth to runtime code.", "expected_relationships": ["eval.bookings.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_businesses": []}

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

shipping-benchmark-011 - Critical

Question: Investigate why audit history says shipment SHP-2026-0011-A completed while its business record says Failed.

Expected answer: {"scenario_id": "shipping-benchmark-011", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "audit_history_inconsistency", "subcategory": "audit_history_inconsistency", "difficulty": "hard", "question": "Investigate why audit history says shipment SHP-2026-0011-A completed while its business record says Failed.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-011/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-011/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-011/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-011/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-2026-0011-A"], "expected_root_cause_concepts": ["business and audit states conflict for the same correlation"], "expected_tables": ["container_assignments", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.container_assignments", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-111", "SHP-2026-0011-A", "business and audit states conflict for the same correlation"], "acceptable_fix_concepts": ["reconcile history from the authoritative transaction boundary"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_3"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing audit history inconsistency without exposing ground truth to runtime code.", "expected_relationships": ["eval.container_assignments.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer man"]}

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225849-6C40CBCF Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=SHP-2026-0011-A, business_identifier=SHP-2026-0011-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=audit, possible_table_or_column=history, possible_table_or_column=says, possible_table_or_column=shipment, possible_column=SHP, possible_table_or_column=completed, possible_table_or_column=while, possible_table_or_column=its, possible_table_or_column=business AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why audit history says shipment SHP-2026-0011-A completed while its business record says Failed. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: audit_history - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: shipment_milestones, shipments, transport_work_orders, invoices, dangerous_goods,

vw_shipping_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 19eec4fd-2e15-4480-a2c4-27a764982d73 - database_name: eval_shipping - schema_name: default - database_engine: mysql - connection_string_database: eval_shipping - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|19eec4fd-2e15-4480-a2c4-27a764982d73|eval_shipping - Discovered tables: audit_history, batch_runs, bills_of_lading, bookings, carrier_assignments, conta

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in audit_history: No rows returned - SQL - Summarize analytical results in shipment_milestones: No rows returned - SQL - Summarize analytical results in shipments: No rows returned - SQL - Prove requested entity exists in audit_history: No rows returned - SQL - Prove requested entity exists in shipment_milestones: No rows returned - SQL - Prove requested entity exists in shipments: No rows returned - SQL - Inspect relevant rows in audit_history: No rows returned - SQL - Inspect relevant rows in shipment_milestones: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

shipping-benchmark-012 - Critical

Question: Investigate why shipment SHP-2026-0012-A failed validation after a reference lookup.

Expected answer: {"scenario_id": "shipping-benchmark-012", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "missing_reference_data", "subcategory": "missing_reference_data", "difficulty": "easy", "question": "Investigate why shipment SHP-2026-0012-A failed validation after a reference lookup.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-012/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-012/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-012/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-012/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-2026-0012-A"], "expected_root_cause_concepts": ["required active reference data is absent from discovered relationships"], "expected_tables": ["container_events", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.container_events", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-112", "SHP-2026-0012-A", "required active reference data is absent from discovered relationships"], "acceptable_fix_concepts": ["restore governed reference data before replaying the workflow"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_4"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing missing reference data without exposing ground truth to runtime code.", "expected_relationships": ["eval.container_events.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated appli"]

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

shipping-benchmark-013 - Critical

Question: Investigate why workflow procedure processing left shipment SHP-2026-0013-A in the wrong status.

Expected answer: {"scenario_id": "shipping-benchmark-013", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "stored_procedure_defect", "subcategory": "stored_procedure_defect", "difficulty": "hard", "question": "Investigate why workflow procedure processing left shipment SHP-2026-0013-A in the wrong status.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-013/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-013/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-013/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-013/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-2026-0013-A"], "expected_root_cause_concepts": ["procedure definition targets the workflow object but the observed transition is incorrect"], "expected_tables": ["shipment_milestones", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.shipment_milestones", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-113", "SHP-2026-0013-A", "procedure definition targets the workflow object but the observed transition is incorrect"], "acceptable_fix_concepts": ["version and test the procedure correction before controlled replay"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_5"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing stored procedure defect without exposing ground truth to runtime code.", "expected_relationships": ["eval.shipment_milestones.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another"]}

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225906-0947AB6E Investigation Mode: INVESTIGATION Detected Intent: PROCESS_FLOW_BREAK Response Type: insufficient_evidence Extracted Entities: exact_id_or_code=SHP-2026-0013-A, business_identifier=SHP-2026-0013-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=workflow, possible_table_or_column=procedure, possible_table_or_column=processing, possible_table_or_column=left, possible_table_or_column=shipment, possible_column=SHP, possible_table_or_column=in, possible_table_or_column=the, possible_table_or_column=wrong, possible_table_or_column=status AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: PROCESS_FLOW_BREAK - User Goal: Answer the user question with database evidence: Investigate why workflow procedure processing left shipment SHP-2026-0013-A in the wrong status. - Working Hypothesis: The business process should move from the current step to the next expected step. ## Stage 2 - Discover Context - Target Object: shipment_milestones - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: shipments, transport_work_orders, invoices, dangerous_goods, customs_holds, vw_shipping_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 19eec4fd-2e15-4480-a2c4-27a764982d73 - database_name: eval_shipping - schema_name: default - database_engine: mysql - connection_string_database: eval_shipping - metadata_cache_key: 678264ff-8b4f-47

a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|19eec4fd-2e15-4480-a2c4-27a764982d73|eval_shipping - Discovered tables: audit_history, batch_runs, bills_of_lading, bookings,

Root cause: Configuration issue

Evidence collected: - Supplied business key found in returned evidence: SHP-2026-0013-A. - Affected rows were returned by safe SQL evidence.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

shipping-benchmark-014 - Critical

Question: Investigate why updating shipment SHP-2026-0014-A produced no consistent audit outcome.

Expected answer: {"scenario_id": "shipping-benchmark-014", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "trigger_failure", "subcategory": "trigger_failure", "difficulty": "hard", "question": "Investigate why updating shipment SHP-2026-0014-A produced no consistent audit outcome.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-014/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-014/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-014/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-014/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-2026-0014-A"], "expected_root_cause_concepts": ["trigger metadata is relevant but the expected correlated audit transition is absent"], "expected_tables": ["transport_work_orders", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.transport_work_orders", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-114", "SHP-2026-0014-A", "trigger metadata is relevant but the expected correlated audit transition is absent"], "acceptable_fix_concepts": ["repair trigger behavior and reconcile missing audit events"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_6"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing trigger failure without exposing ground truth to runtime code.", "expected_relationships": ["eval.transport_work_orders.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest conten"]

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225918-42F92AFE Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=SHP-2026-0014-A, business_identifier=SHP-2026-0014-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=updating, possible_table_or_column=shipment, possible_column=SHP, possible_table_or_column=produced, possible_table_or_column=no, possible_table_or_column=consistent, possible_table_or_column=audit, possible_table_or_column=outcome AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why updating shipment SHP-2026-0014-A produced no consistent audit outcome. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: shipment_milestones - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: audit_history, shipments,

transport_work_orders, invoices, dangerous_goods, vw_shipping_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ### Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 19eec4fd-2e15-4480-a2c4-27a764982d73 - database_name: eval_shipping - schema_name: default - database_engine: mysql - connection_string_database: eval_shipping - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|19eec4fd-2e15-4480-a2c4-27a764982d73|eval_shipping - Discovered tables: audit_history, batch_runs, bills_of_lading, bookings, carrier_assignments, container_assignments, container_events, container_mas

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in shipment_milestones: 2 row(s) returned - SQL - Summarize analytical results in shipments: 3 row(s) returned - SQL - Summarize analytical results in audit_history: 3 row(s) returned - SQL - Prove requested entity exists in shipment_milestones: No rows returned - SQL - Prove requested entity exists in audit_history: No rows returned - SQL - Prove requested entity exists in shipments: No rows returned - SQL - Inspect relevant rows in shipment_milestones: No rows returned - SQL - Inspect relevant rows in audit_history: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

shipping-benchmark-015 - Critical

Question: Investigate why batch processing failed for shipment SHP-2026-0015-A while neighboring items completed.

Expected answer: {"scenario_id": "shipping-benchmark-015", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "batch_processing_failure", "subcategory": "batch_processing_failure", "difficulty": "medium", "question": "Investigate why batch processing failed for shipment SHP-2026-0015-A while neighboring items completed.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-015/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-015/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-015/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-015/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-2026-0015-A"], "expected_root_cause_concepts": ["the correlated batch or integration evidence isolates one failed item"], "expected_tables": ["voyages", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.voyages", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-115", "SHP-2026-0015-A", "the correlated batch or integration evidence isolates one failed item"], "acceptable_fix_concepts": ["restart only the failed item with checkpoint protection"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_7"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing batch processing failure without exposing ground truth to runtime code.", "expected_relationships": ["eval.voyages.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application f

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

shipping-benchmark-016 - Critical

Question: Investigate intermittent competing updates for shipment SHP-2026-0016-A that produced inconsistent terminal states.

Expected answer: {"scenario_id": "shipping-benchmark-016", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "concurrency_race_condition", "subcategory": "concurrency_race_condition", "difficulty": "expert", "question": "Investigate intermittent competing updates for shipment SHP-2026-0016-A that produced inconsistent terminal states.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-016/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-016/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-016/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-016/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-2026-0016-A"], "expected_root_cause_concepts": ["same-correlation events at overlapping timestamps show a lost-update race"], "expected_tables": ["bills_of_lading", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.bills_of_lading", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-116", "SHP-2026-0016-A", "same-correlation events at overlapping timestamps show a lost-update race"], "acceptable_fix_concepts": ["use optimistic concurrency and retry on version conflict"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_8"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing concurrency race condition without exposing ground truth to runtime code.", "expected_relationships": ["eval.bills_of_lading.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected"]}

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Evidence collection

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Query the expected entity across correlated tables and retain row-level evidence IDs.

shipping-benchmark-017 - Critical

Question: Investigate why shipment SHP-2026-0017-A reports completion although downstream changes were rolled back.

Expected answer: {"scenario_id": "shipping-benchmark-017", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "transaction_rollback", "subcategory": "transaction_rollback", "difficulty": "hard", "question": "Investigate why shipment SHP-2026-0017-A reports completion although downstream changes were rolled back.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-017/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-017/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-017/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-017/cleanup.sql"}

"evaluation_scenarios/shipping/shipping-benchmark-017/cleanup.sql", "expected_response_type":
"confirmed_root_cause", "expected_entities": ["SHP-2026-0017-A"], "expected_root_cause_concepts":
["exception and audit evidence show the unit of work rolled back"], "expected_tables": ["shipments",
"integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status",
"EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.shipments",
"eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence":
["EVAL-SHIPPING-117", "SHP-2026-0017-A", "exception and audit evidence show the unit of work rolled back"],
"acceptable_fix_concepts": ["make the status transition atomic with downstream writes"], "prohibited_claims":
["objects or evidence not returned by the database", "production impact inferred from synthetic data"],
"critical_failure_rules": ["fabricated_evidence", "invented_database_object",
"wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence",
"unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1,
"active": true, "expected_procedures": ["eval.usp_shipping_workflow_1"], "expected_functions":
["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [],
"business_description": "A synthetic shipping production-support incident testing transaction rollback without
exposing ground truth to runtime code.", "expected_relationships": ["eval.shipments.CorrelationId ->
eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational
evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content",
"unrelated application framework tables"],

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225945-BB982FD1 Investigation Mode:
INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type:
multiple_possible_causes Extracted Entities: exact_id_or_code=SHP-2026-0017-A,
business_idenfier=SHP-2026-0017-A, possible_table_or_column=Investigate, possible_table_or_column=why,
possible_table_or_column=shipment, possible_column=SHP, possible_table_or_column=reports,
possible_table_or_column=completion, possible_table_or_column=although,
possible_table_or_column=downstream, possible_table_or_column=changes, possible_table_or_column=were,
possible_table_or_column=rolled, possible_table_or_column=back AI-assisted reasoning: Disabled Reason:
AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable
Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 -
Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require
live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal:
GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate
why shipment SHP-2026-0017-A reports completion although downstream changes were rolled back. - Working
Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target
Object: damage_reports - Target Reason: Selected from metadata/question matches plus SQL evidence on
those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined
(No non-primary natural key column was identified.) - Supporting Objects: shipment_milestones, shipments,
repair_orders, transport_work_orders, invoices, vw_shipping_operations_1 - Upstream Objects: None discovered
- Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read
path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id:
2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 19eec4fd-2e15-4480-a2c4-27a764982d73 -
database_name: eval_shipping - schema_name: default - database_engine: mysql -
connection_string_database: eval_shipping - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f
847a6a-9db2-4ae5-b144-71598ec8fb87|19eec4fd-2e15-4480-a2c4-27a764982d73|eval_shipping - Discovered
tables: audit_history, batch_runs, bills_of_lading

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in damage_reports: 1 row(s) returned - SQL -
Summarize analytical results in shipment_milestones: 2 row(s) returned - SQL - Summarize analytical results in
shipments: 4 row(s) returned - SQL - Prove requested entity exists in damage_reports: No rows returned - SQL -
Prove requested entity exists in shipment_milestones: No rows returned - SQL - Prove requested entity exists in
shipments: 1 row(s) returned - SQL - Inspect relevant rows in damage_reports: No rows returned - SQL - Inspect
relevant rows in shipment_milestones: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

shipping-benchmark-018 - Critical

Question: Investigate why replaying shipment SHP-2026-0018-A repeated a previously accepted operation.

Expected answer: {"scenario_id": "shipping-benchmark-018", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "idempotency_issue", "subcategory": "idempotency_issue", "difficulty": "expert", "question": "Investigate why replaying shipment SHP-2026-0018-A repeated a previously accepted operation.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-018/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-018/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-018/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-018/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-2026-0018-A"], "expected_root_cause_concepts": ["replay created a second correlated operation without an idempotency guard"], "expected_tables": ["bookings", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.bookings", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-118", "SHP-2026-0018-A", "replay created a second correlated operation without an idempotency guard"], "acceptable_fix_concepts": ["enforce a unique idempotency key across retries"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_2"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing idempotency issue without exposing ground truth to runtime code.", "expected_relationships": ["eval.bookings.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content", "unrelated application framework tables"], "required_

Actual answer: ENTITY_NOT_FOUND: No exact or safe partial database match confirmed the supplied identifier.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

shipping-benchmark-019 - Critical

Question: Investigate why shipment SHP-2026-0019-A displays Completed while failure evidence remains open.

Expected answer: {"scenario_id": "shipping-benchmark-019", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "incorrect_business_status", "subcategory": "incorrect_business_status", "difficulty": "easy", "question": "Investigate why shipment SHP-2026-0019-A displays Completed while failure evidence remains open.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-019/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-019/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-019/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-019/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-2026-0019-A"], "expected_root_cause_concepts": ["the business status contradicts active correlated failure evidence"], "expected_tables": ["container_assignments",

"integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.container_assignments", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-119", "SHP-2026-0019-A", "the business status contradicts active correlated failure evidence"], "acceptable_fix_concepts": ["derive status from authoritative workflow completion"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_3"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing incorrect business status without exposing ground truth to runtime code.", "expected_relationships": ["eval.container_assignments.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows from another correlation", "expected-answer manifest content"]

Actual answer: Investigation Complete. Investigation ID: INV-20260716-230005-8AE739FE Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=SHP-2026-0019-A, business_idenfier=SHP-2026-0019-A, possible_table_or_column=Investigate, possible_table_or_column=why, possible_table_or_column=shipment, possible_column=SHP, possible_table_or_column=displays, possible_table_or_column=Completed, possible_table_or_column=while, possible_table_or_column=failure, possible_table_or_column=evidence, possible_table_or_column=remains, possible_table_or_column=open AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Investigate why shipment SHP-2026-0019-A displays Completed while failure evidence remains open. - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: shipment_milestones - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: shipments, transport_work_orders, invoices, dangerous_goods, customs_holds, vw_shipping_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 19eec4fd-2e15-4480-a2c4-27a764982d73 - database_name: eval_shipping - schema_name: default - database_engine: mysql - connection_string_database: eval_shipping - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|19eec4fd-2e15-4480-a2c4-27a764982d73|eval_shipping - Discovered tables: audit_history, batch_runs, bills_of_lading, bookings, carrier_assignments, container_

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in shipment_milestones: No rows returned - SQL - Summarize analytical results in shipments: No rows returned - SQL - Summarize analytical results in transport_work_orders: No rows returned - SQL - Prove requested entity exists in shipment_milestones: No rows returned - SQL - Prove requested entity exists in shipments: No rows returned - SQL - Prove requested entity exists in transport_work_orders: No rows returned - SQL - Inspect relevant rows in shipment_milestones: No rows returned - SQL - Inspect relevant rows in shipments: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

shipping-benchmark-020 - Critical

Question: Investigate the end-to-end failure of shipment SHP-2026-0020-A across business, integration, exception, and audit records.

Expected answer: {"scenario_id": "shipping-benchmark-020", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "multi_table_investigation", "subcategory": "multi_table_investigation", "difficulty": "medium", "question": "Investigate the end-to-end failure of shipment SHP-2026-0020-A across business, integration, exception, and audit records.", "baseline_script": "evaluation_scenarios/shipping/shipping-benchmark-020/baseline_reset.sql", "setup_script": "evaluation_scenarios/shipping/shipping-benchmark-020/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-benchmark-020/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-benchmark-020/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-2026-0020-A"], "expected_root_cause_concepts": ["correlated multi-table evidence traces the failure to one interrupted transition"], "expected_tables": ["container_events", "integration_messages", "exceptions", "audit_history"], "expected_columns": ["BusinessKey", "Status", "EventTime", "CorrelationId", "Details"], "expected_database_objects": ["eval.container_events", "eval.integration_messages", "eval.exceptions", "eval.audit_history"], "required_evidence": ["EVAL-SHIPPING-120", "SHP-2026-0020-A", "correlated multi-table evidence traces the failure to one interrupted transition"], "acceptable_fix_concepts": ["repair the transition and reconcile each correlated downstream state"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "wrong_business_entity_investigated", "confirmed_root_cause_without_supporting_evidence", "unsafe_remediation", "test_data_or_expected_answer_leaked_into_application_prompt"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_4"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "A synthetic shipping production-support incident testing multi table investigation without exposing ground truth to runtime code.", "expected_relationships": ["eval.container_events.CorrelationId -> eval.integration_messages.CorrelationId", "shared BusinessKey and timestamp sequence across operational evidence"], "evidence_exclusions": ["rows fro

Actual answer: AMBIGUOUS_ENTITY: Multiple plausible records matched. Select a candidate or request human review before investigation continues.

Root cause: Reasoning

Evidence collected: No supporting-evidence section persisted.

Suggested fix: Enable evidence-grounded AI reasoning and prohibit confirmation without supporting rows.

shipping-pilot-001 - Critical

Question: Delivery is complete for shipment SHP-5001; why is the empty-return work order missing?

Expected answer: {"scenario_id": "shipping-pilot-001", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "missing_workflow_output", "difficulty": "medium", "question": "Delivery is complete for shipment SHP-5001; why is the empty-return work order missing?", "baseline_script": "evaluation_databases/shipping/sql/02_seed.sql", "setup_script": "evaluation_scenarios/shipping/shipping-pilot-001/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-pilot-001/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-pilot-001/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-5001"], "expected_root_cause_concepts": ["required downstream work item was not created after a completed upstream transition"], "expected_tables": ["shipments", "transport_work_orders", "exceptions"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_shipping_operations_1"], "required_evidence": ["EVAL-SHIPPING-001", "SHP-5001", "Delivered", "missing EMPTY-SHP-5001"], "acceptable_fix_concepts": ["transactional workflow completion"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object",

"confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_1"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225618-161FE745 Investigation Mode: INVESTIGATION Detected Intent: MISSING_DATA Response Type: insufficient_evidence Extracted Entities: exact_id_or_code=SHP-5001, business_identifier=SHP-5001, possible_table_or_column=Delivery, possible_table_or_column=is, possible_table_or_column=complete, possible_table_or_column=for, possible_table_or_column=shipment, possible_column=SHP, possible_table_or_column=why, possible_table_or_column=the, possible_table_or_column=empty, possible_table_or_column=return, possible_table_or_column=work, possible_table_or_column=order AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: MISSING_DATA - User Goal: Answer the user question with database evidence: Delivery is complete for shipment SHP-5001; why is the empty-return work order missing? - Working Hypothesis: A downstream record should have been generated but is missing. ## Stage 2 - Discover Context - Target Object: transport_work_orders - Target Reason: Selected from main problem phrase 'work order missing' (missing); parent and secondary-cause terms do not override the target. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: empty_return_instructions, shipment_milestones, shipments, repair_orders, truck_movements, integration_messages - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 19eec4fd-2e15-4480-a2c4-27a764982d73 - database_name: eval_shipping - schema_name: default - database_engine: mysql - connection_string_database: eval_shipping - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|19eec4fd-2e15-4480-a2c4-27a764982d73|eval_shipping - Discovered tables: audit_history, batch_runs, bills_of_lading, bookings, carrier_assignments, container_assignments, container_events, contain

Root cause: Configuration issue

Evidence collected: - No confirming rows were returned for the reported issue.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

shipping-pilot-002 - Critical

Question: Why does container CONT-5002 have a duplicate terminal gate event?

Expected answer: {"scenario_id": "shipping-pilot-002", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "duplicate_integration", "difficulty": "hard", "question": "Why does container CONT-5002 have a duplicate terminal gate event?", "baseline_script": "evaluation_databases/shipping/sql/02_seed.sql", "setup_script": "evaluation_scenarios/shipping/shipping-pilot-002/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-pilot-002/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-pilot-002/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["CONT-5002"], "expected_root_cause_concepts": ["integration retry inserted the same business event without an idempotency guard"], "expected_tables": ["container_events", "container_assignments", "integration_messages"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_shipping_operations_1"],

"required_evidence": ["EVAL-SHIPPING-002", "CONT-5002", "GATE-CONT-5002 count=2"],
"acceptable_fix_concepts": ["idempotency key"], "prohibited_claims": ["objects or evidence not returned by the
database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence",
"invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1,
"active": true, "expected_procedures": ["eval.usp_shipping_workflow_2"], "expected_functions":
["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [],
"business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [],
"required_business_objects": [], "required_workflow": [], "expected_recommendation": [],
"unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false,
"human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [],
"estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225628-8A62B053 Investigation Mode: INVESTIGATION Detected Intent: DUPLICATE_DATA Response Type: insufficient_evidence Extracted Entities: exact_id_or_code=CONT-5002, business_identifier=CONT-5002, possible_table_or_column=Why, possible_table_or_column=does, possible_table_or_column=container, possible_column=CONT, possible_table_or_column=have, possible_table_or_column=duplicate, possible_table_or_column=terminal, possible_table_or_column=gate, possible_table_or_column=event AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: DUPLICATE_DATA - User Goal: Answer the user question with database evidence: Why does container CONT-5002 have a duplicate terminal gate event? - Working Hypothesis: There should be only one valid record for the affected business key. ## Stage 2 - Discover Context - Target Object: terminals - Target Reason: Selected from main problem phrase 'duplicate terminal gate' (duplicate); parent and secondary-cause terms do not override the target. - Business Key: BusinessKey (Inferred from duplicate evidence grouped by a non-primary business column.) - Supporting Objects: container_events, gate_transactions, container_master, container_assignments, reefer_settings - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 19eec4fd-2e15-4480-a2c4-27a764982d73 - database_name: eval_shipping - schema_name: default - database_engine: mysql - connection_string_database: eval_shipping - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|19eec4fd-2e15-4480-a2c4-27a764982d73|eval_shipping - Discovered tables: audit_history, batch_runs, bills_of_lading, bookings, carrier_assignments, container_assignments, container_events, container_master, customers, customs_holds, customs_releases, damage_reports, dangerous_goods, demurrage_d

Root cause: Configuration issue

Evidence collected: - No confirming rows were returned for the reported issue.

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

shipping-pilot-003 - Critical

Question: Why was the wrong carrier selected for empty work order WO-5003?

Expected answer: {"scenario_id": "shipping-pilot-003", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "incorrect_selection", "difficulty": "hard", "question": "Why was the wrong carrier selected for empty work order WO-5003?", "baseline_script": "evaluation_databases/shipping/sql/02_seed.sql", "setup_script": "evaluation_scenarios/shipping/shipping-pilot-003/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-pilot-003/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-pilot-003/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["WO-5003"], "expected_root_cause_concepts": ["selection used

historical assignment instead of the active work item"], "expected_tables": ["transport_work_orders", "carrier_assignments", "audit_history"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_shipping_operations_1"], "required_evidence": ["EVAL-SHIPPING-003", "WO-5003", "Historical", "MoveType=Empty"], "acceptable_fix_concepts": ["active assignment filtering"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_3"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225637-FD927C8F Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=WO-5003, business_identifier=WO-5003, possible_table_or_column=Why, possible_table_or_column=was, possible_table_or_column=the, possible_table_or_column=wrong, possible_table_or_column=carrier, possible_table_or_column=selected, possible_table_or_column=for, possible_table_or_column=empty, possible_table_or_column=work, possible_table_or_column=order, possible_column=WO AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Why was the wrong carrier selected for empty work order WO-5003? - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: transport_work_orders - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: carrier_assignments, repair_orders, empty_return_instructions, truck_movements, rail_movements - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 19eec4fd-2e15-4480-a2c4-27a764982d73 - database_name: eval_shipping - schema_name: default - database_engine: mysql - connection_string_database: eval_shipping - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|19eec4fd-2e15-4480-a2c4-27a764982d73|eval_shipping - Discovered tables: audit_history, batch_runs, bills_of_lading, bookings, carrier_assignments, container_assignments, container_events, container_master, customers, customs_holds, cu

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in transport_work_orders: 2 row(s) returned - SQL - Summarize analytical results in carrier_assignments: 2 row(s) returned - SQL - Summarize analytical results in repair_orders: 1 row(s) returned - SQL - Prove requested entity exists in transport_work_orders: 1 row(s) returned - SQL - Prove reported condition on transport_work_orders.TransportWorkOrdersId: No rows returned - SQL - Prove requested entity exists in carrier_assignments: No rows returned - SQL - Prove reported condition on carrier_assignments.TransportWorkOrdersId: No rows returned - SQL - Prove reported condition on carrier_assignments.CarrierAssignmentsId: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

shipping-pilot-004 - Critical

Question: Why does shipment SHP-5004 remain In Transit after import discharge?

Expected answer: {"scenario_id": "shipping-pilot-004", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "root_cause", "subcategory": "stalled_status", "difficulty": "medium", "question": "Why does shipment SHP-5004 remain In Transit after import discharge?", "baseline_script": "evaluation_databases/shipping/sql/02_seed.sql", "setup_script": "evaluation_scenarios/shipping/shipping-pilot-004/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-pilot-004/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-pilot-004/cleanup.sql", "expected_response_type": "confirmed_root_cause", "expected_entities": ["SHP-5004"], "expected_root_cause_concepts": ["integration failure prevented the terminal status transition"], "expected_tables": ["shipments", "shipment_milestones", "integration_messages"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_shipping_operations_1"], "required_evidence": ["EVAL-SHIPPING-004", "SHP-5004", "Import Discharge", "Failed"], "acceptable_fix_concepts": ["retry and reconciliation"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_4"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225646-78EBE332 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=SHP-5004, business_identifier=SHP-5004, possible_table_or_column=Why, possible_table_or_column=does, possible_table_or_column=shipment, possible_column=SHP, possible_table_or_column=remain, possible_table_or_column=In, possible_table_or_column=Transit, possible_table_or_column=after, possible_table_or_column=import, possible_table_or_column=discharge AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Why does shipment SHP-5004 remain In Transit after import discharge? - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: shipments - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: shipments, transport_work_orders, invoices, dangerous_goods, customs_holds, vw_shipping_operations_1 - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 19eec4fd-2e15-4480-a2c4-27a764982d73 - database_name: eval_shipping - schema_name: default - database_engine: mysql - connection_string_database: eval_shipping - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|19eec4fd-2e15-4480-a2c4-27a764982d73|eval_shipping - Discovered tables: audit_history, batch_runs, bills_of_lading, bookings, carrier_assignments, container_assignments, container_events, container_master, customers, customs_holds, customs_releases, damage

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in shipment_milestones: 2 row(s) returned - SQL - Summarize analytical results in shipments: 3 row(s) returned - SQL - Summarize analytical results in transport_work_orders: 2 row(s) returned - SQL - Prove requested entity exists in shipment_milestones: No rows

returned - SQL - Prove requested entity exists in shipments: 1 row(s) returned - SQL - Prove requested entity exists in transport_work_orders: No rows returned - SQL - Inspect relevant rows in shipment_milestones: No rows returned - SQL - Inspect relevant rows in shipments: 1 row(s) returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.

shipping-pilot-005 - Critical

Question: Is there enough evidence to confirm container CONT-5005 was delayed?

Expected answer: {"scenario_id": "shipping-pilot-005", "domain": "shipping", "database_engine": "sqlserver", "database_version": "Azure SQL Database / SQL Server 2022", "category": "evidence_safety", "subcategory": "insufficient_evidence", "difficulty": "hard", "question": "Is there enough evidence to confirm container CONT-5005 was delayed?", "baseline_script": "evaluation_databases/shipping/sql/02_seed.sql", "setup_script": "evaluation_scenarios/shipping/shipping-pilot-005/inject.sql", "verification_script": "evaluation_scenarios/shipping/shipping-pilot-005/verify.sql", "cleanup_script": "evaluation_scenarios/shipping/shipping-pilot-005/cleanup.sql", "expected_response_type": "insufficient_evidence", "expected_entities": ["CONT-5005"], "expected_root_cause_concepts": [], "expected_tables": ["container_assignments", "container_events", "shipment_milestones"], "expected_columns": ["BusinessKey", "Status", "CorrelationId", "Details"], "expected_database_objects": ["eval.vw_shipping_operations_1"], "required_evidence": ["EVAL-SHIPPING-005", "CONT-5005", "planned SLA timestamp unavailable"], "acceptable_fix_concepts": ["collect additional timestamp evidence"], "prohibited_claims": ["objects or evidence not returned by the database", "production impact inferred from synthetic data"], "critical_failure_rules": ["fabricated_evidence", "invented_database_object", "confirmed_root_cause_without_supporting_evidence"], "scenario_version": 1, "active": true, "expected_procedures": ["eval.usp_shipping_workflow_5"], "expected_functions": ["eval.fn_shipping_active_status"], "expected_triggers": ["eval.tr_bookings_audit"], "expected_jobs": [], "business_description": "Legacy evaluation scenario", "expected_relationships": [], "evidence_exclusions": [], "required_business_objects": [], "required_workflow": [], "expected_recommendation": [], "unsafe_recommendations": [], "expected_confidence_range": [0.0, 1.0], "expected_human_review": false, "human_review_conditions": [], "expected_ai_judge_category_scores": {}, "expected_citations": [], "estimated_duration_seconds": 60, "estimated_token_usage": 2000, "tags": []}

Actual answer: Investigation Complete. Investigation ID: INV-20260716-225659-A0B17043 Investigation Mode: INVESTIGATION Detected Intent: GENERAL_DATABASE_QUESTION Response Type: multiple_possible_causes Extracted Entities: exact_id_or_code=CONT-5005, business_idenfier=CONT-5005, possible_table_or_column=Is, possible_table_or_column=there, possible_table_or_column=enough, possible_table_or_column=evidence, possible_table_or_column=to, possible_table_or_column=confirm, possible_table_or_column=container, possible_column=CONT, possible_table_or_column=was, possible_table_or_column=delayed AI-assisted reasoning: Disabled Reason: AI_REASONING_ENABLED=false Evidence package sent: No LLM evidence validation: Not applicable Evidence citations: Not applicable Professional report files have been generated for download. ## Stage 1 - Understand the Question - Investigation Mode: INVESTIGATION - Mode Rationale: Question appears to require live evidence, metadata, safe SQL, evidence gate, and root-cause reasoning. - Investigation Goal: GENERAL_DATABASE_QUESTION - User Goal: Answer the user question with database evidence: Is there enough evidence to confirm container CONT-5005 was delayed? - Working Hypothesis: The database evidence should explain the reported behavior. ## Stage 2 - Discover Context - Target Object: container_assignments - Target Reason: Selected from metadata/question matches plus SQL evidence on those matched candidates; returned rows alone cannot select an affected object. - Business Key: Not determined (No non-primary natural key column was identified.) - Supporting Objects: container_master, container_assignments, reefer_settings, gate_transactions, empty_return_instructions - Upstream Objects: None discovered - Downstream Objects: None discovered - Write Path: No write path confirmed - Read Path: No procedure read path confirmed - Documents Used: No matching uploaded documents found ## Metadata Debug - workspace_id: 2f847a6a-9db2-4ae5-b144-71598ec8fb87 - connection_id: 19eec4fd-2e15-4480-a2c4-27a764982d73 - database_name: eval_shipping - schema_name: default - database_engine: mysql - connection_string_database: eval_shipping - metadata_cache_key: 678264ff-8b4f-47a5-89dc-a90b9e371dde|2f847a6a-9db2-4ae5-b144-71598ec8fb87|19eec4fd-2e15-4480-a2c4-27a764982d73|ev

al_shipping - Discovered tables: audit_history, batch_runs, bills_of_lading, bookings, carrier_assignments, container_assignments, container_events, container_master, customers, customs_holds, customs_

Root cause: Configuration issue

Evidence collected: - SQL - Summarize analytical results in container_events: 17 row(s) returned - SQL - Summarize analytical results in container_master: 1 row(s) returned - SQL - Summarize analytical results in container_assignments: 1 row(s) returned - SQL - Prove requested entity exists in container_events: No rows returned - SQL - Prove requested entity exists in container_master: No rows returned - SQL - Prove requested entity exists in container_assignments: 1 row(s) returned - SQL - Inspect relevant rows in container_events: No rows returned - SQL - Inspect relevant rows in container_master: No rows returned

Suggested fix: Enable and verify the production-equivalent reasoning and evidence feature flags.